Copenhagen Business School

Bachelor Thesis

The Effect of Sentiment on Public Equities in Sweden

Investor Sentiment and Cross-Sectional Return Predictability in the Swedish Equity Market

Author	Jonathan Stausholm Carlsen
Student number	170414
Programme	BSc in Economics and Business Administration
Supervisor	Jimmy Martinez-Correa
Course	Behavioral Finance
Term	Spring 2026
Date	May 2026
Pages	40
Words	xxx
Characters	xxx

Contents

Abstract

1 Introduction, Motivation and Purpose

In the efficient-market framework, persistent return predictability requires compensation for risk, market frictions, or information that prices have yet to incorporate (Fama, 1970, 1991). Similarly, the capital asset pricing model links expected returns to exposure to systematic risk (Sharpe, 1964). Within this benchmark, a systematic relation between investor sentiment and future cross-sectional returns requires either a risk-based interpretation or a pricing effect that arbitrage removes only slowly.

Behavioural finance provides a competing interpretation. If investors hold heterogeneous or biased beliefs, and if arbitrage is limited by short-sale constraints, financing risk, or model uncertainty, sentiment-driven demand can move prices away from fundamental values. Sentiment should have its strongest price effects among firms whose values are difficult to estimate and whose mispricing is costly to arbitrage. Baker and Wurgler formalize this prediction by arguing that sentiment should matter most for small, young, volatile, unprofitable, non-dividend-paying, intangible, distressed, or otherwise speculative firms (Baker and Wurgler, 2006, 2007). *The empirical problem is whether return differences across Swedish equities are consistent with rational risk compensation alone, or whether they also contain patterns consistent with sentiment-driven mispricing.*

Swedish evidence has mainly focused on constructing a local sentiment index or studying broad market-level relations. Zhang and Zhang construct a Swedish sentiment index using economic sentiment, consumer confidence, turnover, IPO activity, IPO returns, and an equity-to-debt proxy (Zhang and Zhang, 2023). Their evidence establishes the Swedish proxy family used to measure local sentiment. A cross-sectional test remains open for Swedish firms with different exposure to valuation uncertainty and limits to arbitrage.

Investor sentiment is not directly observed, and the proxies used to measure it can also capture macroeconomic and financial conditions. Survey confidence, turnover, IPO activity, financing composition, and dividend-valuation premia may reflect optimism, but they may also reflect business-cycle conditions, liquidity, discount rates, risk appetite, or firms' financing opportunities. The empirical design therefore treats sentiment as a latent market-state variable measured with error rather than as a directly observed behavioral shock. For this reason, the analysis compares a raw sentiment index with a macro-adjusted index and interprets the results as predictive evidence, not as clean causal identification of sentiment-driven mispricing.

The empirical test constructs a Sweden-specific sentiment measure and links it to monthly return differences across characteristic-sorted portfolios. Beginning-of-period sentiment enters as a predictive sentiment state for subsequent relative returns. The cross-sectional design extends Swedish sentiment research from market-level index relations toward firm characteristics that theory identifies as sentiment-prone.

1.1 Research Question

> Does beginning-of-period investor sentiment predict relative future returns in the Swedish public equity market, and is this predictive relation stronger for stocks with subjective valuations and greater limits to arbitrage?

1.1.1 Hypotheses

The research question implies testable predictions from the cross-sectional sentiment framework in Baker and Wurgler (2006) and Baker and Wurgler (2007). If sentiment is reflected in relative prices, beginning-of-period sentiment should forecast later return differences most clearly among firms whose valuations are more subjective and whose mispricing is harder to arbitrage. Stambaugh et al. (2012) sharpen this prediction by linking high sentiment to overpricing that is difficult to correct when pessimistic investors face short-sale constraints. Applied to the Swedish setting, these arguments imply that lagged sentiment should predict relative future returns, that the predictive relation should be strongest for Swedish stocks with higher valuation uncertainty or stronger limits to arbitrage, and that the negative predictive relation should appear more clearly in the sentiment-prone leg than in the sentiment-insensitive leg.

- **H1.** Consistent with the cross-sectional sentiment framework in Baker and Wurgler (2006), beginning-of-period investor sentiment predicts relative future returns in the Swedish equity cross-section.
- **H2.** Consistent with Baker and Wurgler (2006) and Baker and Wurgler (2007), this predictive relation is stronger for stocks with subjective valuations or limits to arbitrage.
- **H3.** Consistent with the short-sale asymmetry mechanism in Stambaugh et al. (2012), the negative predictive relation after high beginning-of-period sentiment is stronger for the sentiment-prone portfolio leg than for the sentiment-insensitive leg.
- **H0.** Beginning-of-period investor sentiment has no predictive relation with relative future returns in the Swedish equity cross-section.

2 Background

909 companies are listed in Sweden as of 2025, with a total market capitalisation of nearly USD 1.3 trillion and market capitalisation equal to 194 percent of GDP. Among European peer countries, only the United Kingdom has more listed companies (OECD, 2026). That breadth makes Sweden empirically suitable for a cross-sectional sentiment test because the analysis compares firms across size, age, risk, profitability, dividend policy, tangibility, liquidity, turnover, idiosyncratic volatility, and sales growth.

The Swedish market also contains a clear split between established main-market firms and smaller growth-market firms. The OECD identifies Nasdaq Stockholm, Nordic Growth Market, and Spotlight Stock Market as the three main trading venues. Nasdaq Stockholm accounts for more than 99 percent of market capitalisation, while First North, NGM Nordic SME, and Spotlight contain many smaller and growth-oriented companies. Regulated markets are concentrated in industrial and financial companies, while MTFs have larger technology and health-care representation (OECD, 2026). The listed-firm distribution matches the prediction in Baker and Wurgler (2006) that sentiment effects should be more visible among firms with subjective valuations and stronger limits to arbitrage.

Sweden has also had unusually active public equity financing. From 2014 to 2025, the OECD reports 1,114 new listings and 615 delistings, giving a net increase of 499 listed companies. The median Swedish IPO size from 2000 to 2025 was USD 9 million, the smallest among the peer countries covered by the OECD (OECD, 2026). The large number of public firms outside the largest listed companies makes characteristic-sorted portfolio tests more informative than an aggregate index-only analysis.

SCB's shareholder statistics use a broader quoted-company classification. Statistics Sweden reports 935 quoted companies on Swedish marketplaces at the end of December 2025, including 128 firms on OMX Stockholm Large Cap, 132 on Mid Cap, 103 on Small Cap, 346 on First North, 128 on Spotlight, and 92 on NGM Nordic SME (Statistics Sweden, 2026). The empirical universe is narrower because it applies common-equity, Swedish-ISIN, positive-price, and primary-security filters. The filtered Sweden panel contains 169,320 monthly security observations from January 2000 to December 2025 and 1,587 securities over the full sample. In December 2025, the filtered panel contains 900 selected primary common-equity lines.

2.1 OMXS30 Performance and Market Concentration

OMXS30 is Nasdaq Stockholm's leading blue-chip index and consists of 30 large and actively traded stocks. Nasdaq describes the index as suitable for derivatives and exchange-traded products because the underlying shares have high liquidity, and the index is revised twice a year (Nasdaq, 2026). The index serves as a large-cap benchmark for the Swedish market, but it is narrower than the empirical sample because it excludes most small, young, and growth-market firms.

The monthly OMXS30 series used for the background analysis runs from January 2009 to December 2025. Over that window, the adjusted index level rises from 617.38 to 2,882.97, corresponding to a cumulative price return of 367.0 percent and an annualized price return of 9.5 percent. The same period contains several sharp market stress episodes. The largest peak-to-trough monthly drawdown is the 2021–2022 decline, where the index falls 24.4 percent from December 2021 to September 2022 and recovers its prior monthly high in February 2024. The next largest drawdowns are the 2015–2016 and 2011 episodes, both at 21.7 percent. The Covid-19 shock is shorter and sharper, with a 16.9 percent decline from January 2020 to March 2020 and recovery by September 2020.

The largest one-month drop in the same monthly series is March 2020, when OMXS30 falls 11.17 percent. Other large monthly declines occur in August 2011 at 10.44 percent, May 2019 at 9.88 percent, March 2025 at 8.46 percent, and June 2022 at 8.32 percent. The sample contains both slow valuation cycles and abrupt market-wide stress. The drawdown episodes are reported in table 13.

Market value in the filtered 2025 sample is concentrated among a small group of large companies. The ten largest companies account for 42.6 percent of sample market value, while the five largest account for 28.0 percent. Investor, Atlas Copco, and Volvo alone account for 20.8 percent after aggregating multiple listed share classes at company level. The largest companies are listed in table 14. The concentration explains why an OMXS30-style large-cap index is not the same empirical object as the full Swedish cross-section. The index is dominated by liquid blue-chip firms, while the empirical analysis tests whether sentiment predicts relative returns across the much broader Swedish equity sample.

Table 1: GICS Sector Composition, Sweden Sample and S&P 500

GICS sector	Sweden sample, Dec. 2025 (%)	S&P 500, Dec. 2025 (%)
Information technology	8.3	34.4
Financials	23.4	13.4
Industrials	39.0	8.2
Consumer discretionary	7.6	10.4
Health care	5.5	9.6
Communication services	3.9	10.6
Consumer staples	3.4	4.7
Real estate	4.9	1.8
Materials	3.9	1.8
Energy	0.0	2.8
Utilities	0.0	2.3

Note. Sweden values are market-capitalisation shares in the filtered Swedish equity sample using the latest available GICS sector classification. The Swedish sample also contains 0.1 percent unclassified market value, which is omitted from the table. S&P 500 values are index-level GICS sector weights reported by S&P Dow Jones Indices as of December 31, 2025 (S&P Dow Jones Indices, 2026).

The U.S. benchmark is dominated by information technology and communication services, whereas the Swedish sample is more heavily exposed to industrials and financials. Sector composition affects the distribution of valuation uncertainty. A technology-heavy benchmark contains many firms whose value depends on growth options, intangible capital, and long-horizon cash-flow expectations. The Swedish sample combines a large number of small growth-oriented firms with aggregate market value concentrated in industrial and financial companies. The expected strength and location of sentiment effects may differ even if the underlying mechanism in Baker and Wurgler (2006) is the same.

At the end of 2025, SCB reports SEK 12,835 billion in market value for companies quoted on Swedish marketplaces. Households held 11.7 percent of that value, while foreign investors held 37.4 percent and financial corporations held 32.4 percent (Statistics Sweden, 2026).[1] The U.S. market has a broader household-investor channel. Federal Reserve data report USD 47,186 billion in household directly held corporate equities, while the Investment Company Institute reports that 56 percent of U.S. households owned regulated funds and that nearly 130 million individuals owned regulated funds (Board of Governors of the Federal Reserve System, 2026; Investment Company Institute, 2026). JPMorganChase Institute evidence further shows that retail investment flows rose sharply from 2023 to early 2025, that early-2025 flows exceeded the pandemic-era peak in dollars transferred, and that

[1]The Swedish 2025 ownership figures characterise the current market setting. The full 2000–2025 sample includes earlier ownership regimes and lower retail participation. Euroclear Sweden reports that 2025 was a record year for Swedish share ownership, with more than 2.8 million unique shareholders at year-end and an increase of approximately 47,300 shareholders during the year (Euroclear Sweden, 2026).

younger and lower-income individuals now participate at much higher rates than in the mid-2010s (Wheat and Eckerd, 2025).

The U.S. market has a more direct retail sentiment channel than Sweden. When individual investors are both numerous and active, sentiment can enter prices through marginal trading demand alongside long-run ownership shares. Sweden's smaller direct household ownership share points to a weaker retail-driven aggregate channel if professional and foreign investors correct mispricing more quickly. Retail participation can still affect prices through funds, attention-driven trading, IPO demand, and the marginal pricing of smaller and harder-to-arbitrage firms. Kumar and Lee (2006) show that retail trades are systematically correlated and that retail sentiment explains return comovement especially among small, low-priced, low-institutional-ownership, and costly-to-arbitrage stocks. Barber and Odean (2008) show that individual investors are especially prone to buy attention-grabbing stocks. The Swedish setting points to sentiment effects that may be weaker at the aggregate market level and more visible in the cross-section among firms where retail attention and arbitrage frictions are more important.

3 Literature Overview

Miller's heterogeneous-expectations model shows that, when short selling is restricted, prices can be set by the optimistic investors willing to hold the available supply, with weaker influence from pessimistic investors outside the market (Miller, 1977). Blume and Easley add that, in incomplete markets, traders with inaccurate beliefs can survive through wealth dynamics, so competition can preserve heterogeneous beliefs (Blume and Easley, 2006). Uncertainty widens the dispersion of valuations. A young, volatile, newly listed, or hard-to-value firm can become overpriced even when the average forecast is moderate, because pessimistic investors can avoid the stock more easily than they can sell it short. The first empirical implication is that sentiment should be most visible where disagreement is large and negative views are costly to express.

Later limits-to-arbitrage work makes this mechanism dynamic. Noise-trader risk implies that rational arbitrageurs may avoid correcting mispricing when sentiment can push prices further away from fundamentals before convergence occurs (De Long et al., 1990). Shleifer and Vishny similarly argue that financing, agency, and performance-evaluation constraints allow recognized mispricing to survive (Shleifer and Vishny, 1997). These papers sharpen Miller's asymmetry. Underpricing is easier to correct because investors can buy undervalued securities. Overpricing is harder to correct because it requires short selling, leverage, or a costly contrarian position. The cross-sectional prediction is that optimistic sentiment should produce relative overpricing in securities with high valuation uncertainty and high arbitrage risk.

Baker and Wurgler define investor sentiment as a market-wide propensity to speculate and test whether this state predicts the cross-section of stock returns (Baker and Wurgler, 2006, 2007). The definition is deliberately operational. It requires a time-varying demand component that is weakly tied to fundamentals and that affects stocks differently depending on their valuation difficulty and arbitrage frictions. The empirical contrast is between sentiment-prone firms and more bond-like firms. Baker and Wurgler identify the sentiment-prone side as small, young, volatile, unprofitable, non-dividend-paying, intangible, distressed, or high-growth firms. The bond-like side consists of firms with more stable earnings, dividends, tangible assets, and longer histories.

In Baker and Wurgler's U.S. evidence, the predictive relation is conditional on the state of sentiment. After low sentiment, speculative firms tend to earn relatively high subsequent returns. After high sentiment, the same types of firms tend to earn relatively low subsequent returns (Baker and Wurgler, 2006). The return relation varies with sentiment because sentiment changes the relative valuation of firms whose cash flows are harder to anchor. The Swedish empirical object is the return spread between firms with different sentiment sensitivity.

Baker and Wurgler construct a composite sentiment index from several noisy proxies, including the closed-end fund discount, detrended turnover, IPO volume, average first-day IPO returns, the equity share in new issues, and the dividend premium (Baker and Wurgler, 2006, 2007). Each proxy has a separate economic interpretation. Turnover captures speculative trading activity, IPO proxies capture hot-issue market conditions, the equity share captures issuance timing, and the dividend premium captures relative demand for safer dividend-paying firms. Baker and Wurgler's earlier evidence that the equity share in new issues predicts low future aggregate returns gives issuance behavior a separate market-timing foundation (Baker and Wurgler, 2000). The composite index extracts the common component across sentiment proxies that are individually contaminated but jointly informative about speculative demand.

Baker and Wurgler first estimate a preliminary principal component from current and lagged proxy values, select the timing of each proxy by its correlation with the preliminary index, and then estimate the final index from the selected six proxies (Baker and Wurgler, 2006). Their raw index loads negatively on the closed-end fund discount and dividend premium and positively on turnover, IPO volume, IPO first-day returns, and the equity share. The first principal component explains 49 percent of the variance in the raw proxies and 53 percent in the macro-orthogonalized proxies. Those figures show that the proxy set contains a sizeable common component, while the usual joint-hypothesis problem stays in view.

Qiu and Welch compare changes in consumer confidence and closed-end fund discounts with UBS/Gallup investor optimism (Qiu and Welch, 2006). They find that changes in Michigan consumer confidence correlate strongly with changes in direct investor optimism, while changes in closed-end fund discounts are weakly related to the same optimism measure. For the Swedish setting, their evidence supports survey-based sentiment proxies such as economic sentiment and consumer confidence and weakens a mechanical preference for U.S.-specific market proxies when those proxies are unavailable or less natural in a smaller European market.

Sentiment proxies can move with fundamentals as well as investor demand. Baker and Wurgler residualize their sentiment proxies against macroeconomic conditions before constructing a macro-adjusted sentiment index (Baker and Wurgler, 2006). Finter, Niessen-Ruenzi and Ruenzi apply the same principle in Germany, where the country-specific sentiment indicator is built from macro-adjusted proxies before PCA (Finter et al., 2012). Macro adjustment reduces business-cycle contamination and leaves residual variation that is more plausibly interpreted as sentiment. In Sweden, survey confidence, turnover, IPO activity, and capital-structure proxies can all respond to macroeconomic conditions.

Sentiment changes and sentiment levels answer different empirical questions. Baker and Wurgler use sentiment changes to study contemporaneous co-movement between sentiment and stock returns, while sentiment levels predict subsequent return reversals (Baker and Wurgler, 2007). Short-run sentiment movements are the relevant regressors for contemporaneous price pressure. Lagged sentiment levels are the relevant sentiment states for overpricing and later reversal. Ding, Mazouz and Wang formalize the same timing distinction in a multi-asset setting. Sentiment-prone assets react more strongly to short-run sentiment movements, while persistent high sentiment lowers their future expected returns because prices have already been pushed upward (Ding et al., 2019).

Stambaugh, Yu and Yuan sharpen the return-predictability test by focusing on overpricing. They study 11 anomaly strategies and show that anomaly profits are stronger after high sentiment, with the pattern coming primarily from the short legs (Stambaugh et al., 2012). The result follows Miller's short-sale asymmetry. If optimistic sentiment creates overpricing, subsequent returns should be weakest among securities identified as likely overpriced. Their later work on idiosyncratic volatility extends the mechanism. High idiosyncratic volatility increases arbitrage risk, and the negative idiosyncratic-volatility return relation is strongest among overpriced stocks (Stambaugh et al., 2015). In the Swedish tests, idiosyncratic volatility proxies for the difficulty of correcting sentiment-driven mispricing.

Persistent sentiment predictors can generate misleading return-predictability results if many portfolios and horizons are searched. Stambaugh, Yu and Yuan address this concern directly by comparing investor sentiment with random persistent predictors and show that the coherent anomaly pattern is difficult to reproduce by chance (Stambaugh et al., 2014). Swedish evidence is more persuasive when signs line up across theoretically related portfolios, when predictability concentrates among difficult-to-arbitrage firms, and when long-leg and short-leg results match the overpricing mechanism.

Baker, Wurgler and Yuan show that sentiment indices can be constructed outside the United States and that country sentiment can be decomposed into global and local components (Baker et al., 2012). Local sentiment is especially relevant for within-country cross-sectional returns, while global sentiment is more important for broad market returns. A Swedish sentiment index can be treated as a local sentiment state for Swedish cross-sectional tests, while global and Nordic spillover channels remain separate extensions unless they are explicitly modeled.

Finter, Niessen-Ruenzi and Ruenzi construct a German sentiment indicator from survey, options, trading, fund-flow, IPO, and issuance-related proxies for German stocks from 1993 to 2006 (Finter et al., 2012). Their macro-adjusted PCA index explains contemporaneous return spreads between sentiment-sensitive and sentiment-insensitive stocks, especially for high-volatility, high-idiosyncratic-risk, young, small, non-dividend-paying, and unprofitable firms. However, they find only weak evidence that the index predicts future return spreads. The German evidence prevents the Swedish thesis from importing the U.S. reversal evidence as a foregone conclusion. In a European market, sentiment may be measurable and economically meaningful even if predictive reversals are weaker, shorter-lived, or more dependent on specification.

Shoaib studies Nordic sentiment and U.S. spillovers, which provides regional context for local and foreign sentiment channels (Shoaib, 2019). Zhang and Zhang construct a Swedish investor sentiment index from economic sentiment, consumer confidence, turnover, IPO volume, IPO returns, and an equity-to-debt proxy over January 2014 to December 2022 (Zhang and Zhang, 2023). Their empirical test is mainly a market-level relation with Swedish broad indices, leaving relative returns across sentiment-sensitive firms for further analysis.

Prior work explains why sentiment should affect prices, how a latent sentiment index can be measured, why predictability should be strongest among hard-to-value and hard-to-arbitrage firms, and why local sentiment is relevant outside the United States. The Swedish cross-sectional evidence remains less developed. The present analysis addresses that gap by constructing a Sweden-specific sentiment measure and testing whether lagged sentiment predicts return differences between Swedish firms with different exposure to valuation uncertainty and limits to arbitrage.

4 Data

The empirical design proceeds from raw Swedish market, accounting, sentiment, macroeconomic, and IPO data to monthly sentiment-conditioned return tests. Investor sentiment is latent, accounting measures are observed with delay, and return predictability must be evaluated using information available before the return is measured. The method separates sample construction, source resolution, point-in-time accounting, characteristic construction, sentiment-index construction, portfolio formation, and regression testing.

4.1 Empirical Scope and Unit of Observation

Market and accounting data cover Swedish listed firms where Compustat market and fundamental records are available, with the broad data backbone running from January 2000 to December 2025. The main return and portfolio unit is the security-month, while accounting measures are linked at the company level. The issuer identifier, company_id, is used for issuer-level fundamentals, while the share-class identifier, security_id, is used for prices, returns, and portfolio assignment.

company_id is defined as country code plus gvkey, and security_id is defined as country code plus gvkey plus iid. These identifiers determine how market data, accounting data, characteristics, and portfolios are joined. Because the sentiment literature tests cross-sectional predictions across assets, return patterns are interpreted at the security-month level while company-level accounting information is used where fundamentals are issuer-specific.

The final regression sample is determined by the overlap between sentiment proxies, macro controls, factor controls, firm characteristics, and future return windows, so each empirical table reports its own observation count.

4.2 Data Sources

The analysis combines Swedish market, accounting, sentiment, IPO, interest-rate, and macroeconomic data into monthly measures. Daily market data provide trading-day prices, shares, turnover, total-return factors, and IPO first-close matching. A monthly market file supplies the preferred post-2007 monthly price source. Quarterly fundamentals provide accounting fields and filing dates. Survey sentiment is drawn from the economic and consumer sentiment workbook. IPO offer prices and dates are drawn from the Bloomberg IPO file. Interest-rate controls are drawn from Swedish 3-month interbank and 10-year yield files. Macro controls are drawn from CPI, PPI, industrial production, and employment series. The constructed measures are grouped into sentiment proxies, firm characteristics, factor controls, and macro controls.

Survey confidence supplies ESI and CCI, IPO records supply $NIPO$ and $RIPO$, market trading data supply $TURN$, accounting and market values supply $DIVP$ and the firm characteristics, interest-rate data supply the 3-month risk-free proxy and the 10-year yield, and macroeconomic series supply the controls used in the baseline and expanded orthogonalization tests.

4.3 Sample Construction and Security Universe

The security universe is formed from Swedish daily prices, monthly prices, and quarterly fundamentals, with company and security identifiers assigned before returns and characteristics are constructed. The first filter is applied at the security-month level. A security-month is eligible when the source record has a valid ISIN, the issue-type code identifies common equity using 0, EQ, or COM, and the month-end price is strictly positive. If a source file has no usable issue-type or ISIN metadata, the construction records this as an unfiltered metadata case rather than silently removing the whole input.

The second filter keeps one selected security line per issuer-month. When several eligible lines exist for the same company_id and month-end date, the construction first prefers primary issue identifiers, then Swedish ISIN lines, then the largest month-end market equity, highest month-end trading volume, longest observed security history, preferred monthly source records, and finally security_id for deterministic tie-breaking. The selected line defines the monthly common-equity universe used for returns, portfolio sorts, and factor controls. Daily observations used for liquidity, idiosyncratic volatility, and daily factor construction are then restricted to security-months that survive this monthly universe filter.

The resulting filtered Sweden panel contains 169,320 monthly security observations from January 2000 to December 2025 and 1,587 securities over the full sample. The usable regression sample is narrower because each test also requires the relevant lagged firm characteristic, future holding-period return, sentiment index observation, factor controls, and any robustness-specific controls. The monthly composition of the filtered market panel is reported in figure 3.

4.4 Market Data Construction

Monthly market observations combine daily and monthly market records into one security-month panel. The resolved panel supplies adjusted returns, market equity, liquidity measures, factor inputs, and portfolio holding-period returns. The detailed daily/monthly source-resolution rule is reported in section A.1.1.

4.5 Accounting Data and Point-in-Time Fundamentals

Accounting information is converted from quarterly fundamental records into monthly point-in-time snapshots. For each month-end, the panel assigns the latest filing that would have been observable by that date, so accounting measures enter portfolio and characteristic construction only after they could have been known. The filing-date hierarchy and fallback lags are documented in section A.1.2.

Point-in-time accounting snapshots prevent book equity, profitability, dividends, investment, and growth measures from using information unavailable at portfolio formation. The characteristics used to identify hard-to-value and difficult-to-arbitrage firms follow Baker and Wurgler (2006) and are measured using information observable at the monthly portfolio date.

The raw Compustat extracts also impose limits on which accounting characteristics from Baker and Wurgler (2006) can be measured in Sweden. The detailed input-field coverage table is reported in table 12. In short, book-equity, total-assets, earnings, revenue, and tangible fixed-asset fields are usable, while research and development expense, deferred-tax adjustment fields, and the external-finance fields needed for a paper-faithful EF/A measure are not sufficiently covered. Consequently, RD/A is excluded and EF/A is not used as a main characteristic.

These coverage limits define the Swedish empirical specification. The analysis keeps the Baker-Wurgler cross-sectional logic, while the retained sentiment proxies and firm characteristics are restricted to measures that can be constructed consistently for Swedish equities. The methodology therefore defines each retained sentiment proxy, firm characteristic, factor control, and macro control before the return tests are estimated.

5 Methodology

Figure 4 provides a compact overview of the empirical construction and testing pipeline, from raw Swedish data inputs to sentiment-index construction, characteristic-sorted portfolios, predictive regressions, and hypothesis interpretation.

5.1 Firm Characteristics

Baker and Wurgler (2006) use firm characteristics to operationalise two unobserved economic dimensions: subjective valuation and limits to arbitrage. Size, age, volatility, profitability, dividend policy, asset tangibility, growth opportunities, and distress are not sentiment measures themselves. They classify firms according to how difficult their cash flows are to value and how costly mispricing is likely to be to correct. The characteristics therefore convert a market-wide sentiment state into testable cross-sectional portfolio spreads.

The Swedish characteristic set follows this operational logic, but it does not use every characteristic in Baker and Wurgler (2006). The retained measures are those that can be constructed consistently from the Swedish market and accounting data: market equity, age, total risk, book-to-market, profitability, dividend status, asset tangibility, sales growth, illiquidity, turnover, and Fama-French three-factor idiosyncratic volatility. Research and development intensity, external finance over assets, and a paper-identical distress construction are excluded because the required fields are not sufficiently covered in the Swedish extracts. The trading-friction and idiosyncratic-volatility measures extend the Baker-Wurgler characteristic logic with later limits-to-arbitrage evidence from Stambaugh et al. (2012) and Stambaugh et al. (2015). Each subsection below defines the implemented measure, the accounting or market data used, and the direction of the sentiment-prone spread.

5.1.1 Market Equity

$ME_{i,t}$ is measured as month-end price multiplied by shares outstanding and scaled to market equity. The small-firm portfolio is the sentiment-prone side of the size sort. Small firms are usually followed by fewer investors, have less diversified operations, and can be more expensive to arbitrage. In the Swedish portfolio tests, the directional size spread is coded as small firms minus large firms.

5.1.2 Age

$age_{i,t}$ is measured as the number of months since the first observed listing month divided by twelve. Young firms are the sentiment-prone side because their operating histories are shorter and their valuations rely more heavily on expectations about future growth. The directional age spread is coded as young firms minus old firms.

5.1.3 Total Risk

$risk_{i,t}$ is the rolling standard deviation of monthly percentage returns over the previous twelve months, with at least nine monthly observations required. High total risk is used as a proxy for valuation uncertainty and disagreement. The risk spread is coded as high-risk firms minus low-risk firms.

5.1.4 Book-to-Market

$BE/ME_{i,t}$ is measured as repaired point-in-time book equity divided by market equity. Book equity must be positive and market equity must be positive. The characteristic is retained as a valuation diagnostic because book-to-market is closely related to the dividend-premium and valuation channels in the sentiment index. The main reported tests emphasize the validated sentiment-prone characteristics rather than treating $BE/ME_{i,t}$ as an independent source of evidence.

5.1.5 Profitability and Unprofitable Firms

Earnings are measured from the trailing point-in-time accounting snapshot. $E^+/BE_{i,t}$ scales positive trailing earnings by book equity, with positive book equity required. $UNPROFITABLE_{i,t}$ equals one when trailing earnings are non-positive and equals zero when earnings are positive. Low profitability and non-positive earnings identify firms with weaker cash-flow anchors. The unprofitable-firm spread is coded as unprofitable firms minus profitable firms, while the continuous profitability spread is coded as low profitability minus high profitability.

5.1.6 Dividend-Payer Status

$D\_PAYER_{i,t}$ records regular dividend-payer status from the point-in-time fundamental layer, with daily dividend evidence used as a fallback where available. $NON\_D\_PAYER_{i,t}$ equals one minus $D\_PAYER_{i,t}$ when dividend status is observed. Non-dividend payers are the sentiment-prone side because their valuation depends more on distant cash flows and less on current payout evidence. The dividend-status spread is coded as non-dividend payers minus dividend payers. Dividend-status evidence receives a mechanical-overlap caveat because $SENT_t^{\perp}$ includes $DIVP_t$.

5.1.7 Asset Tangibility

$PPE/A_{i,t}$ is measured as gross property, plant, and equipment divided by total assets and scaled by 100, with positive total assets required. Low tangibility is the sentiment-prone side because firms with fewer observable hard assets have weaker balance-sheet anchors for valuation. The tangibility spread is coded as low tangibility minus high tangibility.

5.1.8 Sales Growth

$GS_{i,t}$ follows the growth-opportunity characteristic in Baker and Wurgler (2006). It is measured as the year-over-year change in trailing-twelve-month revenue divided by prior-year trailing-twelve-month revenue,

$$GS_{i,t} = \frac{REVTTM_{i,t} - REVTTM_{i,t-12}}{REVTTM_{i,t-12}}. \tag{1}$$

The Swedish sales-growth construction uses revtq as the single production revenue input. $GS_{i,t}$ is computed only when prior-year trailing revenue is positive and current trailing revenue is non-negative. High sales growth is the sentiment-prone side because high-growth firms rely more heavily on distant cash-flow expectations and growth opportunities. The sales-growth spread is coded as high sales growth minus low sales growth.

5.1.9 Illiquidity

$ILLIQ_{i,t}$ is an Amihud-style monthly illiquidity measure. It is computed as the monthly average of absolute cleaned daily returns divided by daily traded value in millions, with at least five usable daily observations required. High illiquidity is the sentiment-prone side because trading frictions make mispricing harder to correct. The illiquidity spread is coded as high illiquidity minus low illiquidity.

5.1.10 Share Turnover

$XTURN_{i,t}$ is measured as monthly traded shares divided by month-end shares outstanding. Low turnover is coded as the sentiment-prone side in the portfolio tests. The interpretation is empirically ambiguous because turnover can proxy both speculative participation and the ability of arbitrage capital to enter or exit the stock. The turnover spread is treated as a trading-friction characteristic rather than a pure measure of sentiment sensitivity.

5.1.11 Idiosyncratic Volatility

$IVOL_{i,t}^{FF3}$ is estimated from daily returns within each security-month. The residual-volatility measure follows the logic of the Fama-French three-factor model, where excess stock returns are decomposed into exposure to the market excess return, a size factor, and a book-to-market factor (Fama and French, 1993). The fitted component captures return variation associated with common market, size, and value factors. The residual component captures firm-specific return variation left after those factor exposures have been removed.

Daily excess returns are regressed on the Swedish daily market, size, and value factors,

$$r_{i,d} - RF_d = \alpha_{i,t} + \beta_{i,t} MKT\_RF\_D_d + s_{i,t} SMB\_D_d + h_{i,t} HML\_D_d + \varepsilon_{i,d}. \tag{2}$$

The monthly idiosyncratic-volatility characteristic is the standard deviation of the residuals $\varepsilon_{i,d}$. High $IVOL_{i,t}^{FF3}$ firms form the sentiment-prone side of the sort, and the spread is coded as high idiosyncratic volatility minus low idiosyncratic volatility.

Stambaugh et al. (2015) show that idiosyncratic volatility is a limits-to-arbitrage characteristic, not only a risk descriptor. Their argument is that arbitrageurs cannot diversify away the position-level idiosyncratic risk that arises when correcting stock-specific mispricing. They show that the IVOL-return relation depends on mispricing direction. Among overpriced stocks, high-IVOL firms earn much lower subsequent benchmark-adjusted returns than low-IVOL firms. Among underpriced stocks, the positive high-IVOL effect is weaker. The asymmetry is consistent with short-sale constraints because correcting overpricing requires shorting, while correcting underpricing can be done by buying.

Stambaugh et al. (2015) also show that investor sentiment strengthens the IVOL effect in the predicted direction. After high sentiment, the negative relation between IVOL and future returns becomes stronger among overpriced stocks. The finding makes $IVOL_{i,t}^{FF3}$ directly relevant for identifying the part of the Swedish cross-section where sentiment-driven overpricing should be most difficult to arbitrage away. The available Swedish data do not observe short interest, securities lending constraints, or stock-level institutional ownership. Idiosyncratic volatility is an indirect proxy for arbitrage risk rather than a direct measure of short-sale constraints.

5.2 Winsorization, Decile Assignment, and Portfolio Sorting Characteristics

Continuous characteristics are winsorized before portfolio assignment so that extreme observations do not determine the cross-sectional breakpoints. Market equity, total risk, book-to-market, profitability scaled by book equity, dividend scaled by book equity, asset tangibility, sales growth, illiquidity, turnover, and idiosyncratic volatility are winsorized within month at the 1st and 99th percentiles. Age is winsorized within fiscal year because it changes mechanically with calendar time rather than with monthly market conditions. The winsorized values remain economically interpretable, but the portfolio sorts are less sensitive to isolated data errors, stale prices, and extreme accounting ratios.

Decile assignments use Swedish observations as the reference universe in each formation period. For each month, local breakpoints are computed from eligible Swedish common-equity observations and securities are assigned to deciles from 1 to 10. The lowest decile contains the lowest values of the sorting characteristic and the highest decile contains the highest values. The analysis tests whether sentiment predicts relative returns inside the Swedish cross-section, so the relevant benchmark is the Swedish distribution of size, risk, tangibility, sales growth, turnover, and idiosyncratic volatility.

Zero and non-positive accounting values carry economic meaning for earnings, dividends, and tangible assets. Non-positive earnings identify unprofitable firms, zero dividends identify non-dividend payers, and zero tangible assets identify firms with little observable asset backing. Binary characteristics are separated from continuous decile characteristics. Non-dividend-payer status and unprofitable status enter as binary portfolio sorts, while size, risk, age, profitability, tangibility, sales growth, illiquidity, turnover, and idiosyncratic volatility enter through decile assignments. For diagnostic accounting ratios, observations with non-positive earnings, zero dividends, or zero tangible assets are assigned to a separate zero category before the positive observations are ranked.

Portfolio formation uses one-month lagged sorting signals. At month t, each security is assigned to a portfolio using the characteristic value observed at $t-1$, and the return measured over the following holding period is then attached to that portfolio. Lagged assignment aligns the portfolio sort with information available before the return is realized and prevents the realized return month from determining the characteristic assignment.

The main empirical spreads are coded in the direction implied by the sentiment hypothesis. The sentiment-prone leg is formed from the side of the characteristic distribution that Baker and Wurgler identify as harder to value or harder to arbitrage. The sentiment-insensitive leg is formed from the more stable or less sentiment-sensitive side. For continuous characteristics, the directional spread uses the relevant extreme deciles: low-minus-high spreads are $P01 - P10$, while high-minus-low spreads are $P10 - P01$. Binary characteristics are unchanged because they naturally form two groups. Small firms are compared with large firms, high-risk firms with low-risk firms, high-idiosyncratic-volatility firms with low-idiosyncratic-volatility firms, low-profitability firms with high-profitability firms, low-tangibility firms with high-tangibility firms, high-growth firms with low-growth firms, unprofitable firms with profitable firms, non-dividend payers with dividend payers, illiquid firms with liquid firms, and low-turnover firms with high-turnover firms. With this coding, a negative coefficient on lagged sentiment means that the sentiment-prone leg earns lower future returns relative to the sentiment-insensitive leg after high sentiment.

5.3 Composite Sentiment Index

Investor sentiment is estimated as a latent market-wide component extracted from several noisy sentiment proxies. The top-down approach in Baker and Wurgler (2006) and Baker and Wurgler (2007) measures sentiment through common variation in proxies related to survey confidence, trading activity, IPO markets, financing choices, and the relative valuation of dividend-paying firms. Brown and Cliff (2004) provide a related factor-extraction approach for combining sentiment measures. $SENT^{\perp}$ is the preferred macro-adjusted index. $SENT$ is the raw comparison index.

5.3.1 Survey Confidence

ESI_t and CCI_t enter because survey confidence can contain direct information about optimism. Qiu and Welch (2006) compare changes in Michigan consumer confidence with UBS and Gallup investor optimism, a survey that asks investors about expected stock-market performance over the next year. The strong correlation between the two survey series supports the use of confidence indicators as sentiment proxies. The Swedish index combines these survey proxies with market-based proxies available at monthly frequency.

5.3.2 Trading Activity and Financing Composition

$TURN_t$ captures the liquidity and trading-activity channel. Behavioral models link trading to heterogeneous beliefs and noise-trader demand, because optimistic investors can trade aggressively when beliefs diverge and arbitrage is limited (De Long et al., 1990; Miller, 1977). Baker and Stein (2004) argue that market liquidity can proxy for sentiment when short-sale constraints make pessimistic views harder to trade on than optimistic views. High turnover is interpreted as stronger optimistic participation in the market. Baker and Wurgler (2000) show that the equity share in new issues predicts low subsequent aggregate returns, which motivates the use of financing composition as a sentiment-related proxy. ED_RATIO_t measures this issuance-composition channel in the Swedish sentiment proxy set.

5.3.3 IPO Market Conditions

$NIPO_t$ and $RIPO_t$ capture hot-issue-market conditions. Baker and Wurgler use IPO volume and first-day IPO returns as sentiment proxies, drawing on the IPO-cycle literature in which IPO activity and initial returns move with market conditions (Ibbotson and Jaffe, 1975; Ritter, 1991; Baker and Wurgler, 2006). Lowry and Schwert (2002) and Benveniste et al. (2003) show that IPO returns also contain sequential-learning and information-spillover effects, so the IPO proxies are interpreted as market-state proxies with sentiment, valuation news, and information-production content.

IPO counts and IPO returns have different missing-data meanings. $NIPO_t$ is measured on the full monthly paper-sample calendar and equals zero in months with no IPO activity. Raw $RIPO_t$ is defined only when at least one first-day IPO return is observed, and RIPO_OBSERVED_FLAG records whether a monthly IPO-return observation contributes to the raw series. Standardized $RIPO_t$ is neutral-imputed only after standardization when PCA requires a balanced matrix.

5.3.4 Dividend Premium

$DIVP_t$ follows the dividend-premium channel in Baker and Wurgler (2006). It is constructed as the log difference between value-weighted market-to-book ratios of regular dividend payers and non-payers. Dividend payers are typically larger, more profitable, and less growth-oriented firms (Fama and French, 2001). A lower dividend premium is consistent with stronger relative demand for speculative non-payers (Baker and Wurgler, 2004, 2006).

Raw proxy summary statistics and standardized proxy time series are reported in table 22 and figure 5.

5.3.5 PCA Construction Trace

PCA construction follows the staged procedure used by Zhang and Zhang (2023), applied to the Swedish proxy set with the added dividend-premium channel. The first stage estimates a preliminary PCA on current and one-month-lagged versions of each proxy. The preliminary score determines the timing of each proxy family. The selected proxies are then used to construct the raw comparison index, $SENT$. The same selected proxy timing fixes the inputs for the macro-adjusted $SENT^{\perp}$ construction.

Retaining components until cumulative explained variance exceeds 85 percent is a methodological deviation from Baker and Wurgler (2006), who construct their sentiment index from the first principal component after selecting proxy timing. The 85 percent retention rule follows the CICSI construction in Yi and Mao (2009), where several sentiment proxies are combined through a cumulative-variance PCA threshold for the Chinese stock market. Later Chinese sentiment research applies the same standard when constructing composite sentiment indices, including the investor sentiment measures in Guo et al. (2026). The same 85 percent threshold is used in the preliminary PCA, the raw selected-proxy PCA, and the macro-residualized PCA. A first-component robustness comparison is reported in table 20.

The preliminary PCA uses fourteen candidate proxies, consisting of current and lagged values of the seven proxy families. Retained components explain 86.0 percent of standardized proxy variance. Table 15 documents the retained components used for timing selection and component interpretation. The first component is mainly a financing and dividend-valuation component, while the second component captures trading activity. Survey confidence and IPO-market proxies enter through later components. The full eigenvector matrix is reported in table 16.

For each proxy candidate j, the construction converts the retained component eigenvectors into a score coefficient,

$$w_j = \sum_{k=1}^{K} \ell_{jk} \frac{\lambda_k}{\sum_m \lambda_m}, \tag{3}$$

where ℓ_{jk} is the loading of variable j on retained component k, λ_k is the eigenvalue of component k, and K is the number of retained components. The preliminary index is the weighted sum of standardized proxy candidates, $\widehat{SENT}_t^{pre} = \sum_j w_j z_{j,t}$. Its role is timing selection.

Lead-lag selection compares the correlation between each current or lagged proxy and the preliminary PCA index. Table 17 selects ESI_{t-1}, CCI_{t-1}, $TURN_{t-1}$, $NIPO_t$, $RIPO_t$, ED_RATIO_{t-1}, and $DIVP_t$. The correlation gap shows how strong each timing decision is. The dividend-premium timing is almost indifferent, with a gap of 0.002 between current and lagged values. Survey confidence has a clearer lagged timing. $RIPO_t$ contributes weak timing information, with a selected correlation of 0.001.

After timing is fixed, the selected raw proxies enter a second PCA. Five retained components explain 89.9 percent of standardized variance, with the first component explaining 25.9 percent. Table 18 shows that the raw index is most closely tied to survey confidence and dividend valuation. ESI_{t-1} and CCI_{t-1} have the largest raw score coefficients and the highest correlations with $SENT_t$. Before macro adjustment, the raw index is a confidence-and-valuation composite.

5.4 Macro-Adjusted Sentiment

The macro-adjusted sentiment path applies the selected proxy timing from the raw PCA stage and then removes linear macroeconomic variation before extracting the final common component. For each selected proxy $X_{j,t}$, the residualized proxy is estimated from

$$X_{j,t} = a_j + b_{1j}\widetilde{CPI}_t + b_{2j}\widetilde{PPI}_t + b_{3j}\widetilde{IP}_t + b_{4j}\widetilde{EM}_t + rX_{j,t}, \tag{4}$$

where the tilde denotes normalized macro controls and $rX_{j,t}$ is the residualized proxy. PCA is then applied to the standardized residuals $z(rX_{j,t})$. Raw $RIPO_t$ remains missing outside months with observed IPO first-day returns. Standardized $RIPO_t$ and residualized $rRIPO_t$ receive neutral imputation after standardization when PCA requires a balanced matrix.

The final orthogonalized composite index is constructed as

$$\begin{aligned} SENT_t^{\perp} &= 0.115\, z(rESI_{t-1}) + 0.016\, z(rCCI_{t-1}) + 0.250\, z(rTURN_{t-1}) \\ &\quad + 0.150\, z(rNIPO_t) + 0.064\, z(rRIPO_t) + 0.199\, z(rED\_RATIO_{t-1}) + 0.224\, z(rDIVP_t). \end{aligned} \tag{5}$$

The coefficients are PCA score coefficients from the retained macro-residualized components. A larger coefficient means that the standardized residualized proxy receives more direct weight in the final index, while the correlation with the final index measures how closely that proxy moves with the constructed sentiment state over time.

The macro-residualized PCA has a stronger first component than the raw selected-proxy PCA. The first component explains 31.2 percent of residualized proxy variance, and five retained components explain 87.4 percent. Table 19 gives the final composition of $SENT_t^{\perp}$. The largest final coefficients belong to turnover, the dividend premium, and the equity-to-debt ratio. The same three proxies also have the highest correlations with the final index.

Table 2 compares the raw and macro-orthogonalized indices proxy by proxy. Macro adjustment shifts index influence away from survey confidence and toward market-based residual variation. The correlation between economic sentiment and the index falls from 0.699 to 0.127, while the correlation between turnover and the index rises from 0.166 to 0.756. The dividend premium and equity-to-debt ratio also become more central after residualization. The correlation between $SENT_t$ and $SENT_t^{\perp}$ is 0.548, so $SENT^{\perp}$ preserves part of the raw common component while changing its composition.

Table 2: $SENT$ and $SENT^{\perp}$ composition compared

Proxy family	Raw coef.	Raw corr.	Orth. coef.	Orth. corr.	Corr. change	Interpretation
ESI_{t-1}	0.262	0.699	0.115	0.127	-0.572	Survey influence falls
CCI_{t-1}	0.203	0.668	0.016	0.044	-0.624	Survey influence falls
$TURN_{t-1}$	0.121	0.166	0.250	0.756	0.590	Trading residual becomes central
$NIPO_t$	0.101	0.226	0.150	0.221	-0.005	IPO volume remains moderate
$RIPO_t$	0.058	0.093	0.064	0.138	0.045	IPO returns remain weak
ED_RATIO_{t-1}	0.122	0.445	0.199	0.754	0.309	Financing residual becomes central
$DIVP_t$	0.179	0.534	0.224	0.773	0.239	Dividend valuation becomes central

Note. Correlation change equals the macro-orthogonalized correlation minus the raw correlation. Orthogonalized coefficients omit the residual prefix to keep proxy families comparable across columns.

5.5 Risk-Free Rate and Factor Controls

5.5.1 Theoretical Role of Factor Controls

Baker and Wurgler (2006) estimate predictive regressions with market, size, value, and momentum controls to test whether sentiment-conditioned return patterns survive standard return-factor adjustment. Their controls combine the market excess return, the Fama-French size and value factors, and a momentum factor (Fama and French, 1993; Carhart, 1997). The Swedish regressions follow the same logic. $RMRF_t$, SMB_t, HML_t, and UMD_t absorb broad market movements and common return components associated with firm size, book-to-market, and prior returns. The coefficient on lagged sentiment is then interpreted as predictive variation in the directional spread after conditioning on these factor returns.

Factor controls strengthen the empirical interpretation without turning the test into causal identification. A significant sentiment coefficient after factor adjustment is less likely to reflect ordinary exposure to the market, size, value, or momentum factors. It remains a predictive association between lagged sentiment and later return spreads.

5.5.2 Swedish Construction From Available Data

Baker and Wurgler use $RMRF$, SMB, HML, and UMD as factor controls in their multivariate predictive regressions (Baker and Wurgler, 2006). Their Table V first reports sentiment-only regressions and then re-estimates the same long-short portfolio regressions with the market excess return, the Fama-French size and value factors, and momentum included. The role of the four factor controls is to test whether the sentiment coefficient remains after absorbing standard return variation associated with broad market exposure, firm size, book-to-market, and prior returns. Their specification omits SMB when the dependent portfolio is the size spread and omits HML when the dependent portfolio is the book-to-market spread, because those controls would mechanically overlap with the dependent return. The Swedish regressions use the same factor-adjustment logic with Swedish factor controls and the final set of sentiment-prone directional spreads.

The factor controls are constructed from the same Swedish data as the sentiment-conditioned portfolios, using the resolved monthly market panel, common-equity screen, point-in-time fundamentals, and Swedish interest-rate series. The controls therefore share the sample, return units, and portfolio timing used in the main regressions. The full construction of $RMRF$, SMB, HML, and UMD, including breakpoints and value-weighting, is reported in section A.2.1. The dependent spread and factor controls are measured over the same 1-, 3-, 6-, and 12-month forward horizons.

5.6 Idiosyncratic Volatility Extension

The IVOL extension estimates monthly idiosyncratic volatility from daily excess returns. The Fama-French three-factor model represents excess stock returns as a linear function of the market excess return, a size factor, and a value factor (Fama and French, 1993). The model works by estimating a stock's factor exposures and then treating the residual as return variation that is not explained by those common factors. The Swedish implementation applies that structure at daily frequency within each security-month. It first constructs daily Swedish factors by converting the monthly risk-free proxy into daily rates and estimating daily market, size, and value factor returns. $IVOL_{i,t}^{FF3}$ is the monthly standard deviation of residuals from the daily three-factor regression, requiring at least 15 valid daily observations.

$IVOL_{i,t}^{FF3}$ proxies for arbitrage risk in the sentiment-conditioned tests. The empirical results use this residual-volatility measure as the main IVOL specification.

5.7 Portfolio Test Design

The portfolio tests translate the cross-sectional prediction in Baker and Wurgler (2006) into monthly Swedish return series. Each month, eligible securities are sorted using one-month lagged characteristics. Continuous characteristics are assigned to decile portfolios, while binary characteristics form two portfolios directly. Returns are measured after portfolio formation, so the dependent returns use information available before the holding-period return is realized. Equal-weighted portfolio returns are used because the theory concerns cross-sectional differences in sentiment sensitivity rather than the value-weighted performance of the largest Swedish firms.

The first empirical design follows the long-short portfolio logic in Baker and Wurgler (2006). Baker and Wurgler form characteristic portfolios for size, age, volatility, profitability, dividend policy, asset tangibility, growth opportunities, and distress. Their central prediction is that sentiment should affect stocks most strongly when valuation is subjective and arbitrage is costly. In their U.S. tests, subsequent returns are relatively low after high sentiment for small, young, volatile, unprofitable, non-dividend-paying, and otherwise speculative firms. The directional spreads code each spread as the return on the sentiment-prone leg minus the return on the sentiment-insensitive leg. A negative coefficient on lagged sentiment means that high sentiment predicts lower future returns for the stocks expected to be most sensitive to sentiment.

The spread design is a broad cross-sectional test. It asks whether the relative return between two characteristic groups moves in the direction predicted by sentiment theory. For continuous characteristics, the dependent spread is formed from the sentiment-prone extreme decile minus the sentiment-insensitive extreme decile. If the sentiment-prone side is low values of the characteristic, the spread is $P01 - P10$. If the sentiment-prone side is high values, the spread is $P10 - P01$. Binary characteristics use non-dividend payers minus dividend payers or unprofitable firms minus profitable firms. The spread combines both sides of the comparison into one dependent return and tests whether the cross-section tilts away from sentiment-prone firms after high sentiment. The leg decomposition then separates whether the effect comes from unusually weak returns on the sentiment-prone firms, unusually strong returns on the sentiment-insensitive firms, or both.

The second empirical design decomposes each directional spread into its two portfolio legs. For continuous characteristics, these legs are the same extreme deciles used in the spread construction, $P01$ and $P10$, with the ordering determined by the expected sentiment-prone side. For binary characteristics, the legs are the two binary groups. Stambaugh et al. (2012) combine market-wide sentiment with Miller's short-sale asymmetry. Their argument is that overpricing should be more common than underpricing when short-sale impediments prevent pessimistic investors from fully expressing negative views. In their anomaly tests, long-short anomaly profits are stronger after high sentiment, the short legs have lower subsequent returns after high sentiment, and the long legs show little relation to sentiment. The sentiment-prone leg is the closest analogue to the overpriced short leg in Stambaugh, Yu, and Yuan.

The two designs answer related but different questions. The spread test based on Baker and Wurgler (2006) asks whether Swedish sentiment predicts relative returns across characteristic-sorted portfolios. The Stambaugh-style leg test asks where that predictability comes from. Evidence is strongest when a characteristic has a negative spread coefficient and the sentiment-prone leg also has a negative coefficient. A spread driven mainly by the sentiment-insensitive leg still shows predictability, but it gives weaker support for the overpricing interpretation because the Stambaugh mechanism predicts that high sentiment should mainly depress future returns on the overpriced, difficult-to-arbitrage side.

5.8 Regression Design

5.8.1 Predictive Object

The empirical regressions estimate the predictive relations defined in the portfolio test design. The dependent return is a characteristic-sorted portfolio return, either a sentiment-prone-minus-sentiment-insensitive directional spread or one of its two component legs (Baker and Wurgler, 2006; Li, 2026).

Ordinary least squares is used because the empirical object is a linear predictive relation between a lagged sentiment state and future portfolio returns. The coefficient on lagged sentiment has a direct quantitative interpretation in percentage points. It gives the expected change in the future directional spread associated with a one-unit increase in sentiment, conditional on the included factor controls. The regression unit is a characteristic-horizon time series, with separate models estimated for each characteristic and return horizon.

5.8.2 Directional-Spread Regression

The primary spread test based on Baker and Wurgler (2006) uses the compounded future return on the sentiment-prone leg minus the compounded future return on the sentiment-insensitive leg. For characteristic j and horizon h, the estimated model is

$$\begin{aligned} R^{P-S}_{j,t:t+h-1} = {} & \alpha_j + \beta_j SENT^{\perp}_{t-1} + \gamma_j RMRF_{t:t+h-1} + \delta_j SMB_{t:t+h-1} \\ & + \eta_j HML_{t:t+h-1} + \theta_j UMD_{t:t+h-1} + \varepsilon_{j,t+h}. \end{aligned} \tag{6}$$

$R^{P-S}_{j,t:t+h-1}$ is measured in percentage points and compounds monthly returns from month t through month $t+h-1$. $SENT^{\perp}_{t-1}$ is the lagged macro-orthogonalized sentiment index. The factor controls are compounded over the same future horizon as the dependent return. The estimated coefficient β_j is the conditional sentiment-predictability coefficient. A negative β_j means that high sentiment predicts lower future returns for the sentiment-prone side of characteristic j relative to the sentiment-insensitive side.

For continuous characteristics, $R^{P-S}_{j,t:t+h-1}$ is the difference between the sentiment-prone extreme decile portfolio and the sentiment-insensitive extreme decile portfolio. Thus, low-minus-high characteristics use $P01 - P10$, while high-minus-low characteristics use $P10 - P01$. For binary characteristics, $R^{P-S}_{j,t:t+h-1}$ is the return on the sentiment-prone binary group minus the return on the safer binary group. A mechanical $P10 - P01$ diagnostic is retained for comparisons that require raw high-minus-low ordering.

The same regression is also estimated with $SENT_{t-1}$ in place of $SENT^{\perp}_{t-1}$. That comparison isolates the empirical role of macro orthogonalization. A stronger coefficient for $SENT^{\perp}_{t-1}$ means that the macro-adjusted component of the index carries more return-predictive information than the raw proxy combination.

5.8.3 Leg-Decomposition Regression

The Stambaugh-style leg decomposition estimates the same predictive equation separately for the sentiment-prone leg and the sentiment-insensitive leg. For each leg ℓ, the model is

$$\begin{aligned} R^{\ell}_{j,t:t+h-1} = \alpha_{j,\ell} + \beta_{j,\ell} SENT^{\perp}_{t-1} + \gamma_{j,\ell} RMRF_{t:t+h-1} + \delta_{j,\ell} SMB_{t:t+h-1} \\ + \eta_{j,\ell} HML_{t:t+h-1} + \theta_{j,\ell} UMD_{t:t+h-1} + \varepsilon_{j,\ell,t+h}. \end{aligned} \tag{7}$$

The leg regressions identify whether spread predictability comes from weak subsequent returns on the sentiment-prone side, strong subsequent returns on the sentiment-insensitive side, or both. For continuous characteristics, the leg regressions use the same extreme $P01$ and $P10$ portfolios as the directional spread. The Stambaugh-style test is therefore concentrated on the firms most likely to be overpriced under the sentiment mechanism. The theoretically cleaner overpricing interpretation requires a negative coefficient on the sentiment-prone leg after high sentiment. A spread result driven mainly by the sentiment-insensitive leg is still predictive evidence, but it is less directly aligned with the short-leg asymmetry in Stambaugh et al. (2012).

5.8.4 Sentiment-Only and Factor-Adjusted Comparison

The factor-control robustness check estimates the predictive spread regression twice on the same complete-case sample. The sentiment-only specification is

$$R^{P-S}_{j,t:t+h-1} = \alpha_j + \beta_j SENT^{\perp}_{t-1} + \varepsilon_{j,t+h}. \tag{8}$$

The factor-adjusted specification adds $RMRF$, SMB, HML, and UMD. The comparison separates standalone explanatory power from conditional predictability. The sentiment-only R^2 shows how much variation in the directional spread is explained by lagged sentiment alone. The factor-adjusted R^2 shows how much variation is explained after standard return factors enter the model. The change in β_j shows whether the sentiment relation becomes weaker or stronger once common market, size, value, and momentum variation is absorbed.

5.8.5 Non-Overlapping Horizon Regressions

The non-overlapping robustness test re-estimates the directional-spread regression on calendar-offset subsamples to check whether the main results survive when adjacent multi-month return windows share less of the same future monthly returns. Reported statistics include median coefficients, sign consistency, and significance shares across offsets; the exact offset rule is described in section A.2.2.

5.8.6 Sentiment-Innovation Regression

The sentiment-innovation test removes first-order persistence from $SENT^{\perp}$ before prediction. The residual from an AR(1) model of the index replaces the sentiment level in the directional-spread regression. The test asks whether unexpected changes in sentiment predict future cross-sectional return spreads, while the baseline level regression asks whether persistent high- and low-sentiment states predict future reversals. The exact innovation equation is reported in section A.2.2.

5.8.7 Auxiliary Regressions

Two auxiliary OLS regressions enter before the final predictive tests. The macro-orthogonalized sentiment index residualizes each selected sentiment proxy against Swedish macroeconomic controls,

$$Z_{k,t} = a_k + b_{1k}CPI_t + b_{2k}PPI_t + b_{3k}IP_t + b_{4k}EM_t + e_{k,t}. \tag{9}$$

The residuals $e_{k,t}$ are standardized and passed into the PCA index construction. The residualization step reduces business-cycle contamination in the sentiment proxies before the return-predictability tests are estimated.

Idiosyncratic volatility is estimated from daily stock-level excess returns within each security-month. The three-factor residual-volatility model is

$$r_{i,d} - RF_d = \alpha_{i,m} + \beta_{i,m}MKT\_RF_d + s_{i,m}SMB_d + h_{i,m}HML_d + \varepsilon_{i,d}. \tag{10}$$

The monthly characteristic $IVOL^{FF3}_{i,m}$ is the standard deviation of the fitted residuals $\varepsilon_{i,d}$. The daily OLS regression constructs a stock-level measure of arbitrage risk for the subsequent characteristic sorts.

5.8.8 HAC-Newey-West and Bootstrap t-statistics

The main inference uses Newey-West/HAC standard errors because the regressions use monthly returns, persistent sentiment, and overlapping multi-month forward return windows (Newey and West, 1987). The lag length is at least $h - 1$ for a horizon of h months. Moving-block bootstrap t-statistics are reported as a robustness check, following the concern in Baker and Wurgler (2006) that persistent sentiment can affect finite-sample inference. The detailed covariance and bootstrap implementation is described in section A.2.2.

5.9 Diagnostics and Validation

Section A.2.3 documents the source-overlap, duplicate-key, coverage, factor-leg, IVOL-status, and sentiment-index checks used to validate the empirical pipeline. These diagnostics separate data-quality issues from unresolved limitations, since measurement error in a constructed panel can otherwise be mistaken for asset-pricing evidence.

5.10 Methodological Limitations and Deliberately Secondary Elements

Global, Nordic, and U.S. sentiment spillovers are relevant in the literature and belong to later extensions beyond the Sweden-first design. The empirical design is interpreted as return-predictability evidence. A causal interpretation would require a separate specification with a credible identification strategy.

6 Investigation, Analysis and Results

Directional spreads based on Baker and Wurgler (2006) provide the broad evidence, while the Stambaugh-style leg decomposition gives a stricter test of where predictability comes from. High sentiment predicts lower subsequent returns for sentiment-prone Swedish portfolios in selected parts of the cross-section, especially total risk, idiosyncratic volatility, size, low tangibility, and dividend status. Profitability, illiquidity, turnover, and sales growth do not support a broad sentiment relation across all firm characteristics. The leg evidence is narrower still. Total risk, idiosyncratic volatility, and size give the cleanest joint support, while low tangibility and dividend status are stronger as directional-spread results than as Stambaugh-style sentiment-prone-leg results.

6.1 Sentiment Index and Portfolio Setup

The final sentiment index is $SENT_t^{\perp}$, a macro-orthogonalized Swedish PCA index that combines survey confidence, turnover, IPO activity, IPO first-day returns, the equity-to-debt proxy, and the dividend premium. The raw comparison index, $SENT_t$, uses the same proxy set before macro residualization. The raw and orthogonalized series have a correlation of 0.548, so macro adjustment changes the empirical sentiment state materially.

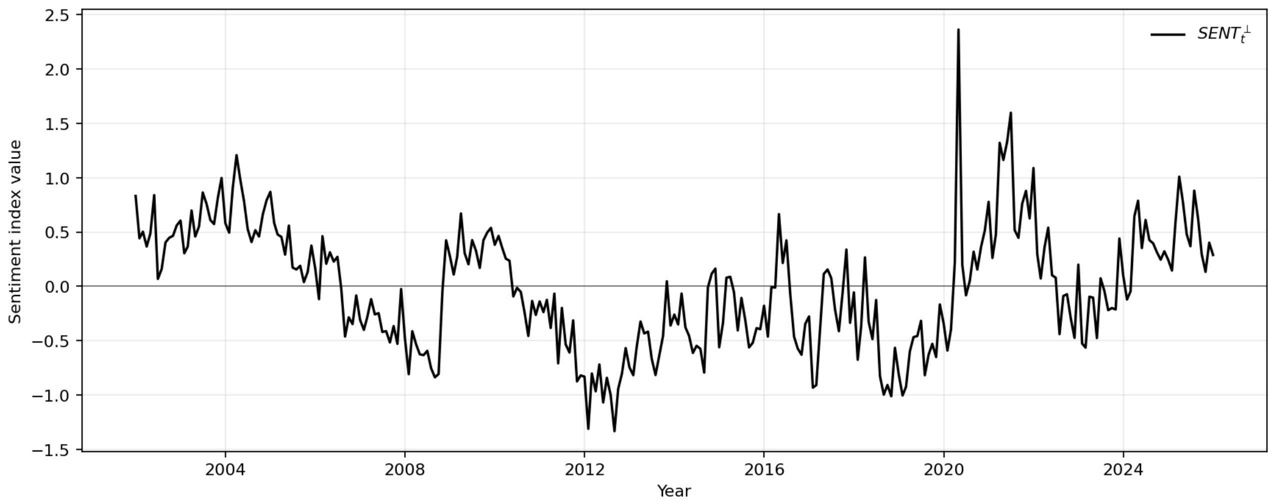

Figure 1: Macro-orthogonalized Swedish dividend-premium sentiment index

Note. The figure plots $SENT_t^{\perp}$.

$SENT_t^{\perp}$ is noisy at monthly frequency, but the series still contains persistent regimes. Sentiment is mostly below zero during 2011–2013 and again around 2018–2019, while 2021 contains a prolonged high-sentiment period. The predictive regressions use these beginning-of-period sentiment states to predict future characteristic-spread returns. $SENT^{\perp}$ enters the regressions as a market sentiment state rather than as a smooth business-cycle series.

The largest observation occurs in the Covid crisis window, peaking in April 2020 after the March 2020 market shock. The spike is too large to read as ordinary optimism alone, and it likely reflects an extreme combination of survey movements, market activity, and macro-residualized proxy behavior during an unusual market episode. The spike is retained in $SENT^{\perp}$ because the Covid period is an economically real stress episode rather than a coding error. Figure 8 and table 30 report the Covid-winsorized robustness check for the Sibley-expanded index.

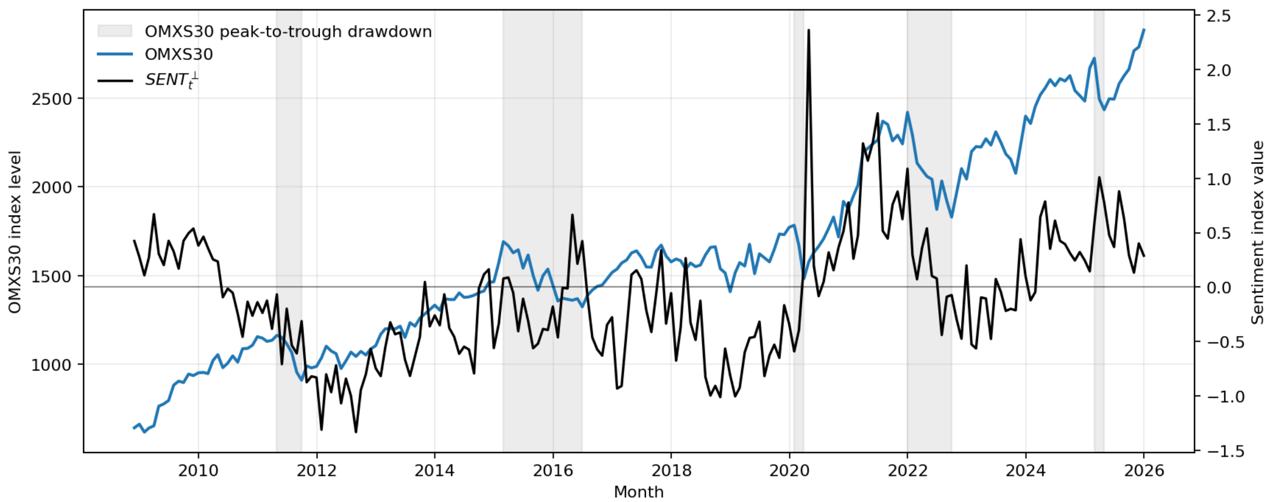

Figure 2: OMXS30 and macro-orthogonalized sentiment during major drawdowns

Note. The blue line is the OMXS30 adjusted index level on the left axis. The black line is $SENT_t^{\perp}$ on the right axis. Grey areas mark the five largest peak-to-trough OMXS30 drawdowns in the monthly OMXS30 series.

Figure 2 places $SENT_t^{\perp}$ against large-cap Swedish market stress. Elevated sentiment is followed by weak aggregate returns around the 2021–2022 market decline. Other drawdowns occur when $SENT_t^{\perp}$ is moderate, and some high-sentiment observations are followed by no immediate large OMXS30 decline. The aggregate index relation is weaker than the cross-sectional prediction tested below.

OMXS30 is a large-cap benchmark made up of liquid, established Swedish firms. The mechanism in Baker and Wurgler (2006) predicts larger sentiment effects in firms whose values are harder to estimate and harder to arbitrage, not necessarily in a blue-chip index dominated by larger firms with more durable cash-flow profiles. The sentiment index moves through important Swedish stress periods, but the return-predictability evidence comes from the characteristic-sorted cross-section. Selected standardized proxy panels around the same drawdowns are reported in figure 6.

Table 3: Sentiment-index summary statistics

Index	Obs.	Mean	Std. dev.	Min.	Median	Max.
$SENT$	289	-0.009	0.519	-1.379	-0.006	4.205
$SENT^{\perp}$	289	-0.014	0.558	-1.333	-0.043	2.361
$\Delta SENT^{\perp}$	288	-0.002	0.356	-2.163	-0.012	2.143

Note. $SENT$ and $SENT^{\perp}$ have a correlation of 0.548.

The final directional portfolio design uses eleven characteristics: ME, age, risk, $IVOL^{FF3}$, non-dividend-payer status, unprofitable status, $E + BE$, PPE/A, GS, $ILLIQ$, and $XTURN$. BE/ME remains a diagnostic value sort rather than a main directional sentiment-prone test.

Table 4: Directional portfolio definitions

Firm characteristic	Sentiment-prone leg	Directional spread
ME	Small firms	Low minus high
$risk$	High total volatility	High minus low
$IVOL^{FF3}$	High three-factor idiosyncratic volatility	High minus low
Age	Young firms	Low minus high
Non-dividend payer	Non-payers	Non-payers minus payers
Unprofitable	Non-positive earnings firms	Unprofitable minus profitable
$E + BE$	Low profitability	Low minus high
PPE/A	Low tangibility	Low minus high
GS	High sales growth	High minus low
Illiquidity	High Amihud illiquidity	High minus low
Turnover	Low share turnover	Low minus high

Note. For continuous characteristics, low minus high means $P01 - P10$; high minus low means $P10 - P01$. Binary characteristics use their two natural groups. BE/ME is retained only as a diagnostic value sort because the sentiment-prone side is not monotonic and the dividend-premium proxy is valuation-related.

The directional spread is always the return on the sentiment-prone side minus the return on the sentiment-insensitive side. A negative coefficient on lagged sentiment means that the sentiment-prone portfolio earns lower subsequent returns after high sentiment. Negative coefficients are the hypothesis-consistent sign in the spread tests based on Baker and Wurgler (2006). The later leg-decomposition tables estimate the two component legs separately.

The sign convention turns the research question into a direct test. H1 requires lagged sentiment to predict future relative returns, while H2 requires the negative relation to be strongest in the portfolios with subjective valuations or limits to arbitrage. H3 requires the sentiment-prone leg to carry stronger negative predictability than the sentiment-insensitive leg.

Table 23 and figure 7 shows that several spreads load on the same underlying part of the Swedish cross-section. The IVOL spread is strongly related to total risk, and size is also closely related to illiquidity and dividend status. The sales-growth spread is only weakly correlated with the main risk and size spreads. The results describe a connected cluster of small, risky, illiquid, non-dividend-paying firms plus a more separate growth-opportunity characteristic, rather than eleven independent tests.

The primary inferential family is defined before interpreting the results as the 44 factor-adjusted $SENT^{\perp}$ directional-spread regressions, covering eleven characteristics across four return horizons. The raw and Sibley-expanded indices, leg decompositions, non-overlapping windows, and innovation regressions are treated as comparisons and robustness checks rather than additional primary tests. The restriction keeps the main inference tied to the research question and reduces the risk that the interpretation is selected from the full set of reported specifications.

6.2 Baker-Wurgler Style Directional Spreads

The Baker-Wurgler-style spread tests give selective support for sentiment-related return predictability in Sweden. Beginning-of-period $SENT^{\perp}$ predicts lower future returns for several sentiment-prone legs relative to their sentiment-insensitive counterparts, but the evidence is concentrated in specific firm characteristics. The strongest results are total risk, $IVOL^{FF3}$, size, low tangibility, and non-dividend-payer status. Unprofitable firms and young firms provide weaker horizon-dependent support. Low profitability, sales growth, illiquidity, and low turnover do not support the same interpretation.

6.2.1 Main Factor-Adjusted Evidence

Each coefficient measures the percentage-point change in a future directional spread associated with a one-unit increase in lagged sentiment. Every spread is defined as the sentiment-prone leg minus the sentiment-insensitive leg, so a negative coefficient means that high sentiment predicts lower future returns for the sentiment-prone leg relative to the sentiment-insensitive leg. The $SENT^{\perp}$ estimates across all horizons are reported in table 5. Stars on coefficients mark HAC significance at the 10, 5, and 1 percent levels.

Total risk provides the strongest in table 5. The coefficient for total risk (σ) is negative and significant in every horizon, growing from -2.60 at 1 month to -19.65 at 12 months. $IVOL^{FF3}$ follows the same pattern, with significant negative coefficients from 1 to 12 months and a 12-month coefficient of -18.42. These two characteristics give the cleanest support for H2 because high total risk and high idiosyncratic volatility can be interpeted as proxies for valuation uncertainty and arbitrage risk.

The small-minus-large coefficient for size and age is negative in every horizon, but the evidence

Table 5: Overview of $SENT^{\perp}$ predictive regression coefficients by horizon

Characteristic	1 month		3 months		6 months		12 months	
	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t
Panel A. Size, Age, and Risk								
ME	-1.34*	-1.90	-2.44	-1.38	-6.55**	-2.39	-14.04***	-3.15
Age	-0.38	-0.80	-1.06	-0.79	-2.86	-1.23	-8.37**	-2.29
σ	-2.60***	-3.50	-6.49***	-3.56	-12.08***	-4.30	-19.65***	-4.43
$IVOL^{FF3}$	-2.24***	-2.69	-5.60***	-2.97	-11.74***	-4.05	-18.42***	-3.75
Panel B. Profitability and Dividend Policy								
$E+BE$	-0.44	-0.64	0.43	0.34	0.41	0.18	0.08	0.02
$Unprofitable$	-0.54	-0.96	-1.11	-1.36	-2.50*	-1.84	-4.86***	-3.04
$Nonpayer$	-0.63**	-2.24	-1.88**	-2.46	-3.58***	-2.88	-4.96***	-2.80
Panel C. Tangibility and Growth Opportunities								
PPE/A	-1.27***	-2.99	-4.46***	-3.86	-8.34***	-4.21	-15.27***	-4.83
GS	-0.80	-1.33	-0.88	-0.64	-1.41	-0.55	-3.21	-0.61
Panel D. Trading Frictions								
$ILLIQ$	-0.30	-0.67	-0.03	-0.03	-1.32	-0.65	-2.58	-0.72
$XTURN$	-0.11	-0.16	1.04	0.62	1.91	0.67	4.46	1.23

Note. Each coefficient is from the factor-adjusted predictive regression using lagged $SENT^{\perp}$. Coefficients are percentage-point effects on future sentiment-prone-minus-insensitive directional-spread returns. HAC t-statistics use Newey-West standard errors. Negative coefficients are theory-consistent for all listed spreads. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

becomes clearly supported only at 6 and 12 months. The 12-month coefficient of -14.04 means that high $SENT^{\perp}$ predicts lower future returns for small firms relative to large firms over the following year. Young firms also have negative coefficients in every horizon, but only the 12-month estimate is statistically supported. The age result therefore reinforces the long-horizon pattern, while size is the more statistically significant characteristic within this group.

Unprofitable firms have negative coefficients in all horizons and become statistically supported at 6 and 12 months, which is consistent with the Baker-Wurgler prediction that weak cash-flow anchors make valuation more sentiment sensitive. Low profitability measured by $E+BE$ gives the opposite reading. Its coefficients turn positive from 3 to 12 months and are statistically weak. Dividend status gives stronger evidence. Non-dividend payers have negative and statistically supported coefficients in every horizon, reaching -4.96 at 12 months. This supports H2, although the result is interpreted cautiously because $DIVP$ is one of the sentiment proxies used to construct $SENT^{\perp}$.

Tangibility is one of the most stable results in the section. Low-tangibility firms have significant negative coefficients at every horizon, with the 12-month coefficient reaching -15.27. This is consistent with the view that firms with fewer hard assets are harder to value and more exposed to sentiment-related repricing. Sales growth has the expected negative sign in every horizon, but the HAC t-statistics are weak throughout, making it statistically insignificant.

Trading frictions are the weakest part of the directional-spread evidence. Illiquidity has the expected negative sign in every horizon, but the coefficients are small and insignificant at the 10% level. The findings related to turnover are more problematic because the coefficients turn positive from 3 to 12 months. The 12-month coefficient is 4.46, which is inconsistent with the expected reversal pattern. The trading-friction evidence therefore does not support H2 in the same way as risk, $IVOL^{FF3}$, size, low tangibility, and dividend status.

6.2.2 Raw Versus Macro-Adjusted Sentiment

The evidence depends strongly on using $SENT^{\perp}$ rather than raw $SENT$. At the 12-month horizon, the average coefficient is -7.89 for $SENT^{\perp}$ and -4.00 for $SENT$. The share of characteristics with a theory-consistent 5% result is 0.64 for $SENT^{\perp}$ at 12 months, compared with 0.09 for $SENT$. The raw-versus-orthogonalized horizon summary is reported in table 24, and the full horizon tables comparing $SENT$ and $SENT^{\perp}$ are reported in tables 25 to 28.

Raw $SENT$ gives strong evidence for low tangibility and some short-horizon evidence for low turnover and sales growth, but it does not reproduce the strongest risk and IVOL pattern. At 12 months, the raw total-risk and $IVOL^{FF3}$ coefficients are negative but statistically weak, with HAC t-statistics of -1.09 and -1.07. The same characteristics under $SENT^{\perp}$ have HAC t-statistics of -4.43 and -3.75. The Swedish Baker-Wurgler-style evidence therefore comes mainly from the macro-adjusted sentiment state rather than from the raw common proxy component.

This differs from Baker and Wurgler (2006), where raw and orthogonalized sentiment generally preserve the same message for the central speculative-firm spreads. In the Swedish results, $SENT$ appears more closely tied to survey confidence and broad confidence-cycle variation, while $SENT^{\perp}$ loads more strongly on residualized market-based proxies such as $DIVP$, turnover, and financing composition. The Swedish evidence resembles Baker-Wurgler most clearly for risk, $IVOL^{FF3}$, size, and dividend status, but the pattern is narrower and more dependent on macro adjustment.

6.2.3 Multiple Testing and Factor-Control Robustness

The 44 primary $SENT^{\perp}$ spread tests combine eleven characteristics and four horizons. Table 6 treats those tests as one family and applies a Benjamini-Hochberg false-discovery-rate correction to the one-sided expected-sign HAC p-values. This correction addresses the risk that isolated significant coefficients arise from testing many related characteristics and horizons.

Table 6: Multiple-testing diagnostic for primary $SENT^{\perp}$ spread tests

Horizon	HAC 5% count	FDR 10% count	FDR 5% count	Characteristics surviving FDR 5%
1 month	5	5	4	Total risk, low tangibility, $IVOL^{FF3}$, non-dividend payer
3 months	4	4	4	Low tangibility, total risk, $IVOL^{FF3}$, non-dividend payer
6 months	6	6	5	Total risk, low tangibility, $IVOL^{FF3}$, non-dividend payer, small firms
12 months	7	7	7	Low tangibility, total risk, $IVOL^{FF3}$, small firms, unprofitable, non-dividend payer, young firms

Note. The primary family contains the 44 factor-adjusted $SENT^{\perp}$ directional-spread tests, defined by eleven characteristics and four horizons. HAC 5% count reports unadjusted one-sided expected-sign HAC significance. FDR columns apply the Benjamini-Hochberg false-discovery-rate procedure to one-sided expected-sign normal p-values computed from the HAC t-statistics across the full 44-test family.

The false-discovery-rate diagnostic supports the main hierarchy but narrows the interpretation. The risk, $IVOL^{FF3}$, low-tangibility, and non-dividend-payer results survive the 5 percent FDR threshold at the shorter horizons, while small firms, unprofitable firms, and young firms enter the surviving set only at longer horizons. Low profitability, illiquidity, sales growth, and turnover do not survive the correction. The correction therefore supports the clustered interpretation of H2 rather than a broad statement that sentiment-related predictability appears across all tested characteristics.

Table 7 separates sentiment-only evidence from factor-adjusted evidence. The coefficient on lagged sentiment is a conditional cross-sectional predictability estimate after contemporaneous market, size, value, and momentum factor returns are absorbed over the same holding horizon.

Table 7: Robustness to factor controls

Horizon	Sent.-only coef.	Factor-adj. coef.	Change	Sent.-only R^2	Factor-adj. R^2	Change R^2
1 month	-0.27	-0.97	-0.69	0.003	0.290	0.287
3 months	0.07	-2.04	-2.12	0.010	0.336	0.326
6 months	-1.35	-4.37	-3.01	0.017	0.385	0.369
12 months	-5.24	-7.89	-2.65	0.042	0.478	0.436

Note. Each row averages the eleven directional-spread regressions for $SENT^{\perp}$. Sentiment-only and factor-adjusted regressions use the same complete-case sample. The factor-adjusted specification adds $RMRF$, SMB, HML, and UMD.

Lagged $SENT^{\perp}$ explains little spread variation on its own, with average R^2 below 0.05 at every horizon. The factor-adjusted R^2 is much higher, reaching 0.478 at 12 months, which means that market, size, value, and momentum returns account for a large part of the time-series movement in the directional spreads. The sentiment coefficient nevertheless becomes more negative after factor adjustment at every horizon. $SENT^{\perp}$ therefore retains incremental predictive content after standard return factors have been absorbed.

The Baker-Wurgler-style spread evidence gives a qualified answer to the research question. Beginning-of-period $SENT^{\perp}$ predicts relative future returns in the Swedish cross-section, which supports H1. The support for H2 is concentrated in risk, $IVOL^{FF3}$, size, low tangibility, and dividend status. Profitability, illiquidity, turnover, and sales growth keep the conclusion from becoming a general statement that sentiment predicts all theoretically sentiment-prone Swedish spreads.

6.3 Stambaugh-Style Leg Decomposition

Stambaugh et al. (2012) use leg-level anomaly returns to distinguish broad spread predictability from an overpricing mechanism. Their central result is that high sentiment predicts especially weak subsequent returns on the short legs of anomaly strategies, where short-sale impediments make overpricing harder to eliminate. The leg decomposition separates each directional spread into a sentiment-prone leg and a sentiment-insensitive leg. The sentiment-prone leg is the closest analogue to the Stambaugh short leg because it is the side expected to be most exposed to high-sentiment overpricing.

Tables 8 and 9 separate the leg evidence. A directional spread can become negative because the sentiment-prone leg underperforms, because the sentiment-insensitive leg outperforms, or because both happen at the same time. The Stambaugh-style prediction is strongest when the sentiment-prone leg has a negative coefficient and the sentiment-insensitive leg does not carry most of the spread. A positive or statistically dominant sentiment-insensitive leg can still create a predictive directional spread, but it weakens the interpretation that high sentiment mainly overprices the hard-to-arbitrage side.

Total risk and $IVOL^{FF3}$ provide the clearest Stambaugh-style evidence. Their sentiment-prone leg coefficients are negative and statistically supported at every horizon in table 8. At 12 months, the sentiment-prone leg coefficients are -15.56 for total risk and -13.43 for $IVOL^{FF3}$. The sentiment-insensitive legs are also positive and statistically supported in table 9, so the spreads are reinforced by both legs rather than generated only by weak returns on the sentiment-prone side. The Stambaugh interpretation remains strongest for these two characteristics because the negative sentiment-prone leg is large, persistent, and statistically supported.

Size and age give a second layer of support. The small-firm sentiment-prone leg turns statistically supported at 6 and 12 months, with coefficients of -6.61 and -15.28. The young-firm sentiment-prone leg follows the same longer-horizon pattern, with coefficients of -3.70 and -8.85. Their sentiment-insensitive legs are close to zero and statistically weak. The evidence therefore points more directly to weak future returns on the sentiment-prone side than to a spread mechanically produced by strong returns in the sentiment-insensitive leg.

Table 8: Overview of $SENT^{\perp}$ sentiment-prone leg coefficients by horizon

Characteristic	1 month		3 months		6 months		12 months	
	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t
Panel A. Size, Age, and Risk								
ME	-1.16	-1.61	-2.34	-1.30	-6.61**	-2.31	-15.28***	-3.07
Age	-0.54	-1.55	-1.27	-1.22	-3.70**	-2.08	-8.85**	-2.53
σ	-1.79***	-3.18	-4.72***	-3.24	-8.70***	-3.84	-15.56***	-3.50
$IVOL^{FF3}$	-1.54**	-2.28	-4.01**	-2.40	-8.52***	-2.97	-13.43**	-2.24
Panel B. Profitability and Dividend Policy								
$E+BE$	0.34	0.91	1.49	1.58	1.59	0.97	2.96	1.05
$Unprofitable$	0.37	0.49	1.80	0.97	3.45	0.85	11.32	1.08
$Nonpayer$	-0.17	-0.62	-0.72	-1.05	-1.52	-1.52	-1.84	-1.22
Panel C. Tangibility and Growth Opportunities								
PPE/A	-0.18	-0.41	-1.07	-0.96	-1.44	-0.82	-3.15	-1.01
GS	-0.41	-1.00	-1.01	-1.04	-2.73	-1.56	-4.97	-1.57
Panel D. Trading Frictions								
$ILLIQ$	0.05	0.11	0.56	0.58	-0.38	-0.21	-1.25	-0.32
$XTURN$	-0.31	-0.83	-1.17	-1.04	-2.71	-1.43	-4.42	-1.21

Note. Each coefficient is from the factor-adjusted leg regression using lagged $SENT^{\perp}$. Coefficients are percentage-point effects on future returns for the sentiment-prone leg of each directional spread. Negative coefficients are theory-consistent under the Stambaugh-style overpricing interpretation. HAC t-statistics use Newey-West standard errors. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

The sentiment-insensitive leg regressions change the interpretation of several strong spread results. Total risk and $IVOL^{FF3}$ have positive and statistically supported sentiment-insensitive leg coefficients as well as negative sentiment-prone leg coefficients. Their spread results therefore come from both sides of the characteristic comparison, although the sentiment-prone leg evidence remains strong enough to support the Stambaugh mechanism. Low tangibility is different. The 12-month low-tangibility spread is one of the strongest Baker-Wurgler-style results, but the leg decomposition shows that the high-tangibility sentiment-insensitive leg carries most of the effect. The low-tangibility sentiment-prone leg coefficient is -3.15 with a HAC t-statistic of -1.01, while the high-tangibility sentiment-insensitive leg is 12.60 with a HAC t-statistic of 5.82. Non-dividend payer evidence has the same structure in weaker form. Its sentiment-prone leg coefficient is negative but weak, while the dividend-payer sentiment-insensitive leg is positive and statistically supported.

Table 9: Overview of $SENT^{\perp}$ sentiment-insensitive leg coefficients by horizon

Characteristic	1 month		3 months		6 months		12 months	
	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t	Coef.	HAC t
Panel A. Size, Age, and Risk								
ME	0.17	1.05	0.10	0.26	0.28	0.42	0.62	0.43
Age	-0.16	-0.51	-0.08	-0.09	-0.21	-0.14	-0.20	-0.07
σ	0.81^{**}	2.12	2.07^{**}	2.14	4.58^{***}	2.67	7.78^{***}	3.21
$IVOL^{FF3}$	0.70^{*}	1.76	1.88^{**}	1.96	4.51^{***}	2.68	9.07^{***}	2.91
Panel B. Profitability and Dividend Policy								
$E+BE$	0.78	1.40	1.03	1.10	1.33	0.95	2.97	1.42
$Unprofitable$	0.74	1.49	1.35^{*}	1.94	1.79^{**}	2.22	3.88^{***}	3.33
$Nonpayer$	0.46^{**}	2.23	1.15^{*}	1.90	2.18^{**}	2.23	3.64^{***}	2.60
Panel C. Tangibility and Growth Opportunities								
PPE/A	1.10^{***}	3.24	3.42^{***}	4.15	6.82^{***}	4.73	12.60^{***}	5.82
GS	0.39	0.91	-0.24	-0.22	-1.30	-0.82	-1.83	-0.64
Panel D. Trading Frictions								
$ILLIQ$	0.35^{*}	1.95	0.59	1.31	1.09	1.40	1.99^{*}	1.94
$XTURN$	-0.21	-0.38	-2.03	-1.50	-4.68^{**}	-2.28	-9.64^{***}	-3.50

Note. Each coefficient is from the factor-adjusted leg regression using lagged $SENT^{\perp}$. Coefficients are percentage-point effects on future returns for the sentiment-insensitive leg of each directional spread. Positive coefficients indicate that high sentiment is followed by stronger returns for the more sentiment-insensitive side. HAC t-statistics use Newey-West standard errors. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Sales growth, illiquidity, and low turnover have negative 12-month sentiment-prone leg coefficients, but the evidence is weak. The sales-growth sentiment-prone leg coefficient is -4.97 with a HAC t-statistic of -1.57. Illiquidity is close to zero in statistical terms. Low turnover is harder to interpret because both legs are negative at 12 months, and the sentiment-insensitive leg is much more negative than the sentiment-prone leg. Low-profitability and unprofitable firms fail the strict sentiment-prone leg test. Their 12-month sentiment-prone leg coefficients are positive, which conflicts with the prediction that high sentiment should be followed by lower returns on the sentiment-prone side.

6.3.1 Comparison With Stambaugh, Yu, and Yuan

Stambaugh, Yu, and Yuan find that sentiment-conditioned anomaly returns are driven mainly by the short leg, where overpricing is hardest to correct. The Swedish evidence follows that pattern for risk, $IVOL^{FF3}$, and size. Those characteristics combine negative spread coefficients with negative sentiment-prone-leg coefficients, so they support both the Baker-Wurgler cross-sectional prediction and the stricter Stambaugh overpricing-asymmetry interpretation. Age gives a weaker version of the same pattern.

Low tangibility and dividend status support the Baker-Wurgler spread prediction more clearly than the Stambaugh leg prediction. Their spread coefficients indicate that sentiment predicts relative returns across Swedish characteristic portfolios, but their leg decomposition assigns a large part of the predictive relation to the sentiment-insensitive leg. The results still support H1 because they show cross-sectional predictability, but they provide weaker evidence for H3 because the strict short-leg asymmetry mechanism requires stronger negative predictability in the sentiment-prone leg.

The leg decomposition narrows the empirical answer to the research question. $SENT^{\perp}$ predicts Swedish cross-sectional return differences, which supports H1. Predictability is strongest in characteristics linked to risk, idiosyncratic volatility, and size, which supports H2 for firms where arbitrage risk and valuation uncertainty are most visible. H3 receives narrower support. Risk, $IVOL^{FF3}$, and size fit the predicted sentiment-prone-leg pattern most clearly. Low tangibility and dividend status remain important for H1 and H2, but their leg patterns show that the sentiment-insensitive leg contributes materially. Profitability and turnover are inconsistent with a broad H3 interpretation, while illiquidity and sales growth provide weak support. The Swedish evidence is therefore best read as selective sentiment-related cross-sectional predictability rather than a uniform overpricing-asymmetry pattern.

6.4 Robustness

HAC/Newey-West standard errors and moving-block bootstrap t-statistics adjust the primary inference for heteroskedasticity, serial correlation, and time-series dependence in overlapping monthly return windows. Those corrections improve the coefficient-level inference, but they do not fully address the pattern that the expected-signed t-statistics rise with the return horizon. Coefficients are naturally larger over longer compounded horizons, but stronger t-statistics at longer horizons raise the possibility that persistent sentiment states and repeated holding-period observations make the $SENT^{\perp}$ evidence look stronger than a strictly non-overlapping design would imply.

Two robustness tests address this concern. First, non-overlapping horizon regressions split the sample into h calendar-offset subsamples for each return horizon h,

$$\mathcal{T}_{h,o} = \{t : m(t) - o \equiv 0 \pmod h\}, \qquad o = 0, \ldots, h-1, \tag{11}$$

where $m(t)$ is the monthly time index. The predictive regression is then re-estimated separately in each offset subsample. At the 12-month horizon, this produces twelve regressions for each characteristic, beginning respectively in January, February, and so on. The offset design prevents adjacent observations from mechanically sharing most of the same future holding-period returns. The reported robustness statistics summarize the median coefficient and median expected-signed t-statistic across offsets, together with the fraction of offsets that retain the theory-consistent negative sign.

The 12-month non-overlapping robustness test addresses the horizon where repeated holding-period observations are most severe. The main reversal pattern survives this stricter design. Total risk, three-factor idiosyncratic volatility, low tangibility, small firms, unprofitable firms, young firms, and non-dividend payers all retain negative median coefficients and positive expected-signed median t-statistics. The strongest evidence remains concentrated in risk, idiosyncratic volatility, and low tangibility. Sales growth is correctly signed in most offset samples but weak. Low profitability, illiquidity, and low turnover do not provide comparable support. The non-overlapping regressions reduce the concern that the long-horizon results are mechanically driven by repeated overlapping return windows.

Table 10: Persistence robustness: 12-month non-overlapping horizon regressions

Characteristic	Baseline coef.	Baseline t	Non-overlap coef.	Non-overlap t	Sign share
Total risk	-19.65***	4.43	-23.41***	5.25	1.00
IVOL (FF3)	-18.42***	3.75	-23.72***	5.59	1.00
Low tangibility	-15.27***	4.83	-17.34***	4.31	1.00
Small firms	-14.04***	3.15	-15.53***	3.33	1.00
Unprofitable	-4.86***	3.04	-6.99***	3.15	1.00
Young firms	-8.37**	2.29	-10.02**	2.52	1.00
Non-dividend payer	-4.96***	2.80	-5.27**	2.31	1.00
Sales growth	-3.21	0.61	-3.42	0.44	0.67
Low profitability	0.08	-0.02	-2.06	0.95	0.58
Illiquidity	-2.58	0.72	-3.64	0.80	0.92
Low turnover	4.46	-1.23	4.99	-1.51	0.25

Note. The $SENT^{\perp}$ columns use the overlapping 12-month predictive regressions. The non-overlap columns report the median coefficient and median expected-signed HAC t-statistic across the twelve calendar-offset samples. Sign share is the fraction of offset regressions with the theory-consistent negative coefficient. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Second, sentiment-innovation regressions model the orthogonalized sentiment index as a first-order autoregressive process,

$$SENT_t^{\perp} = a + \rho SENT_{t-1}^{\perp} + u_t, \tag{12}$$

and the residual u_t is interpreted as the unexpected component of sentiment after controlling for its own persistence. The predictive regression is then re-estimated with u_{t-1} in place of $SENT_{t-1}^{\perp}$. The specification asks whether new information in sentiment predicts subsequent cross-sectional return spreads, rather than allowing a long high-sentiment regime to enter the regression repeatedly through a highly persistent level measure. These robustness tests do not replace the level specification in Baker and Wurgler (2006), but they provide a direct check on whether the main results are overstated by overlapping returns or persistent sentiment episodes.

The AR(1) coefficient of $SENT^{\perp}$ is 0.794, with an R^2 of 0.634, confirming that sentiment is highly persistent. When the lagged sentiment level is replaced by the lagged AR(1) innovation, the results weaken substantially. The average expected-signed t-statistic is 1.27 at the 12-month horizon, compared with 2.22 in the overlapping level regression. The results are better interpreted as predictability from persistent high- and low-sentiment regimes than from unexpected month-to-month sentiment shocks.

6.4.1 Extreme-Decile Constituent Reuse

Extreme-decile portfolio regressions may repeatedly measure the same firms rather than a broad cross-section of Swedish equities. Repeated-firm exposure is most relevant for characteristics that are persistent by construction. Large firms tend to remain large, old firms remain old, tangible firms do not rapidly change their asset structure, and highly liquid firms often remain in the liquid tail of the distribution. If the same securities are repeatedly selected into the extreme legs, the predictive regressions may appear broader than they are.

The constituent-reuse diagnostic inspects the firm composition behind the directional extreme-leg portfolios. The diagnostic expands each complete baseline predictive-regression observation into the underlying security-month-leg rows used inside the relevant future return window. It then counts how often companies and securities appear across all directional legs, horizons, and regression windows. The directional legs contain 1,516 unique companies. After applying the complete-case regression requirement, the expanded regression-window exposure still contains 1,461 unique companies.

The regressions reuse persistent characteristic portfolios, not independent firm identities each month. Persistent characteristic portfolios necessarily reuse firms when extreme deciles are formed on slow-moving characteristics such as size, age, tangibility, and liquidity. However, the evidence does not support the stronger concern that the results are driven by a small set of repeatedly sampled firms. The most frequently reused company accounts for only 0.37% of expanded regression-window exposure, while the top 50 companies together account for 13.74%. Continuous portfolio legs are also nearly balanced by construction, whereas binary characteristics are naturally less balanced but still draw on broad firm groups. The diagnostic is a constituent-membership test rather than a return-contribution decomposition, so it cannot prove that recurring firms have no influence on coefficients. The extreme-decile design nevertheless remains a broad characteristic-portfolio design rather than a repeated test on only a small group of companies.

6.4.2 Market-Equity Filtering

Very small firms are theoretically relevant because sentiment effects are expected to be stronger where valuation is more subjective and arbitrage is more difficult. The baseline specification instead relies on the common-equity universe filter, return-cleaning rules, point-in-time accounting construction, stale-fundamental caps, lagged portfolio sorts, and robust inference.

6.4.3 Robustness Interpretation

The robustness evidence is supportive but uneven. The spread evidence supports H1 and H2 for risk, IVOL, size, dividend status, and low tangibility. Stambaugh-style leg evidence supports H3 most clearly for risk, IVOL, and size. Low tangibility is economically and statistically strong in spreads, but its leg pattern shows that the sentiment-insensitive leg contributes materially. Sales growth and illiquidity provide weak evidence for H2, while profitability and turnover are partly inconsistent with H2 and H3. The macro-financial content of $SENT^{\perp}$ remains the central interpretation issue for the discussion.

7 Discussion

7.1 Empirical Interpretation of H1–H3

$SENT^{\perp}$ provides selective evidence of sentiment-related return predictability in the Swedish equity cross-section. The support for H1 is clearest at the 6- and 12-month horizons, where high beginning-of-period sentiment is followed by lower relative returns for several sentiment-prone directional spreads. The pattern is concentrated rather than uniform, with the strongest evidence in total risk, $IVOL^{FF3}$, size, low tangibility, and dividend status.

The results give partial support to H2. The strongest predictive relations appear where the theory in Baker and Wurgler (2006) and Baker and Wurgler (2007) points most directly, namely among firms with subjective valuations and stronger limits to arbitrage. Total risk, idiosyncratic volatility, and size are the cleanest examples because they are directly connected to uncertainty, limited arbitrage capacity, and lower investor coverage. Low tangibility and dividend status also fit the broad Baker-Wurgler spread logic, although their leg patterns are less clean. Profitability, illiquidity, turnover, and sales growth provide weak or conflicting evidence for H2.

The H3 evidence is narrower than the H1 and H2 evidence. A negative directional spread means that the sentiment-prone leg underperforms the sentiment-insensitive leg after high sentiment, but H3 requires more than a negative spread. It requires stronger negative predictability in the sentiment-prone leg itself, consistent with the short-sale asymmetry mechanism in Stambaugh et al. (2012). The leg decomposition fits this mechanism most clearly for total risk, $IVOL^{FF3}$, and size. Low tangibility and dividend status remain important spread results, but their sentiment-insensitive legs contribute materially. These characteristics support cross-sectional predictability more clearly than the stricter overpricing-leg interpretation.

The Swedish setting helps explain why the evidence is concentrated rather than uniform. The market contains many smaller and growth-oriented firms, but household ownership and market structure differ from the U.S. setting behind Baker and Wurgler (2006) and Stambaugh et al. (2012). The evidence therefore looks less like a broad transfer of U.S. sentiment results and more like a Swedish pattern in which sentiment-related predictability appears most strongly among smaller, riskier, and harder-to-arbitrage firms. That interpretation preserves the positive contribution of the results while keeping the uneven characteristic evidence visible.

7.2 Macro-Financial Contamination and Expanded Orthogonalization

Sibley et al. (2012) critique the interpretation of the sentiment index in Baker and Wurgler (2006) rather than only its forecasting performance. Their evidence accepts the starting point that the index predicts cross-sectional returns, but challenges whether the predictive component is necessarily behavioral sentiment. In their abstract, they state that investor sentiment is "strongly correlated with contemporaneous business cycle variables" and that about 63% of the total variation in the index can be explained by such variables. They further conclude that their results "cast doubt on the notion of 'sentiment' as a pure behavioral predictive factor". Their decomposition separates the fitted business-cycle component, $SENTHAT_t$, from the residual component, $SENTRES_t$. In their return-predictability tests, the business-cycle component carries most of the spread and short-leg predictability, while the residual component is much weaker.

The critique applies even to the orthogonalized sentiment index in Baker and Wurgler (2006). Baker and Wurgler residualize their sentiment proxies against a limited macroeconomic control set before extracting the common component. Yet Baker and Wurgler (2007) also acknowledge that sentiment measures may be "contaminated by economic fundamentals". Orthogonalization against a small set of macro controls improves the interpretation of a composite index, while the tested object changes from raw optimism to residual sentiment-related market-state variation.

$SENT^{\perp}$ residualizes the selected Swedish sentiment proxies against CPI, PPI, industrial production, and unemployment. The Sibley-expanded index, $SENT_t^S$, uses the same seven sentiment proxies as $SENT^{\perp}$, including $DIVP_t$, but residualizes them against a broader macro-financial control set before PCA. The expanded controls are unemployment, monthly CPI growth, consumption growth, industrial-production growth, the short rate, the term spread, the value-weighted Swedish market return, monthly market volatility, and the share of zero-return stock-days. These controls enter as $UNEMP$, $dCPI$, $dCONS$, $dIND$, $TBILL$, $TERM$, $VWRETD$, $MKTVOL$, and $PCTZERO$.

Available monthly Swedish data support a narrower control set than the full specification in Sibley et al. (2012). Disposable personal income growth, a recession dummy, the default spread, and aggregate dividend yield are omitted because sufficiently consistent monthly Swedish series are not available for those concepts. The expanded orthogonalization therefore keeps the Sibley logic for the controls that can be measured defensibly and treats the remaining controls as data limitations.

$SENT^{\perp}$ and $SENT^S$ have a correlation of 0.466. The positive correlation shows that they capture a related market state. The moderate magnitude shows that the broader macro-financial controls remove a large part of the $SENT^{\perp}$ common component. The comparison therefore changes the interpretation from a minor robustness check to a test of which conclusions survive a stricter definition of residual sentiment-related variation.

The Sibley-expanded index preserves the expected negative average coefficient at every horizon, but the magnitude and statistical support are much weaker. At the 12-month horizon, the average coefficient falls from -7.89 for $SENT^{\perp}$ to -1.88 for $SENT^S$. The average expected-signed HAC t-statistic falls from 2.22 to 0.73, and the expected-signed bootstrap t-statistic falls from 2.08 to 0.68. The share of characteristics with a theory-consistent 5% HAC result is only 0.09 under $SENT^S$ at every horizon. Characteristic-horizon details are reported in table 29.

Table 11: Sibley-style expanded macro orthogonalization by horizon

Horizon	$SENT^{\perp}$ b	$SENT^{\perp}$ HAC	$SENT^{\perp}$ boot	$SENT^S$ b	$SENT^S$ HAC	$SENT^S$ boot	$SENT^S$ 5%
1 month	-0.97	1.62	1.62	-0.50	0.75	0.68	0.09
3 months	-2.04	1.46	1.45	-0.49	0.24	0.24	0.09
6 months	-4.37*	1.93	1.91	-1.46	0.79	0.75	0.09
12 months	-7.89**	2.22	2.08	-1.88	0.73	0.68	0.09

Note. Each row averages the eleven directional-spread regressions by horizon. HAC and boot columns report expected-signed t-statistics. Positive values support the sentiment-reversal hypothesis. Stars on mean coefficients are based on the expected-signed HAC t-statistic and are descriptive for horizon-level averages.

The characteristic-level evidence shows what survives the stricter macro adjustment. Low tangibility remains the strongest result under $SENT^S$, with a mean expected-signed HAC t-statistic of 2.90. Sales growth and low turnover improve somewhat relative to $SENT^{\perp}$ but remain weak. The main $SENT^{\perp}$ evidence for total risk, $IVOL^{FF3}$, size, non-dividend payers, and unprofitable firms weakens sharply. The mean expected-signed HAC t-statistic falls from 3.37 to 0.43 for $IVOL^{FF3}$, from 2.20 to 0.95 for size, from 2.59 to 0.11 for non-dividend payers, and from 1.80 to 0.09 for unprofitable firms. The strongest interpretation is therefore hierarchical. Low tangibility is the most robust characteristic under expanded macro residualization, while much of the broader $SENT^{\perp}$ evidence depends on the baseline macro adjustment.

7.3 COVID Winsorization as a Stress Test

The Covid-winsorization check in figure 8 and table 30 measures sensitivity to high-leverage crisis observations in the PCA construction. The check leaves $SENT^{\perp}$ unchanged because the Covid shock is treated as a real sentiment episode rather than a data error. Before PCA, the experiment caps only the March 2020 raw ESI_t and CCI_t observations that exceed the four-standard-deviation event-window rule. The baseline $SENT^{\perp}$ series changes little, while the Sibley-expanded series changes materially.

The adjustment changes the 12-month Sibley-expanded average coefficient from -1.88 to -6.98, the expected-signed HAC t-statistic from 0.73 to 1.78, and the 5 percent HAC significance share from 0.09 to 0.36. At the 3- and 6-month horizons, the expected-signed HAC t-statistic rises from 0.24 to 1.21 and from 0.79 to 1.72, respectively. The stress test reduces concern that the results are mechanically dominated by pandemic-period survey outliers. It still leaves the residual index exposed to liquidity conditions, financing constraints, risk appetite, discount-rate variation, and omitted macro-financial state variation.

7.4 Fixed PCA Weights and External Validity

Ung et al. (2024) criticize fixed PCA sentiment indices because proxy loadings may change over time and full-sample PCA weights can embed information that would have been unavailable at the forecast date. Their enhanced sentiment index, STV, estimates principal-component loadings in rolling 36-month windows. The rolling design lets the relative contribution of each proxy vary over time and avoids using future proxy observations when constructing the sentiment value used for prediction. Their empirical comparison is therefore closer to a real-time forecasting exercise than the fixed-weight index design in Baker and Wurgler (2006).

The same limitation applies to $SENT^{\perp}$. The return regressions use lagged sentiment and future Swedish portfolio returns, so the dependent return outcomes do not enter the index construction. However, the PCA weights are estimated using the full proxy sample. The index should therefore be interpreted as an in-sample measure of a historical sentiment-related state, not as a real-time allocation signal or an out-of-sample forecasting model.

Full-sample PCA is a measurement device, not a return-fitting device. The PCA weights are estimated only from sentiment proxies, not from future Swedish return spreads, and the index is not optimized to predict the empirical portfolios. If sentiment is a latent variable measured through noisy proxies, full-sample PCA gives a more stable estimate of the common component. That is defensible when the purpose is historical inference about whether sentiment-related states are associated with later cross-sectional return patterns.

The central limitation is external validity rather than internal return timing. The evidence can support the claim that Swedish sentiment-related states are followed by in-sample cross-sectional return reversals. It cannot by itself show that the same index construction would forecast returns out of sample or work as a real-time trading signal. A rolling or recursive Swedish PCA would be the natural extension, but it would introduce additional choices about window length, component sign anchoring, minimum observations, and proxy availability. In a relatively short monthly Swedish sample, those choices would trade lower look-ahead risk for higher measurement noise.

An appendix implementation of this rolling-window idea is reported in table 21. In this thesis setting, the Swedish STV-style index does not reproduce the main cross-sectional return-spread evidence, which reinforces the interpretation that the baseline index is an in-sample historical measurement device rather than a ready-made real-time forecasting index.

7.5 Factor Controls and Market-Efficiency Interpretation

The weak performance of raw $SENT$ changes the interpretation of the Swedish evidence. In Baker and Wurgler (2006), the raw and macro-orthogonalized indices generally point in the same empirical direction. In the Swedish results, raw $SENT$ gives weaker expected-signed evidence at every horizon, while $SENT^{\perp}$ produces stronger and more consistent negative coefficients for risk, idiosyncratic volatility, dividend status, and size. The PCA diagnostics explain why this difference matters. Raw $SENT$ is most closely related to survey confidence, while $SENT^{\perp}$ loads more strongly on residualized market-based proxies such as turnover, financing composition, and $DIVP$. The pattern fits the critique in Sibley et al. (2012). A sentiment index can predict returns while still containing macro-financial information. The Swedish evidence is strongest when the common component is first stripped of observable macro variation, which makes the result closer to residual sentiment-related market-state predictability than to a broad optimism measure.

The factor-control results change the role assigned to sentiment. Average explanatory power is very low in the sentiment-only regressions and rises sharply after adding $RMRF$, SMB, HML, and UMD. Swedish characteristic spreads therefore move strongly with market, size, value, and momentum variation. Baker and Wurgler (2006) include the same family of controls because the sentiment coefficient should be evaluated after standard return factors have been absorbed. In the asset-pricing tradition of Fama and French (1993) and Carhart (1997), these controls capture common return components that can otherwise be mistaken for predictability.

The factor controls are realized over the same future horizon as the dependent spread return. The regressions are therefore factor-adjusted predictive regressions rather than pure real-time forecasts. The sentiment coefficient measures conditional predictability after absorbing future market, size, value, and momentum return variation over the same holding window. The specification tests whether sentiment-related predictability remains after standard return factors are accounted for, while it is not a standalone trading rule available at the beginning of the holding period.

The evidence is difficult to reconcile with a simple unconditional efficient-market benchmark, but it does not establish a tradable anomaly or definitive market inefficiency after transaction costs, shorting constraints, liquidity, and capacity. Under the efficient-market benchmark in Fama (1970) and Fama (1991), and the systematic-risk logic in Sharpe (1964), predictable cross-sectional return differences should either reflect compensation for risk or disappear through arbitrage. Factor controls remain imperfect, and the Sibley-style tests show that macro-financial contamination remains relevant. The most defensible interpretation is conditional and behavioral in spirit. Swedish sentiment-related states are followed by reversals in selected parts of the cross-section where Baker and Wurgler (2006) and Stambaugh et al. (2012) predict stronger sentiment sensitivity.

7.6 Implications for Identification

The Sibley-expanded results move the interpretation from structural sentiment identification to predictive cross-sectional evidence. $SENT^{\perp}$ predicts Swedish directional spreads in the expected direction, especially for high-risk firms, high-$IVOL^{FF3}$ firms, low-tangibility firms, small firms, and non-dividend payers. The expanded macro adjustment reduces the average 12-month coefficient from -7.89 to -1.88 and lowers the average expected-signed HAC t-statistic from 2.22 to 0.73. A critical reading is that $SENT^{\perp}$ captures a sentiment-related market state with predictive content, while the behavioral component inside that state is only partially isolated.

The main identification problem comes from the economic content of the proxies themselves. Survey confidence, turnover, IPO activity, financing composition, and the dividend premium are plausible sentiment proxies because they move with optimism, speculative participation, hot-issue conditions, issuer timing, and demand for safer dividend-paying firms. The same proxies also move with expected cash flows, discount rates, risk appetite, financing conditions, and market liquidity. Sibley et al. (2012) show that this ambiguity is empirically meaningful in the setting of Baker and Wurgler (2006) because a business-cycle component explains much of the variation and predictive content of the sentiment index. The Swedish correlation of 0.466 between $SENT_t^{\perp}$ and $SENT_t^{S}$ points to the same issue. The expanded controls remove a large part of the $SENT^{\perp}$ sentiment state, and the return evidence weakens with it.

The Sibley-expanded index functions as a stress test of interpretation. A final decomposition of fundamentals and beliefs would require richer identifying variation and additional monthly Swedish data. Disposable income growth, a recession dummy, the default spread, and aggregate dividend yield remain outside the monthly dataset. Some expanded controls may also absorb channels through which sentiment affects prices. Market returns, volatility, liquidity, and term-structure conditions can reflect fundamentals, but they can also be part of the pricing environment in which optimistic demand becomes more or less powerful. The weaker $SENT_t^{S}$ results restrict the interpretation of the $SENT^{\perp}$ evidence and make a purely behavioral reading too strong.

The strongest remaining claim is conditional and cross-sectional. $SENT^{\perp}$ is most informative where the theory predicts larger sentiment-related return differences, namely among firms with subjective valuations or stronger limits to arbitrage. Baker and Wurgler (2006) predict larger return patterns where "valuations are highly subjective and difficult to arbitrage", and the Swedish evidence is strongest for risk, $IVOL^{FF3}$, size, low tangibility, and dividend status. A purely mechanical macro proxy would be expected to map less systematically onto the characteristic families in Baker and Wurgler (2006). The evidence supports sentiment-related predictability in the Swedish cross-section, while the Sibley critique leaves the behavioral purity of the index unresolved.

The leg decomposition narrows the overpricing interpretation further. Stambaugh et al. (2012) argue that high sentiment should mainly predict weak returns on the overpriced side because short-sale impediments make overpricing harder to correct. The leg results fit that logic most clearly for risk, $IVOL^{FF3}$, and size. Low tangibility is strong in the spread regressions, but the sentiment-insensitive leg contributes materially, which makes it weaker evidence for a Stambaugh-style short-leg channel. Profitability, illiquidity, turnover, and sales growth give weaker or conflicting signals. The evidence supports the spread prediction in Baker and Wurgler (2006) more strongly than the stricter Stambaugh overpricing-asymmetry prediction.

The contribution is a Swedish cross-sectional test of sentiment-related return predictability with an explicit Sibley qualification. $SENT^{\perp}$ predicts future return differences among theoretically sentiment-prone Swedish equities, especially in the risk, idiosyncratic-volatility, and size dimensions. The predictive sentiment state combines investor optimism with macro-financial conditions to an extent that remains only partially separated by the identification design. The evidence answers whether the Swedish cross-section contains sentiment-related return patterns, while a structural separation of behavioral beliefs from macro-financial state variation remains outside the identification power of the analysis.

8 Conclusion

Beginning-of-period investor sentiment has a conditional predictive relation with relative future returns in the Swedish equity cross-section. The answer to the research question is affirmative for $SENT^{\perp}$, but the evidence is selective. It appears mainly at 6- and 12-month horizons and is concentrated in total risk, $IVOL^{FF3}$, size, low tangibility, and dividend status.

H1 is supported for $SENT^{\perp}$ after controlling for Swedish market, size, value, and momentum factors. H2 receives partial support because the strongest predictive relations appear where valuation uncertainty and limits to arbitrage are most visible. Low profitability, illiquidity, turnover, and sales growth limit the conclusion. Sentiment-related predictability does not apply across the full characteristic set.

H3 receives narrower support. The leg decomposition gives the clearest sentiment-prone-leg evidence for risk, $IVOL^{FF3}$, and size. Low tangibility and dividend status remain important spread results, but the sentiment-insensitive leg contributes materially. The stricter overpricing-asymmetry pattern is therefore narrower than the broader spread evidence.

$SENT$ performs substantially less well than $SENT^{\perp}$. The raw index is more closely tied to survey confidence, while $SENT^{\perp}$ loads more heavily on residualized market-based sentiment proxies such as turnover, financing composition, and the dividend premium. The evidence is therefore closer to residual sentiment-related market-state predictability than to a broad optimism measure.

Identification remains the main limitation. $SENT^{S}$ shows that part of the predictive content in $SENT^{\perp}$ is sensitive to broader macro-financial residualization, and the full-sample PCA construction makes the index a historical measurement device rather than a real-time forecasting model. The contribution is a Swedish test of cross-sectional sentiment-related return predictability using a constructed sentiment index, characteristic-sorted portfolios, factor-adjusted predictive regressions, leg decomposition, and robustness checks for macro contamination and horizon dependence. The evidence supports conditional association between beginning-of-period sentiment and later Swedish cross-sectional returns, while causal identification and the structural separation of behavioral sentiment, macro-financial state variation, and arbitrage constraints remain unresolved.

9 Perspective

The findings point beyond the Swedish sample in two directions: external validity and mechanism. The external-validity question is which parts of the cross-sectional sentiment framework should travel to other markets. The mechanism question is how local sentiment, external sentiment, investor trading, financing conditions, and arbitrage constraints jointly produce the observed return predictability.

9.1 External Validity and Market Structure

External validity should be evaluated at several levels. The most transferable implication is the cross-sectional prediction in Baker and Wurgler (2006) and Baker and Wurgler (2007). Sentiment-related predictability should be more visible among firms with subjective valuations and stronger limits to arbitrage. The Swedish evidence for risk, $IVOL^{FF3}$, and size fits that broad theoretical pattern and is therefore the part of the result most likely to travel across markets.

The second level is more specific to small open European markets. Sweden combines a concentrated large-cap segment with a broad population of smaller listed and growth-market firms. Similar Nordic or European markets may therefore show the same split between muted aggregate sentiment transmission and stronger cross-sectional predictability in smaller, riskier, and less easily arbitraged firms. Finter et al. (2012) show that European sentiment research can require country-specific measurement, which supports this market-by-market approach.

The least transferable object is the exact Swedish sentiment index. Proxy weights, IPO-market conditions, macro controls, ownership structure, and survey behavior are locally dependent. A German, Danish, Finnish, or U.S. implementation could preserve the same theoretical design while producing a different composite sentiment state. The broader implication is therefore not that Sweden replicates the U.S. evidence mechanically. The findings suggest that the hard-to-value and hard-to-arbitrage prediction is portable, while the specific index construction is market-specific.

9.2 Local, Regional, and Global Sentiment

The next research problem is to decompose the predictive sentiment state. Baker et al. (2012) show that sentiment can contain both global and local components, while Sibley et al. (2012) show that business-cycle information can carry much of the apparent sentiment predictability. A Swedish extension should therefore separate domestic Swedish sentiment from Nordic regional sentiment, U.S. sentiment spillovers, global risk appetite, foreign-investor flows, and macro-financial conditions.

Several channels are empirically distinct. Global sentiment spillovers would appear if international risk appetite predicts Swedish directional spreads even after controlling for local sentiment. Nordic spillovers would appear if sentiment states in similar small open markets move together and predict the same Swedish firm characteristics. U.S. spillovers are especially relevant for speculative, technology, growth, or high-risk firms that may be priced with reference to global risk narratives. Ownership-channel spillovers would require foreign-investor flow or ownership data, while listing-market spillovers would connect IPO activity and small-cap financing conditions across countries.

A natural research design would combine Swedish sentiment proxies with external sentiment indices, foreign-flow measures, and event windows around domestic and international macro announcements. Such a design could distinguish domestic optimism, imported sentiment, and macro-financial news absorption as sources of Swedish return predictability. The findings provide the Swedish cross-sectional pattern that such a decomposition would try to explain.

9.3 From Predictability to Mechanism

The strongest next step is a mechanism design. $SENT^{\perp}$ predicts selected Swedish directional spreads, but direct evidence on trading constraints, investor clienteles, and belief disagreement would be needed to explain why those patterns arise. The short-sale asymmetry mechanism in Stambaugh et al. (2012) and the arbitrage-risk interpretation in Stambaugh et al. (2015) point to short interest, securities-lending supply, borrowing fees, and failed loan availability as the most direct data sources for the arbitrage-constraint channel.

Investor-clientele evidence would require trading or ownership changes by households, institutions, and foreign investors. Retail flow data would test whether high-sentiment states coincide with buying pressure in small, volatile, young, or otherwise sentiment-prone firms. Institutional and foreign ownership changes would show whether professional or international investors absorb or amplify sentiment-related pressure. Analyst disagreement, forecast dispersion, search intensity, news attention, mutual-fund flows, transaction costs, and liquidity measures would add evidence on belief dispersion, attention, and implementability.

Financing-market data would sharpen the IPO and issuance channels. IPO allocation records, first-day trading data, post-listing ownership changes, and small-cap financing conditions would show whether hot-issue activity is connected to investor demand, issuer timing, or broader funding conditions. Event-based designs around sentiment shocks, macro announcements, or issuance waves would help distinguish persistent sentiment states from discrete information events.

9.4 Future Research Priorities

Three extensions follow most directly from the results. The first is a local-versus-global sentiment decomposition. Future research would need to estimate Swedish, Nordic, European, U.S., and global sentiment components jointly and test which component predicts the strongest Swedish characteristics.

The second extension is an investor-type channel. Future work could link sentiment states to household, institutional, and foreign trading flows, then test whether the sentiment-prone firms are bought or sold by the investor groups predicted by behavioral finance. That evidence would connect the Swedish results more directly to the retail-sentiment and attention mechanisms in Kumar and Lee (2006) and Barber and Odean (2008).

The third extension is a direct arbitrage-constraint test in Swedish small caps. Borrowing fees, lending supply, short interest, transaction costs, and liquidity would show whether the firms with stronger sentiment-related reversals are also the firms where mispricing is hardest to correct. That design would move the evidence from characteristic-based inference toward a direct test of the Stambaugh-style channel.

The thesis does not close the question of sentiment-driven mispricing in Sweden. It establishes that sentiment-related market-state variation has structured predictive content in selected parts of the Swedish cross-section. The next step is to identify the investor, trading, and financing channels through which that predictability arises.

A Appendix

A.1 Appendix: Data

A.1.1 Daily and Monthly Market Source Resolution

The monthly security panel is built from both daily and monthly market records. Before 2007, month-end observations are derived from the last valid daily trading record in each month. From January 2007 onward, the monthly workbook is the preferred price source when it provides a valid price. Daily-derived observations are retained as fallback when the monthly source is missing or invalid. Monthly returns are computed from adjusted month-end prices using the total-return factor, and market equity is measured as month-end price multiplied by shares outstanding. The resolved monthly market panel is the return backbone for firm characteristics, factor controls, portfolio sorts, and predictive regressions.

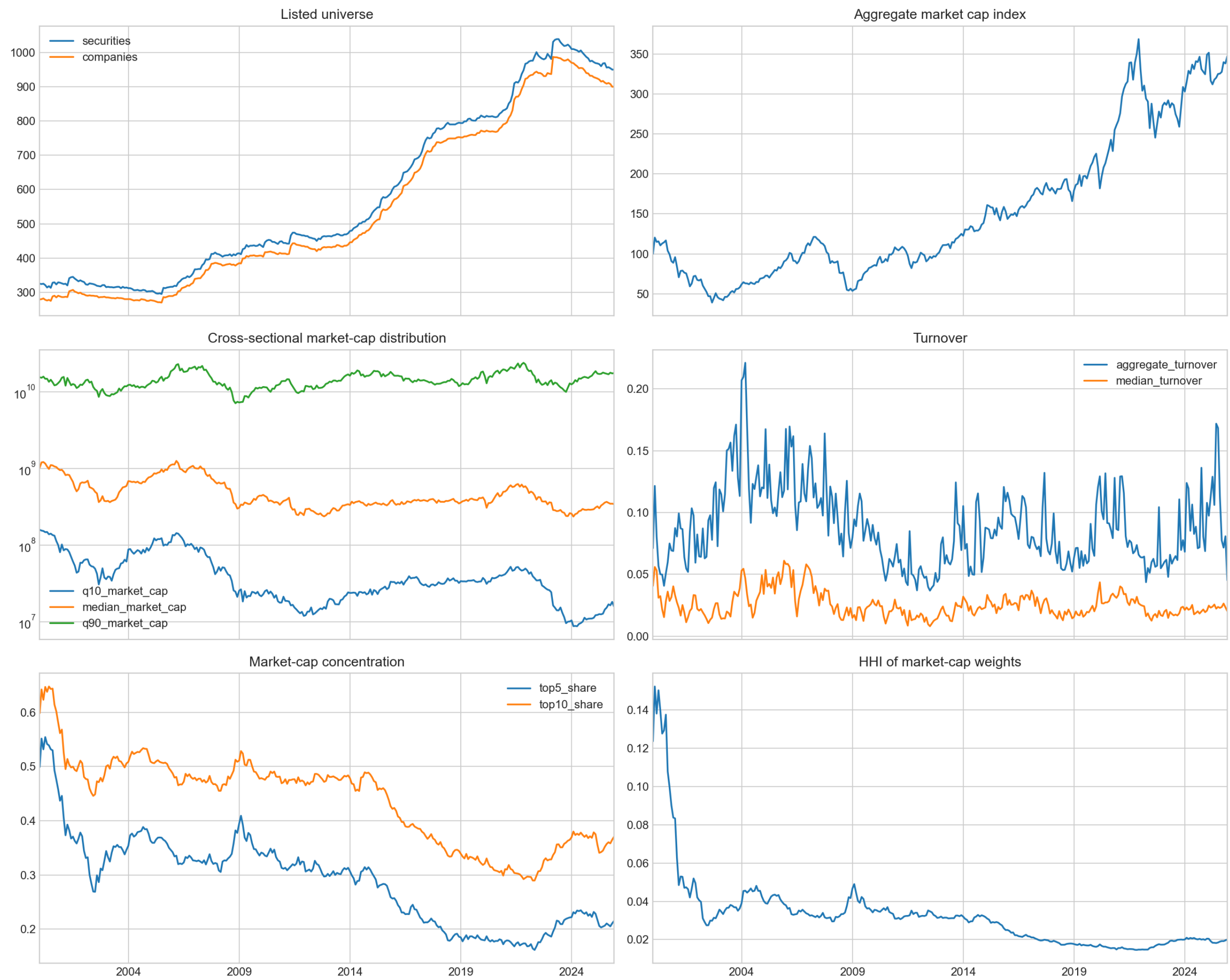

Figure 3: Monthly market-panel composition over time

Note. The figure reports the monthly composition of the filtered Swedish market panel used before empirical test-specific overlap requirements are applied.

A.1.2 Accounting Filing Dates and Point-in-Time Fallback Lags

Quarterly accounting observations are first converted into filing-level records and then into monthly point-in-time snapshots. A report is treated as available on the final filing date where available, otherwise on the preliminary filing date, and otherwise after a fallback lag from fiscal period end. Quarterly observations use a 90-day fallback lag and semiannual observations use a 180-day fallback lag. For each month-end, the snapshot table assigns the latest available filing whose effective month-end is observable by that date. This timing prevents future accounting information from entering portfolio assignments.

A.1.3 Raw Compustat Input-Field Coverage

The raw Compustat extracts impose limits on which accounting characteristics from Baker and Wurgler (2006) can be measured in Sweden. Table 12 reports raw field coverage before any monthly carry-forward is applied. Price, return-factor, and shares-outstanding fields are strongly covered in the market files, while daily trading volume is less complete. Balance-sheet fields needed for total assets, book-equity repair, and liabilities are broadly available, while revenue and tangible fixed-asset fields are usable but incomplete. revtq is used as the single revenue input for the sales-growth characteristic, GS, rather than mixing alternative sales fields. Earnings coverage is strong when income before extraordinary items is used, but the alternative net-income fields are essentially unavailable in the extract. Dividend information is present only for a minority of raw firm-quarter records, which is why dividend-payer status is based on observed dividend history rather than a single-period dividend ratio. External-finance fields are substantially weaker, with sale of common and preferred stock present in fewer than one third of firm-quarters, while repurchases and long-term debt issuance have very low coverage. Research and development expense and deferred-tax adjustment fields are absent from the saved Compustat extract.

Table 12: Raw Compustat input-field coverage

Title	Compustat Code	Raw-row availability	Coverage (%)
Monthly market fields			
Price Close - Monthly	prccm	256,671	96.99%
Trading Volume - Monthly	cshtrm	211,442	79.90%
Daily market, return, and liquidity fields			
Price Close - Daily	prccd	6,307,856	99.99%
Shares Outstanding - Company	cshoc	6,189,018	98.10%
Trading Volume - Daily	cshtrd	4,279,555	67.84%
Total Return Factor	trfd	6,308,333	99.99%
Scale and revenue fields			
Assets - Total	atq	61,856	94.29%
Sales/Turnover (Net)	saleq	49,286	75.13%
Sales/Turnover (Net), YTD	saley	53,105	80.95%
Revenue - Total	revtq	58,650	89.40%
Tangibility and investment fields			
Property Plant and Equipment - Total (Net)	ppentq	52,759	80.42%
Capital Expenditures	capxy	36,080	55.00%
Earnings fields			
Income Before Extraordinary Items	ibq	61,071	93.09%
Net Item - Total	nitq	66	0.1%
Income Before Extraordinary Items, YTD	iby	65,418	99.72%
Net Item - Total, YTD	nity	452	0.69%
Book-equity and balance-sheet fields			
Common/Ordinary Equity - Total	ceqq	51,639	78.72%
Stockholders Equity > Parent > Index Fundamental > Quarterly	seqq	63,087	96.17%
Stockholders Equity - Total	teqq	49,564	75.55%
Liabilities - Total	ltq	61,836	94.26%
Preferred/Preference Stock (Capital) - Total	pstkq	498	0.76%
Retained Earnings	req	34,678	52.86%
Dividend fields			
Cash Dividends	dvy	12,475	19.02%
Dividends - Total	dvty	6,213	9.47%
Dividends - Total, quarterly	dvtq	4,861	7.41%
External-finance fields			
Sale of Common and Preferred Stock	sstky	18,083	27.56%
Purchase of Common and Preferred Stock	prstkcy	926	1.41%
Long-Term Debt - Issuance	dltisy	5	0.01%
Fields not present in the extract			
Research and Development Expense	not present	0	0%
Deferred Taxes and Investment Tax Credit	not present	0	0%

Note. Coverage is measured before point-in-time monthly carry-forward and before portfolio formation. Accounting-field coverage uses the raw quarterly fundamentals denominator of 65,602 firm-quarter records identified by unique gvkey/datadate pairs. Monthly market-field coverage uses 264,645 raw security-month rows from the monthly price extract. Daily market-field coverage uses 6,308,719 raw security-day rows from the daily price extract. Fields marked as not present were not included in the saved WRDS extract used for the analysis.

A.1.4 Swedish Market Background Tables

Table 13: Largest OMXS30 Drawdowns in the Monthly OMXS30 Series

Peak month	Trough month	Recovery month	Peak level	Trough level	Drawdown (%)
Dec. 2021	Sep. 2022	Feb. 2024	2,419.73	1,828.98	-24.4
Feb. 2015	Jun. 2016	Oct. 2019	1,691.03	1,323.58	-21.7
Apr. 2011	Sep. 2011	Jan. 2013	1,162.84	910.17	-21.7
Jan. 2020	Mar. 2020	Sep. 2020	1,783.26	1,482.43	-16.9
Feb. 2025	Apr. 2025	Oct. 2025	2,724.70	2,434.06	-10.7

Note. Drawdowns are computed from monthly adjusted close values. Recovery month is the first later month in which the index reaches or exceeds the previous monthly peak. The monthly OMXS30 series covers January 2009 to December 2025.

Table 14: Largest Companies in the Filtered Sweden Sample, December 2025

Company	Sector	Market cap, SEK bn	Sample share (%)
Investor AB	Financials	1,011.8	8.74
Atlas Copco AB	Industrials	790.1	6.82
Volvo AB	Industrials	601.9	5.20
EQT AB	Financials	449.2	3.88
SEB AB	Financials	393.8	3.40
ASSA ABLOY AB	Industrials	378.7	3.27
Sandvik AB	Industrials	377.1	3.26
Swedbank AB	Financials	363.5	3.14
Saab AB	Industrials	287.0	2.48
Hexagon AB	Information Technology	284.2	2.45

Note. Market capitalisation is aggregated across listed share classes within company name. Sample share is measured relative to the filtered December 2025 Sweden sample.

A.2 Appendix: Methodology

1. Start with the raw inputs
Market data give prices, returns, shares and volume. Compustat gives quarterly accounting data and filing dates. Sentiment inputs cover ESI, CCI, turnover, IPO activity, financing composition and the dividend premium. Macro and rate data cover CPI, PPI, industrial production, employment, the 3-month rate and the 10-year yield.

2. Keep the usable Sweden sample
Keep Swedish listed common equities with usable security-level return observations, positive prices, positive market equity, and the required market, accounting, sentiment and macro data.

3. Keep company and security identifiers separate
company_id = country code + gvkey. security_id = country code + gvkey + iid. Accounting measures are company-level, while prices, returns and portfolio assignments are security-level.

4. Build the monthly analysis panel
Resolve month-end prices. Before 2007, use the last valid daily observation in each month. From 2007 onward, prefer the monthly source and use daily data only as fallback. Then construct adjusted returns, market equity, the risk-free rate, Swedish factors and point-in-time accounting snapshots.

5. Build characteristics and sentiment

Firm characteristics
ME, age, total risk, BE/ME, profitability, dividend-payer status, PPE/A, sales growth, illiquidity, turnover, $IVOL$.

Sentiment proxies
ESI, CCI, $TURN$, $NIPO$, $RIPO$, ED_RATIO, $DIVP$. Residualize the selected proxies against macro controls and use PCA to form $SENT$ and $SENT^{\perp}$.

6. Sort portfolios with lagged characteristics
Use one-month lagged characteristics. Continuous characteristics are sorted into Swedish monthly deciles; binary characteristics form two groups. Directional spreads are coded as sentiment-prone minus sentiment-insensitive, for example small–large, high $IVOL$–low $IVOL$, and low tangibility–high tangibility.

7. Run the predictive tests
Regress future spread returns on lagged $SENT^{\perp}$. Decompose the same tests into sentiment-prone and sentiment-insensitive legs. Add $RMRF$, SMB, HML and UMD over matching 1-, 3-, 6- and 12-month horizons. Use Newey-West/HAC inference and bootstrap robustness.

8. Interpret the results
Test whether high-sentiment states predict lower future returns for sentiment-prone Swedish stocks, and use the spread, leg and robustness evidence to assess H1, H2 and H3.

Figure 4: Empirical construction and testing pipeline.

Note. The figure summarizes how raw Swedish market, accounting, sentiment, IPO, macroeconomic and interest-rate data are converted into the monthly security-level panel used for characteristic-sorted portfolio tests and predictive regressions.

A.2.1 Factor-Construction Details

Monthly factor returns are measured in percentage points and are constructed from the same Swedish security-month panel used in the characteristic sorts. Eligible securities must pass the common-equity screen, have valid monthly returns, and have positive lagged market equity when value-weighted returns are computed. The factor audit table records breakpoints, constituent counts, lagged weight sums, equal-weighted leg returns, value-weighted leg returns, and thin-leg flags.

The Swedish 3-month interbank rate is converted from an annual percentage rate into the monthly risk-free proxy RF_t^{3M}. The value-weighted local market return, $R_t^{M,VW}$, is computed from Swedish common-equity returns using lagged market equity as the portfolio weight. The market excess return is

$$RMRF_t = R_t^{M,VW} - RF_t^{3M}. \tag{13}$$

The 10-year Swedish yield, Y_t^{10Y}, is retained as a macro-financial series, while RF_t^{3M} is the regression risk-free leg.

SMB_t is constructed from monthly Swedish size sorts. Eligible common equities are sorted into low, middle, and high size buckets using the 30th and 70th percentile breakpoints of lagged market equity. The factor return uses the two extreme legs,

$$SMB_t = R_t^{Small,VW} - R_t^{Big,VW}. \tag{14}$$

The middle bucket is retained for audit and breakpoint structure.

HML_t is constructed from monthly Swedish book-to-market sorts. Book equity comes from point-in-time accounting snapshots, while market equity comes from the resolved monthly market panel. The book-to-market signal is lagged before sorting. Eligible common equities are assigned to low, middle, and high book-to-market buckets using 30th and 70th percentile breakpoints. The factor return is

$$HML_t = R_t^{High\ BE/ME,VW} - R_t^{Low\ BE/ME,VW}. \tag{15}$$

UMD_t is constructed from monthly Swedish momentum sorts. The momentum signal compounds prior monthly returns while skipping the most recent month, following the convention that recent one-month reversals should be separated from medium-term momentum. Eligible common equities are sorted into loser, middle, and winner buckets using 30th and 70th percentile breakpoints of the lagged momentum signal. The factor return is

$$UMD_t = R_t^{Winner,VW} - R_t^{Loser,VW}. \tag{16}$$

For horizons of 1, 3, 6, and 12 months, forward factor windows match the compounded portfolio-return window from month t through month $t + h - 1$.

A.2.2 Robustness and Inference Implementation Details

The non-overlapping robustness test re-estimates the directional-spread regression on calendar-offset subsamples. For horizon h, the sample is split into h subsamples according to the month index,

$$m(t) \bmod h = k, \qquad k = 0, 1, \ldots, h - 1. \tag{17}$$

Each offset subsample contains return windows that are separated by the full return horizon. The model is otherwise the same factor-adjusted predictive OLS regression as the primary specification. Reported statistics include median coefficients, median t-statistics, sign-consistency shares, and significance shares across offsets.

The sentiment-innovation test removes first-order persistence from $SENT^{\perp}$ before prediction. The index is modelled as

$$SENT_t^{\perp} = a + \rho SENT_{t-1}^{\perp} + u_t. \tag{18}$$

The residual u_t is the innovation in sentiment, measured as the component of current sentiment left after projecting the index on its own lag. The directional-spread regression is then estimated with u_{t-1} replacing $SENT_{t-1}^{\perp}$. The innovation specification asks whether unexpected changes in sentiment predict future cross-sectional return spreads. The $SENT^{\perp}$ level regression asks whether persistent high- and low-sentiment states predict future reversals.

Standard OLS t-statistics use a covariance estimator that treats regression errors as homoskedastic and serially uncorrelated. The predictive regressions use monthly financial returns, persistent sentiment measures, and multi-month forward return windows. Overlapping returns and persistent sentiment create two econometric problems. Residual variance can change over time, and residuals can be serially correlated because adjacent 3-, 6-, and 12-month return windows share many of the same monthly returns. The coefficient estimate remains an OLS estimate, but the conventional OLS standard error becomes a weak basis for inference when the error covariance matrix is misspecified.

Newey-West inference replaces the conventional OLS covariance matrix with a heteroskedasticity- and autocorrelation-consistent covariance estimator (Newey and West, 1987). Finter et al. (2012) use Newey-West standard errors in their German sentiment regressions, where monthly sentiment and return relations are estimated in time series. Ding et al. (2019) also report Newey-West standard errors and bootstrap inference in sentiment-return regressions, explicitly because the tests involve heteroskedasticity and serial correlation. The point estimate $\hat{\beta}$ is still the OLS coefficient. The difference is the estimated standard error used in the t-statistic,

$$t_{\beta}^{HAC} = \frac{\hat{\beta}}{\widehat{se}_{HAC}(\hat{\beta})}. \tag{19}$$

The HAC standard error is computed from autocovariances of the regression residuals with declining Bartlett weights. The lag length is at least $h-1$ for a horizon of h months, which directly addresses the mechanical overlap in the dependent return windows. Empirical sentiment and anomaly papers use robust time-series inference for the same reason. The return-predictability question concerns whether a coefficient survives serial dependence in monthly portfolio returns. Independent-error inference is too narrow for that question (Stambaugh et al., 2012).

Baker and Wurgler use bootstrap inference in their long-short portfolio regressions because an autocorrelated sentiment index can create finite-sample bias when innovations in sentiment are correlated with innovations in portfolio returns (Baker and Wurgler, 2006). The bootstrap approach follows the same motivation. It treats persistence and time-series dependence as part of the sampling problem and reduces reliance on a single analytical covariance formula.

The moving-block bootstrap resamples blocks of monthly regression observations, re-estimates the OLS regression in each bootstrap sample, and computes the dispersion of the resulting coefficient estimates. The bootstrap t-statistic is

$$t_{\beta}^{Boot} = \frac{\hat{\beta}}{\widehat{se}_{Boot}(\hat{\beta})}. \tag{20}$$

The bootstrap standard error $\widehat{se}_{Boot}(\hat{\beta})$ is the standard deviation of the bootstrap coefficient estimates. Bootstrap t-statistics differ from HAC t-statistics because the standard error is generated from repeated block-resampled regressions. HAC inference corrects the estimated covariance matrix in the original sample. Bootstrap inference asks whether the coefficient remains large relative to the empirical distribution generated by resampling dependent blocks of the data.

A.2.3 Diagnostic Tables and Audit Flags

Diagnostics are part of the empirical design. Market-source overlap diagnostics document the hybrid monthly source resolution. Duplicate-key checks protect the security-month and company-month panels. Coverage audits document characteristic availability. Factor-leg audits record whether the Swedish factor portfolios have usable inputs. IVOL status flags show whether monthly residual-volatility estimates are available or blocked by insufficient inputs. Sentiment-index diagnostics record the selected lead-lag proxy set, PCA coefficients, and macro-orthogonalized proxy weights.

The diagnostics separate data-quality checks from unresolved limitations. Measurement error in a constructed panel can otherwise be mistaken for asset-pricing evidence.

A.2.4 Composite Sentiment Index

Table 15: Preliminary PCA component interpretation for lead-lag treatment

Component	Variance	Cumulative	Largest absolute loadings	Interpretation
Comp1	0.255	0.255	ED_RATIO_{t-1} 0.510, ED_RATIO_t 0.509, $DIVP_t$ 0.453	Financing composition and dividend valuation
Comp2	0.186	0.441	$TURN_t$ 0.523, $TURN_{t-1}$ 0.506, $NIPO_{t-1}$ -0.317	Trading activity and IPO-market state
Comp3	0.136	0.577	ESI_t 0.482, CCI_{t-1} 0.473, CCI_t 0.455	Survey confidence
Comp4	0.106	0.683	$NIPO_t$ 0.604, $NIPO_{t-1}$ 0.540, ESI_{t-1} -0.285	IPO volume timing
Comp5	0.098	0.782	CCI_t -0.481, ESI_t -0.478, ESI_{t-1} 0.471	Survey timing
Comp6	0.041	0.822	$RIPO_t$ 0.555, CCI_{t-1} -0.445, ESI_{t-1} 0.421	IPO-return and survey residual variation
Comp7	0.037	0.860	$RIPO_{t-1}$ 0.600, $NIPO_t$ 0.360, $NIPO_{t-1}$ -0.354	IPO timing residual variation

Note. Largest loadings are selected by absolute value within each retained component. Signs are retained because they show component orientation. The 85% retention threshold follows CICSI-style PCA construction and differs from the first-component construction in Baker and Wurgler (2006).

Table 16: Full preliminary sentiment PCA eigenvectors for lead-lag treatment

Proxy candidate	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6	Comp7
ESI_t	0.0239	-0.1690	0.4822	0.0454	-0.4779	-0.2561	0.0666
ESI_{t-1}	0.0057	-0.1595	0.4295	-0.2846	0.4711	0.4206	0.0647
CCI_t	0.0453	-0.1840	0.4554	0.1684	-0.4807	0.3450	-0.0295
CCI_{t-1}	0.0357	-0.2021	0.4733	-0.1591	0.4419	-0.4450	0.0413
$TURN_t$	-0.1036	0.5228	0.2277	0.2387	0.0920	0.0346	0.1341
$TURN_{t-1}$	-0.0899	0.5062	0.2332	0.2809	0.1040	0.1266	0.2189
$NIPO_t$	0.1538	-0.2916	-0.0574	0.6044	0.1586	0.0419	0.3599
$NIPO_{t-1}$	0.1621	-0.3165	-0.0415	0.5402	0.2506	-0.0080	-0.3544
$RIPO_t$	-0.0243	0.0012	-0.0141	0.0125	0.0853	0.5552	-0.0514
$RIPO_{t-1}$	-0.0200	0.0085	-0.0650	0.0583	0.0273	-0.2294	0.6000
ED_RATIO_t	0.5091	-0.0319	-0.0875	-0.1714	-0.0699	0.1231	0.2846
ED_RATIO_{t-1}	0.5103	-0.0348	-0.0816	-0.1776	-0.0383	0.1293	0.2800
$DIVP_t$	0.4533	0.2746	0.0927	0.0618	0.0360	-0.0953	-0.2776
$DIVP_{t-1}$	0.4487	0.2808	0.1139	0.0433	0.0223	-0.1399	-0.2621

Note. The table reports eigenvectors for the seven retained components from the preliminary PCA on current and one-month-lagged values of the seven sentiment proxies. The retained components explain 86.0% of total standardized proxy variance.

Table 17: Lead-lag selection for sentiment proxies

Proxy	Current corr.	Lag corr.	Corr. gap	Selected timing
ESI	0.003	**0.105**	0.102	Lagged
CCI	0.058	**0.125**	0.067	Lagged
TURN	0.419	**0.448**	0.028	Lagged
NIPO	**0.167**	0.146	0.020	Current
RIPO	**0.001**	-0.040	0.041	Current
ED_RATIO	0.607	**0.614**	0.008	Lagged
DIVP	**0.894**	0.892	0.002	Current

Note. The correlation gap is the selected correlation minus the alternative timing correlation. The selected timing is the version with the higher correlation with the preliminary PCA index.

Table 18: Raw sentiment index composition

Proxy	Score coef. in $SENT_t$	Corr. with $SENT_t$	Corr. rank
ESI_{t-1}	0.262	0.699	1
CCI_{t-1}	0.203	0.668	2
$TURN_{t-1}$	0.121	0.166	6
$NIPO_t$	0.101	0.226	5
$RIPO_t$	0.058	0.093	7
ED_RATIO_{t-1}	0.122	0.445	4
$DIVP_t$	0.179	0.534	3

Note. Score coefficients are summed weighted component coefficients from the selected-proxy PCA. The retained raw components explain 89.9% of standardized selected-proxy variance.

Table 19: Macro-orthogonalized sentiment index composition

Residualized proxy	Score coef. in $SENT_t^{\perp}$	Corr. with $SENT_t^{\perp}$	Corr. rank
$rESI_{t-1}$	0.115	0.127	6
$rCCI_{t-1}$	0.016	0.044	7
$rTURN_{t-1}$	0.250	0.756	2
$rNIPO_t$	0.150	0.221	4
$rRIPO_t$	0.064	0.138	5
rED_RATIO_{t-1}	0.199	0.754	3
$rDIVP_t$	0.224	0.773	1

Note. Score coefficients are summed weighted component coefficients from the residualized selected-proxy PCA. The retained macro-orthogonalized components explain 87.4% of standardized residualized proxy variance.

A.2.5 First-Component PCA Robustness

The baseline macro-orthogonalized index, $SENT^{\perp}$, retains principal components until cumulative explained variance exceeds 85%. This rule is used because the Swedish proxy set does not have a single dominant first component after macro residualization. In Baker and Wurgler (2006), the first principal component explains 49% of the variance in the raw sentiment index and 53% of the variance in the macro-orthogonalized sentiment index. In the Swedish construction, the first residualized component explains only 31.2% of standardized proxy variance, while the retained five-component construction explains 87.4%. A first-component-only index would therefore discard most of the common variation in the residualized proxy set.

The first-component robustness index, $SENT^{\perp,PC1}$, uses the same seven proxies, the same selected timing, and the same Swedish macro controls as $SENT^{\perp}$, but retains only the first principal component. The selected inputs are ESI_{t-1}, CCI_{t-1}, $TURN_{t-1}$, $NIPO_t$, $RIPO_t$, ED_RATIO_{t-1}, and $DIVP_t$. The two indices are strongly but not perfectly related, with a monthly correlation of 0.792. The largest difference occurs in April 2020, where the full retained-component index captures the Covid-period proxy shock more strongly than the first-component-only index.

Table 20: First-component PCA robustness

Measure	$SENT^{\perp}$	$SENT^{\perp,PC1}$
Retained components	5	1
Cumulative variance retained	87.4%	31.2%
Correlation with $SENT^{\perp}$	1.000	0.792
Mean expected-signed HAC t, 1-month horizon	1.62	1.77
Mean expected-signed HAC t, 3-month horizon	1.46	1.87
Mean expected-signed HAC t, 6-month horizon	1.93	2.16
Mean expected-signed HAC t, 12-month horizon	2.22	2.20
Theory-consistent 5% HAC share, 12-month horizon	0.64	0.64

Note. Both indices use the same macro residualization and lead-lag selected proxy set. Expected-signed t-statistics multiply the coefficient t-statistic by the predicted negative sign of the sentiment-prone spread regression.

The first-component results support the baseline rather than replacing it. $SENT^{\perp,PC1}$ produces similar or stronger average expected-signed t-statistics at the 1-, 3-, and 6-month horizons and almost identical average evidence at the 12-month horizon. No baseline-significant directional-spread regression loses 5% HAC significance under the first-component variant. The robustness check therefore shows that the main return-predictability pattern is not mechanically dependent on the multi-component PCA rule. The 85% rule is retained for the thesis baseline because it provides a broader measurement of the latent sentiment-related state and avoids treating a 31.2% first component as sufficient for a multidimensional Swedish proxy set.

A.2.6 Alternative Sentiment-Index Designs

This appendix also compares the thesis baseline, $SENT^{\perp}$, with three alternative sentiment-index designs that were considered during development. The comparison is regenerated on the current eleven-characteristic directional-spread design rather than copied from archived exploratory output. The current design includes GS, excludes the discarded $IVOL^{MKT}$ measure, and uses the same factor-adjusted predictive-regression specification as the main empirical results.

The first alternative, $SENT^{\perp,noDIV}$, removes the dividend-premium proxy and therefore corresponds to the six-proxy macro-orthogonalized Swedish index without the Baker–Wurgler dividend-premium channel. The second alternative, $SENT^{\perp,PLL}$, uses the fixed lead-lag timing from the Swedish sentiment-index paper rather than selecting proxy timing from the present Swedish sample. The third alternative, STV^{SWE}, adapts the enhanced sentiment-index logic in Ung et al. (2024). It uses the same seven available Swedish proxies as the thesis baseline, including $DIVP_t$, but estimates the sentiment component in rolling 36-month windows. Within each window, proxy values are residualized against the Swedish macro controls, standardized, and combined using only the first principal component. The index value for month t is the final-month score from the rolling window ending in t. The purpose of STV^{SWE} is methodological robustness, not replacement of the thesis baseline.

Table 21: Alternative sentiment-index robustness comparison

Measure	$SENT^{\perp}$	$SENT^{\perp,noDIV}$	$SENT^{\perp,PLL}$	STV^{SWE}
Correlation with $SENT^{\perp}$	1.000	0.902	0.851	-0.119
Mean expected-signed HAC t, 1-month horizon	1.62	1.25	1.46	-0.04
Mean expected-signed HAC t, 3-month horizon	1.46	1.08	1.48	0.27
Mean expected-signed HAC t, 6-month horizon	1.93	1.57	1.85	-0.10
Mean expected-signed HAC t, 12-month horizon	2.22	1.90	2.18	-0.57
Theory-consistent 5% HAC share, 12-month horizon	64%	55%	55%	0%

Note. All regressions use the current 11-characteristic directional-spread panel and the same factor controls as the main specification. Expected-signed t-statistics multiply the sentiment coefficient t-statistic by the predicted negative sign of the sentiment-prone spread. STV^{SWE} is a 36-month rolling, macro-residualized first-component PCA index adapted from the enhanced sentiment-index methodology.

The comparison supports the final index choice. Removing the dividend premium weakens average expected-signed evidence at every horizon: at the 12-month horizon, the mean expected-signed HAC t-statistic falls from 2.22 for $SENT^{\perp}$ to 1.90 for $SENT^{\perp,noDIV}$. The paper-lead-lag index is close to the baseline and slightly higher at the 3-month horizon, but it is weaker at the 1-, 6-, and 12-month horizons and has a lower 12-month theory-consistent 5% share. This suggests that the sample-selected timing is not arbitrary, while the paper timing remains a useful benchmark.

The rolling STV^{SWE} index performs poorly in the current cross-sectional design. It is weakly negatively correlated with $SENT^{\perp}$, has negative average expected-signed HAC t-statistics at the 1-, 6-, and 12-month horizons, and produces no 12-month theory-consistent 5% results. This does not reject the rolling-index argument in Ung et al. (2024); rather, it shows that a short-window first-component index designed for time-varying measurement and forecasting does not recover the same Swedish cross-sectional spread evidence. The retained baseline therefore reflects both economic logic and empirical performance: it keeps the dividend-premium channel, lets the Swedish data select proxy timing, and gives the strongest overall evidence in the current directional-spread tests.

A.2.7 Raw Sentiment-Proxy Diagnostics

Table 22: Descriptive statistics for raw sentiment proxy inputs

Sentiment proxy	Mean	Std. dev.	Min.	Max.	Median	Obs.
Economic sentiment index (ESI_t)	0.0007	0.0397	-0.1412	0.4921	-0.0011	299
Consumer confidence index (CCI_t)	0.0008	0.0273	-0.0653	0.1290	-0.0019	299
Turnover ($TURN_t$)	-3.1224	0.3964	-4.0663	-1.9195	-3.1493	299
IPO number ($NIPO_t$)	1.4181	2.6715	0.0000	21.0000	0.0000	299
IPO initial return ($RIPO_t$)	0.1503	0.5412	-0.6511	4.6848	0.0622	130
Equity-to-debt ratio (ED_RATIO_t)	0.2929	0.0791	0.2114	0.5483	0.2558	299
Dividend premium ($DIVP_t$)	-0.3791	0.7499	-2.1240	0.8512	-0.2626	289

Note. Statistics are computed from the unstandardized monthly proxy series before PCA and before macro residualization. $RIPO_t$ is observed only in months with observed IPO initial-return information. $NIPO_t$ equals zero in months without IPO activity. $TURN_t$ and $DIVP_t$ are logarithmic proxy constructions.

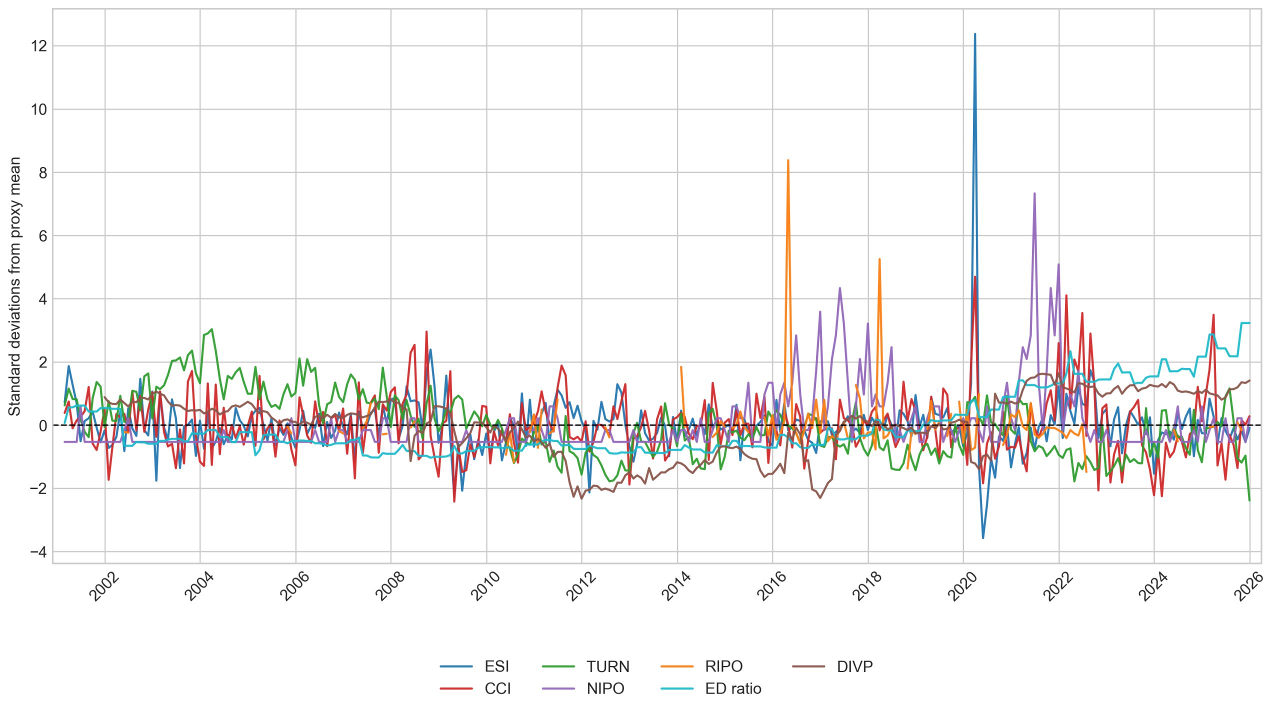

Figure 5: Standardized sentiment proxy inputs over time

Note. Each proxy is standardized by subtracting its sample mean and dividing by its sample standard deviation. The horizontal dashed line marks zero. $RIPO_t$ is plotted only in months with observed IPO initial-return information.

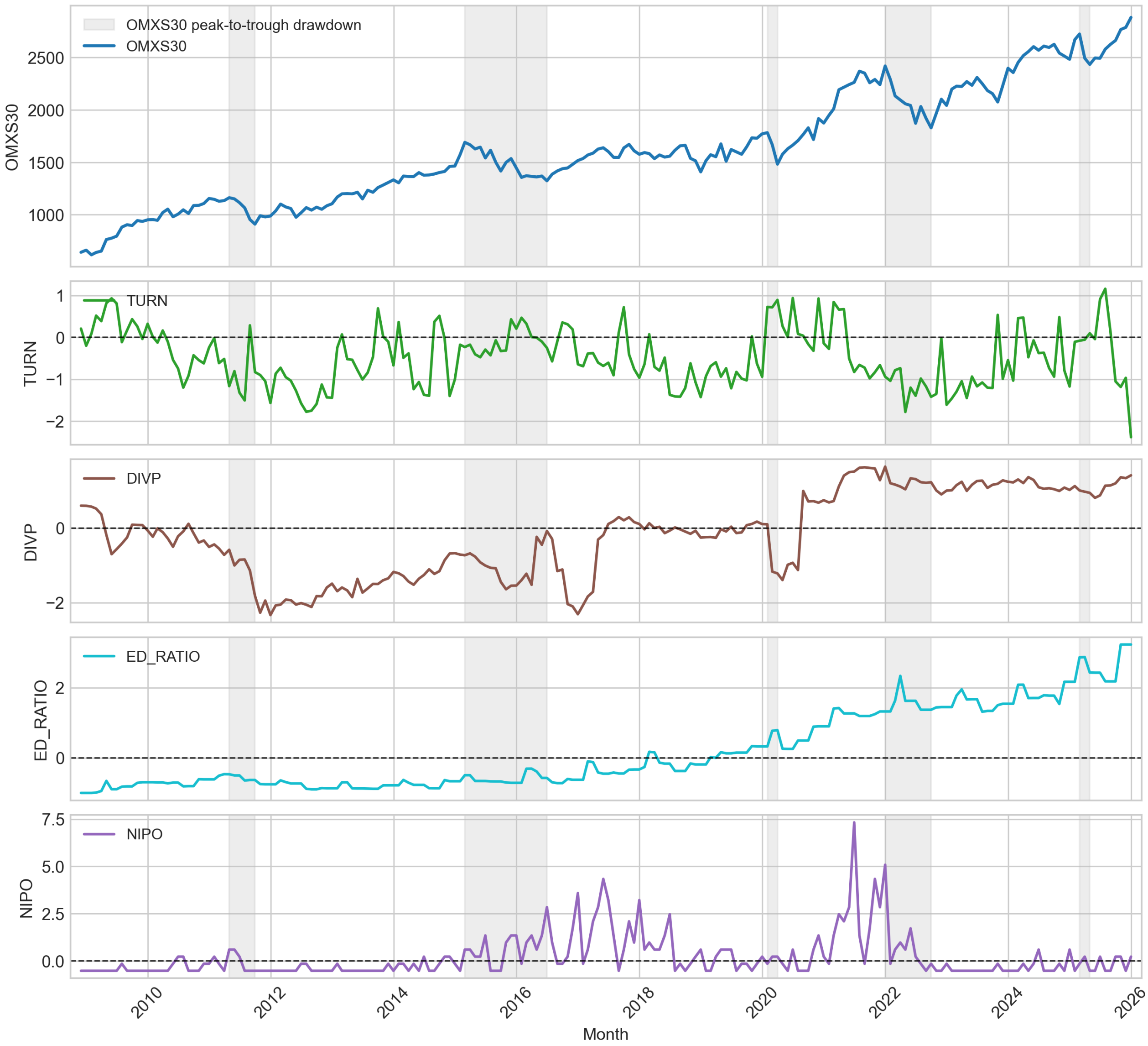

Figure 6: OMXS30 drawdowns and selected standardized sentiment proxies

Note. Grey areas mark the five largest peak-to-trough OMXS30 drawdowns in the monthly OMXS30 series. Proxy series are standardized by subtracting each proxy mean and dividing by its sample standard deviation. The horizontal dashed line in each proxy panel marks zero.

A.3 Appendix: Investigation, Analysis and Results

A.3.1 Baker-Wurgler-Style Directional Spreads

Table 23: Correlations of directional portfolio spreads

	ME	Age	Risk	Non-payer	Unprof.	PPE/A	GS	IVOL FF3	ILLIQ	XTURN	E+BE
ME	1.00										
Age	0.11	1.00									
Risk	0.51	-0.05	1.00								
Non-payer	0.70	0.16	0.61	1.00							
Unprof.	0.39	0.18	0.15	0.40	1.00						
PPE/A	0.26	0.02	0.27	0.22	0.13	1.00					
GS	-0.04	0.09	0.05	-0.06	-0.13	-0.09	1.00				
IVOL FF3	0.62	-0.04	0.81	0.65	0.20	0.20	0.03	1.00			
ILLIQ	0.81	0.14	0.45	0.66	0.32	0.19	-0.06	0.59	1.00		
XTURN	-0.28	0.21	-0.56	-0.47	-0.13	-0.22	0.07	-0.54	-0.07	1.00	
E+BE	0.17	0.02	-0.01	0.18	0.81	0.08	-0.12	0.02	0.11	-0.03	1.00

Note. The table reports pairwise correlations of monthly directional spread returns. Pairwise monthly observations range from 262 to 311.

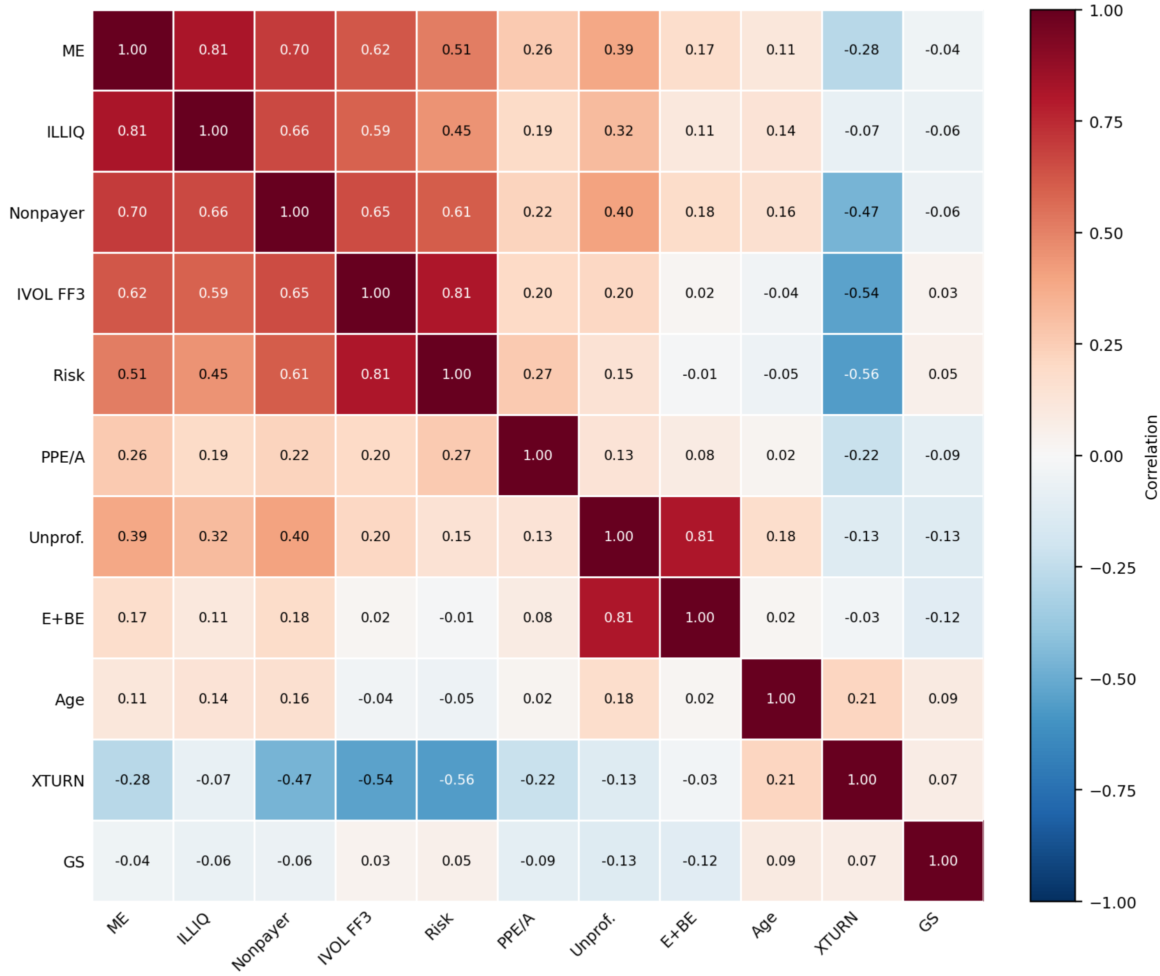

Figure 7: Correlation heatmap of directional spread returns

Note. The figure uses the same pairwise monthly directional-spread correlations as table 23. Characteristics are ordered by a deterministic nearest-correlation path so that related spread-return patterns appear next to each other.

Table 24: Predictive regression summary by return horizon

Horizon	Index	Mean coef.	Mean HAC t	Expected t	Bootstrap t	5% share
1	Orthogonalized	-0.97	-1.62	1.62	1.62	0.36
1	Raw	-0.59	-0.94	0.94	0.86	0.27
3	Orthogonalized	-2.04	-1.46	1.46	1.45	0.36
3	Raw	-1.03	-0.63	0.63	0.61	0.18
6	Orthogonalized	-4.37^{*}	-1.93	1.93	1.91	0.45
6	Raw	-2.63	-1.17	1.17	1.11	0.09
12	Orthogonalized	-7.89^{**}	-2.22	2.22	2.08	0.64
12	Raw	-4.00	-1.06	1.06	1.00	0.09

Note. The expected-signed t-statistic multiplies the coefficient t-statistic by the theoretically expected sign. Positive values support the sentiment-reversal hypothesis. Stars on mean coefficients are based on the absolute mean HAC t-statistic and are descriptive for horizon-level averages.

The detailed horizon tables place $SENT$ and $SENT^{\perp}$ side by side. The $SENT^{\perp}$ estimates across all horizons appear in table 5.

Table 25: Predictive Regressions for Directional Sentiment-Prone Spreads: 1-Month Horizon

Characteristic	Spread	$SENT$ Coef.	$SENT$ HAC t	$SENT^{\perp}$ Coef.	$SENT^{\perp}$ HAC t	$SENT^{\perp}$ Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-0.92	-1.17	-1.34*	-1.90	-1.92
Age	Young minus old	-0.40	-0.91	-0.38	-0.80	-0.83
σ	High minus low	-0.79	-1.19	-2.60***	-3.50	-3.51
$IVOL^{FF3}$	High minus low	-0.68	-0.85	-2.24***	-2.69	-2.69
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	0.28	0.58	-0.44	-0.64	-0.62
$Unprofitable$	Unprofitable minus profitable	-0.03	-0.08	-0.54	-0.96	-0.98
$Nonpayer$	Nonpayer minus payer	0.07	0.27	-0.63**	-2.24	-2.26
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-1.37**	-2.41	-1.27***	-2.99	-2.94
GS	High minus low	-1.10**	-2.18	-0.80	-1.33	-1.28
Panel D. Trading Frictions						
$ILLIQ$	High minus low	0.10	0.20	-0.30	-0.67	-0.66
$XTURN$	Low minus high	-1.68**	-2.56	-0.11	-0.16	-0.16

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent because high sentiment predicts lower subsequent returns for the sentiment-prone side. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Table 26: Predictive Regressions for Directional Sentiment-Prone Spreads: 3-Month Horizon

Characteristic	Spread	$SENT$ Coef.	$SENT$ HAC t	$SENT^{\perp}$ Coef.	$SENT^{\perp}$ HAC t	$SENT^{\perp}$ Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-0.52	-0.34	-2.44	-1.38	-1.39
Age	Young minus old	-1.53	-1.35	-1.06	-0.79	-0.74
σ	High minus low	-1.30	-0.81	-6.49***	-3.56	-3.48
$IVOL^{FF3}$	High minus low	-1.53	-0.85	-5.60***	-2.97	-3.01
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	1.06	1.25	0.43	0.34	0.32
$Unprofitable$	Unprofitable minus profitable	0.41	0.75	-1.11	-1.36	-1.36
$Nonpayer$	Nonpayer minus payer	0.19	0.32	-1.88**	-2.46	-2.48
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-3.47***	-2.85	-4.46***	-3.86	-3.79
GS	High minus low	-1.88	-1.58	-0.88	-0.64	-0.63
Panel D. Trading Frictions						
$ILLIQ$	High minus low	0.72	0.85	-0.03	-0.03	-0.03
$XTURN$	Low minus high	-3.48**	-2.30	1.04	0.62	0.63

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent because high sentiment predicts lower subsequent returns for the sentiment-prone side. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Table 27: Predictive Regressions for Directional Sentiment-Prone Spreads: 6-Month Horizon

Characteristic	Spread	$SENT$		$SENT^{\perp}$		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-3.82	-1.41	-6.55**	-2.39	-2.39
Age	Young minus old	-3.28*	-1.87	-2.86	-1.23	-1.22
σ	High minus low	-4.22*	-1.65	-12.08***	-4.30	-4.25
$IVOL^{FF3}$	High minus low	-3.82	-1.36	-11.74***	-4.05	-4.02
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	0.98	0.63	0.41	0.18	0.18
$Unprofitable$	Unprofitable minus profitable	0.04	0.04	-2.50*	-1.84	-1.76
$Nonpayer$	Nonpayer minus payer	-0.14	-0.14	-3.58***	-2.88	-2.91
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-6.02***	-3.40	-8.34***	-4.21	-4.11
GS	High minus low	-3.70*	-1.77	-1.41	-0.55	-0.54
Panel D. Trading Frictions						
$ILLIQ$	High minus low	-0.20	-0.15	-1.32	-0.65	-0.64
$XTURN$	Low minus high	-4.77*	-1.76	1.91	0.67	0.69

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent because high sentiment predicts lower subsequent returns for the sentiment-prone side. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Table 28: Predictive Regressions for Directional Sentiment-Prone Spreads: 12-Month Horizon

Characteristic	Spread	$SENT$		$SENT^{\perp}$		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-9.14	-1.61	-14.04***	-3.15	-2.84
Age	Young minus old	-6.33*	-1.82	-8.37**	-2.29	-2.11
σ	High minus low	-5.45	-1.09	-19.65***	-4.43	-4.18
$IVOL^{FF3}$	High minus low	-5.15	-1.07	-18.42***	-3.75	-3.76
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	2.08	0.92	0.08	0.02	0.02
$Unprofitable$	Unprofitable minus profitable	-0.99	-0.89	-4.86***	-3.04	-2.90
$Nonpayer$	Nonpayer minus payer	-0.66	-0.41	-4.96***	-2.80	-2.59
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-11.44***	-3.19	-15.27***	-4.83	-4.37
GS	High minus low	-3.88	-1.37	-3.21	-0.61	-0.58
Panel D. Trading Frictions						
$ILLIQ$	High minus low	-0.13	-0.05	-2.58	-0.72	-0.69
$XTURN$	Low minus high	-2.85	-1.10	4.46	1.23	1.14

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent because high sentiment predicts lower subsequent returns for the sentiment-prone side. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

A.4 Appendix: Discussion

A.4.1 Sibley-Expanded Sentiment Robustness

The Sibley-expanded comparison reports the coefficient and HAC t-statistic for each directional-spread regression under $SENT^{\perp}$ and $SENT^{S}$. The comparison column classifies each characteristic-horizon test by whether the expected negative coefficient is significant at the 5 percent HAC level before and after the Sibley-expanded macro adjustment. Negative coefficients support the expected sentiment-reversal pattern because each spread is coded as sentiment-prone minus sentiment-insensitive.

Table 29: Sibley-expanded sentiment robustness by characteristic and horizon

Characteristic	$SENT^{\perp}$		$SENT^{S}$		Baseline → Sibley
	Coef.	HAC t	Coef.	HAC t	
1-month horizon					
Small firms	-1.34*	-1.90	-0.79	-1.09	Never 5%
Young firms	-0.38	-0.80	-0.07	-0.17	Never 5%
Total risk	**-2.60*****	**-3.50**	**-1.14**	**-1.64**	**Loses 5%**
$IVOL^{FF3}$	**-2.24*****	**-2.69**	**-0.62**	**-0.81**	**Loses 5%**
Low profitability	-0.44	-0.64	0.24	0.44	Never 5%
Unprofitable	-0.54	-0.96	-0.15	-0.34	Never 5%
Non-dividend payer	**-0.63****	**-2.24**	**0.07**	**0.38**	**Loses 5%**
Low tangibility	-1.27***	-2.99	-1.55**	-2.50	Stays 5%
Sales growth	-0.80	-1.33	-0.53	-0.97	Never 5%
Illiquidity	-0.30	-0.67	0.15	0.31	Never 5%
Low turnover	-0.11	-0.16	-1.08*	-1.86	Never 5%
3-month horizon					
Small firms	-2.44	-1.38	0.01	0.01	Never 5%
Young firms	-1.06	-0.79	-0.46	-0.48	Never 5%
Total risk	**-6.49*****	**-3.56**	**-1.30**	**-0.96**	**Loses 5%**
$IVOL^{FF3}$	**-5.60*****	**-2.97**	**-0.63**	**-0.41**	**Loses 5%**
Low profitability	0.43	0.34	1.23	1.43	Never 5%
Unprofitable	-1.11	-1.36	0.48	1.22	Never 5%
Non-dividend payer	**-1.88****	**-2.46**	**0.30**	**0.74**	**Loses 5%**
Low tangibility	-4.46***	-3.86	-3.11***	-2.88	Stays 5%
Sales growth	-0.88	-0.64	-0.90	-0.98	Never 5%
Illiquidity	-0.03	-0.03	0.76	1.12	Never 5%
Low turnover	1.04	0.62	-1.82	-1.45	Never 5%
6-month horizon					
Small firms	**-6.55****	**-2.39**	**-3.03**	**-1.27**	**Loses 5%**
Young firms	-2.86	-1.23	-0.54	-0.39	Never 5%
Total risk	**-12.08*****	**-4.30**	**-3.29**	**-1.62**	**Loses 5%**
$IVOL^{FF3}$	**-11.74*****	**-4.05**	**-1.44**	**-0.72**	**Loses 5%**
Low profitability	0.41	0.18	1.62	0.97	Never 5%
Unprofitable	-2.50*	-1.84	0.17	0.26	Never 5%
Non-dividend payer	**-3.58*****	**-2.88**	**-0.20**	**-0.28**	**Loses 5%**
Low tangibility	-8.34***	-4.21	-4.96***	-3.10	Stays 5%
Sales growth	-1.41	-0.55	-1.62	-1.11	Never 5%
Illiquidity	-1.32	-0.65	-0.16	-0.15	Never 5%
Low turnover	1.91	0.67	-2.67	-1.25	Never 5%
12-month horizon					
Small firms	**-14.04*****	**-3.15**	**-6.64**	**-1.46**	**Loses 5%**
Young firms	**-8.37****	**-2.29**	**-1.18**	**-0.52**	**Loses 5%**
Total risk	**-19.65*****	**-4.43**	**-1.63**	**-0.46**	**Loses 5%**
$IVOL^{FF3}$	**-18.42*****	**-3.75**	**0.59**	**0.22**	**Loses 5%**
Low profitability	0.08	0.02	1.27	0.70	Never 5%
Unprofitable	**-4.86*****	**-3.04**	**-1.32**	**-1.50**	**Loses 5%**
Non-dividend payer	**-4.96*****	**-2.80**	**-1.39**	**-1.27**	**Loses 5%**
Low tangibility	-15.27***	-4.83	-8.97***	-3.11	Stays 5%
Sales growth	-3.21	-0.61	-2.49	-1.24	Never 5%
Illiquidity	-2.58	-0.72	1.09	0.62	Never 5%
Low turnover	4.46	1.23	-0.03	-0.01	Never 5%

Note. Coefficients are percentage-point effects on future directional-spread returns. HAC t-statistics use the Newey-West standard errors from the predictive regressions. "Baseline → Sibley" classifies theory-consistent 5 percent HAC significance when moving from $SENT^{\perp}$ to $SENT^{S}$. Coefficient stars mark HAC significance at 10%, 5%, and 1%. Bold rows identify cases where significance is lost under the Sibley-expanded index.

A.4.2 Empirical Consequences

The Covid-winsorized appendix check modifies only the Sibley-expanded robustness index. $SENT^{\perp}$ remains the unadjusted series because the Covid shock is treated as an economically real sentiment episode rather than as a data error. $SENT^{\perp}$ changes little after the Covid adjustment, while $SENT^{S}$ is materially affected by the crisis-period proxy adjustment.

Figure 8: Original and Covid-winsorized sentiment indices

Note. The Covid-winsorized series cap extreme March–April 2020 raw proxy observations before PCA construction. The adjustment is triggered by the March 2020 ESI_t and CCI_t observations. The baseline $SENT^{\perp}$ series remains close to the original series, while $SENT^{S}$ changes materially.

The Covid-winsorized Sibley-expanded index stays significant in 4 characteristic-horizon tests, gains significance in 11, and is never significant in 29 relative to the unadjusted Sibley-expanded index. Relative to $SENT^{\perp}$, 12 tests stay significant, 8 lose significance, 3 gain significance, and 21 are never significant. The adjustment makes the Sibley-expanded index more predictive in several rows, especially at longer horizons, but it does not fully reproduce the $SENT^{\perp}$ cross-sectional pattern.

Table 30: Covid-winsorized Sibley-expanded sentiment regressions by characteristic and horizon

Characteristic	$SENT^S$		$SENT^{S,C}$		Sibley → adj.	Baseline → adj.
	Coef.	HAC t	Coef.	HAC t		
1-month horizon						
Small firms	-0.79	-1.09	-1.29^{*}	-1.78	Never 5%	Never 5%
Young firms	-0.07	-0.17	-0.25	-0.48	Never 5%	Never 5%
Total risk	**-1.14**	**-1.64**	$\mathbf{-1.36^{*}}$	**-1.73**	**Never 5%**	**Loses 5%**
$IVOL^{FF3}$	**-0.62**	**-0.81**	$\mathbf{-1.50^{*}}$	**-1.71**	**Never 5%**	**Loses 5%**
Low profitability	0.24	0.44	0.43	1.05	Never 5%	Never 5%
Unprofitable	-0.15	-0.34	0.36	1.00	Never 5%	Never 5%
Non-dividend payer	**0.07**	**0.38**	**-0.10**	**-0.30**	**Never 5%**	**Loses 5%**
Low tangibility	-1.55^{**}	-2.50	-1.29^{***}	-2.95	Stays 5%	Stays 5%
Sales growth	-0.53	-0.97	-1.10^{*}	-1.79	Never 5%	Never 5%
Illiquidity	0.15	0.31	-0.09	-0.19	Never 5%	Never 5%
Low turnover	-1.08^{*}	-1.86	-0.60	-0.86	Never 5%	Never 5%
3-month horizon						
Small firms	0.01	0.01	-4.07^{**}	-2.36	Gains 5%	Gains 5%
Young firms	-0.46	-0.48	-0.36	-0.24	Never 5%	Never 5%
Total risk	-1.30	-0.96	-4.56^{***}	-2.76	Gains 5%	Stays 5%
$IVOL^{FF3}$	-0.63	-0.41	-4.56^{**}	-2.39	Gains 5%	Stays 5%
Low profitability	1.23	1.43	1.28	1.19	Never 5%	Never 5%
Unprofitable	0.48	1.22	0.32	0.34	Never 5%	Never 5%
Non-dividend payer	**0.30**	**0.74**	**-0.77**	**-0.90**	**Never 5%**	**Loses 5%**
Low tangibility	-3.11^{***}	-2.88	-4.45^{***}	-4.13	Stays 5%	Stays 5%
Sales growth	-0.90	-0.98	-2.72^{**}	-2.03	Gains 5%	Gains 5%
Illiquidity	0.76	1.12	-0.22	-0.18	Never 5%	Never 5%
Low turnover	-1.82	-1.45	0.22	0.11	Never 5%	Never 5%
6-month horizon						
Small firms	-3.03	-1.27	-10.07^{***}	-3.78	Gains 5%	Stays 5%
Young firms	-0.54	-0.39	-2.37	-0.93	Never 5%	Never 5%
Total risk	-3.29	-1.62	-7.40^{***}	-2.59	Gains 5%	Stays 5%
$IVOL^{FF3}$	-1.44	-0.72	-8.27^{***}	-2.60	Gains 5%	Stays 5%
Low profitability	1.62	0.97	2.12	1.00	Never 5%	Never 5%
Unprofitable	0.17	0.26	-0.59	-0.37	Never 5%	Never 5%
Non-dividend payer	**-0.20**	**-0.28**	**-1.70**	**-1.19**	**Never 5%**	**Loses 5%**
Low tangibility	-4.96^{***}	-3.10	-9.31^{***}	-5.58	Stays 5%	Stays 5%
Sales growth	-1.62	-1.11	-4.42^{**}	-2.04	Gains 5%	Gains 5%
Illiquidity	-0.16	-0.15	-2.08	-0.88	Never 5%	Never 5%
Low turnover	-2.67	-1.25	0.13	0.04	Never 5%	Never 5%
12-month horizon						
Small firms	-6.64	-1.46	-20.10^{***}	-4.71	Gains 5%	Stays 5%
Young firms	**-1.18**	**-0.52**	**-6.71**	**-1.55**	**Never 5%**	**Loses 5%**
Total risk	-1.63	-0.46	-15.30^{***}	-3.31	Gains 5%	Stays 5%
$IVOL^{FF3}$	0.59	0.22	-13.69^{**}	-2.53	Gains 5%	Stays 5%
Low profitability	1.27	0.70	3.74	1.22	Never 5%	Never 5%
Unprofitable	**-1.32**	**-1.50**	**-1.27**	**-0.63**	**Never 5%**	**Loses 5%**
Non-dividend payer	**-1.39**	**-1.27**	**-2.77**	**-1.16**	**Never 5%**	**Loses 5%**
Low tangibility	-8.97^{***}	-3.11	-16.49^{***}	-5.71	Stays 5%	Stays 5%
Sales growth	-2.49	-1.24	-7.34^{*}	-1.90	Never 5%	Never 5%
Illiquidity	1.09	0.62	-2.25	-0.53	Never 5%	Never 5%
Low turnover	-0.03	-0.01	5.43	1.26	Never 5%	Never 5%

Note. The table reports directional-spread predictive regressions. $SENT^S$ is the unadjusted Sibley-expanded index, while $SENT^{S,C}$ is the Covid-winsorized Sibley-expanded index. Coefficient stars mark HAC significance at 10%, 5%, and 1%. "Sibley → adj." compares $SENT^S$ with $SENT^{S,C}$. "Baseline → adj." compares $SENT^{\perp}$ with $SENT^{S,C}$. Bold rows identify cases where the Covid-winsorized Sibley index fails to retain a theory-consistent 5% result that was significant in at least one comparison specification.

A.5 Python Code for Empirical Analysis

The following code listings document the post-cleaning Python implementation used for the empirical analysis. The analysis-only notebook script begins from cleaned, analysis-ready inputs in outputs/final_sentiment_sweden and reproduces the sentiment regressions and robustness checks. The portfolio and characteristic modules document the decile assignment and portfolio-sorting code used before the prepared regression panels are created. The remaining helper scripts generate thesis-facing tables and figures from the final analytical outputs.

```
import sys
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from IPython.display import Image, Markdown, display


def find_repo_root(start: Path) -> Path:
    for candidate in [start, *start.parents]:
        if (candidate / "outputs" / "final_sentiment_sweden").exists():
            return candidate
    raise FileNotFoundError("Could not locate repository root with outputs/final_sentiment_sweden.")


REPO_ROOT = find_repo_root(Path.cwd().resolve())
PREPARED_DATA_DIR = REPO_ROOT / "outputs" / "final_sentiment_sweden"
ANALYSIS_OUTPUT_DIR = REPO_ROOT / "outputs" / "sentiment_sweden_analysis_only"
ANALYSIS_OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

SAVE_OUTPUTS = True
BOOTSTRAP_REPETITIONS = 1_000
BOOTSTRAP_SEED = 20260510
PREDICTIVE_HORIZONS = (1, 3, 6, 12)
MAIN_SENTIMENT_REGRESSOR = "SENT_ORTH_DIV_PREMIUM_lag"
CONTROL_COLUMNS = ["RMRF", "SMB", "HML", "UMD"]

SENTIMENT_SPECS = [
    {
        "sentiment_index_family": "dividend_premium_macro_adjusted_fixed_pca",
        "sentiment_index_label": "SENT_ORTH_DIV_PREMIUM",
        "sentiment_level_column": "SENT_ORTH_DIV_PREMIUM",
        "sentiment_regressor": "SENT_ORTH_DIV_PREMIUM_lag",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": True,
    },
    {
        "sentiment_index_family": "dividend_premium_sibley_macro_adjusted_fixed_pca",
        "sentiment_index_label": "SENT_ORTH_SIBLEY_DIV_PREMIUM",
        "sentiment_level_column": "SENT_ORTH_SIBLEY_DIV_PREMIUM",
        "sentiment_regressor": "SENT_ORTH_SIBLEY_DIV_PREMIUM_lag",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": False,
    },
    {
        "sentiment_index_family": "dividend_premium_unadjusted_fixed_pca",
        "sentiment_index_label": "SENT_DIV_PREMIUM",
        "sentiment_level_column": "SENT_DIV_PREMIUM",
        "sentiment_regressor": "SENT_DIV_PREMIUM_lag",
```

```
        "macro_orthogonalized_flag": False,
        "main_specification_flag": False,
    },
]
SENTIMENT_REGRESSORS = [spec["sentiment_regressor"] for spec in SENTIMENT_SPECS]
SENTIMENT_LEVEL_COLUMNS = [spec["sentiment_level_column"] for spec in SENTIMENT_SPECS]

pd.set_option("display.max_columns", 120)
pd.set_option("display.width", 180)

REPO_ROOT, PREPARED_DATA_DIR, ANALYSIS_OUTPUT_DIR


REQUIRED_INPUTS = {
    "sentiment_panel": "final_sentiment_regression_panel_monthly.csv",
    "directional_regression_panel": "final_predictive_regression_panel_directional_spreads.csv",
    "by_leg_regression_panel": "final_predictive_regression_panel_by_leg.csv",
    "sentiment_numeric_summary": "final_sentiment_numeric_summary.csv",
    "pca_lead_lag_selection": "final_sentiment_pca_lead_lag_selection.csv",
    "pca_final_coefficients": "final_sentiment_pca_stage3_weighted_coefficients.csv",
}

missing_inputs = [
    filename for filename in REQUIRED_INPUTS.values()
    if not (PREPARED_DATA_DIR / filename).exists()
]
if missing_inputs:
    raise FileNotFoundError(f"Missing analysis-ready inputs under {PREPARED_DATA_DIR}: {missing_inputs}")


def read_prepared_csv(filename: str, *, date_columns: list[str] | None = None) -> pd.DataFrame:
    frame = pd.read_csv(PREPARED_DATA_DIR / filename)
    for column in date_columns or []:
        if column in frame.columns:
            frame[column] = pd.to_datetime(frame[column], errors="coerce")
    return frame


sentiment_panel = read_prepared_csv("final_sentiment_regression_panel_monthly.csv", date_columns=["month_end_date"])
regression_panel_directional_spreads = read_prepared_csv(
    "final_predictive_regression_panel_directional_spreads.csv",
    date_columns=["month_end_date"],
)
regression_panel_by_leg = read_prepared_csv(
    "final_predictive_regression_panel_by_leg.csv",
    date_columns=["month_end_date"],
)

input_inventory = pd.DataFrame(
    [
        {
            "input": key,
            "file": filename,
            "rows": len(read_prepared_csv(filename)),
            "columns": len(read_prepared_csv(filename).columns),
            "path": str((PREPARED_DATA_DIR / filename).relative_to(REPO_ROOT)),
        }
        for key, filename in REQUIRED_INPUTS.items()
    ]
)
display(input_inventory)
```

```python
display(sentiment_panel.head())
display(regression_panel_directional_spreads.head())


import hashlib
from dataclasses import dataclass
from typing import Iterable

import numpy as np
import pandas as pd


DEFAULT_BOOTSTRAP_REPETITIONS = 1_000
DEFAULT_BOOTSTRAP_SEED = 123
DEFAULT_MIN_BOOTSTRAP_BLOCK_LENGTH = 6


@dataclass(frozen=True)
class RegressionData:
    terms: list[str]
    y: np.ndarray
    x: np.ndarray
    sample: pd.DataFrame


def automatic_newey_west_lags(n_obs: int) -> int:
    """Automatic Newey-West lag rule for monthly predictive regressions."""
    if n_obs <= 1:
        return 0
    return int(np.floor(4 * (n_obs / 100.0) ** (2 / 9)))


def default_bootstrap_block_length(
    n_obs: int,
    *,
    return_horizon_months: int = 1,
    min_block_length: int = DEFAULT_MIN_BOOTSTRAP_BLOCK_LENGTH,
) -> int:
    """Default moving-block bootstrap length, capped to the sample size."""
    if n_obs <= 0:
        return 0
    horizon = max(int(return_horizon_months), 1)
    block_length = max(int(min_block_length), horizon)
    return min(block_length, int(n_obs))


def newey_west_covariance(x: np.ndarray, residuals: np.ndarray, max_lags: int) -> np.ndarray:
    """HAC covariance matrix using Bartlett weights."""
    n_obs, n_params = x.shape
    max_lags = min(max(int(max_lags), 0), max(n_obs - 1, 0))
    x_resid = x * residuals[:, None]
    meat = x_resid.T @ x_resid
    for lag in range(1, max_lags + 1):
        weight = 1.0 - lag / (max_lags + 1.0)
        gamma = x_resid[lag:].T @ x_resid[:-lag]
        meat += weight * (gamma + gamma.T)
    xtx_inv = np.linalg.pinv(x.T @ x)
    finite_sample_correction = n_obs / max(n_obs - n_params, 1)
    return finite_sample_correction * xtx_inv @ meat @ xtx_inv


def prepare_regression_data(
```

```python
    frame: pd.DataFrame,
    y_col: str,
    x_cols: Iterable[str],
    *,
    date_column: str = "month_end_date",
) -> RegressionData:
    """Build aligned numeric arrays for a regression."""
    if date_column in frame.columns:
        frame = frame.sort_values(date_column)
    x_cols = list(x_cols)
    cols = [y_col, *x_cols]
    sample = frame[cols].apply(pd.to_numeric, errors="coerce").dropna()
    terms = ["intercept", *x_cols]
    if sample.empty:
        return RegressionData(
            terms=terms,
            y=np.array([], dtype=float),
            x=np.empty((0, len(terms)), dtype=float),
            sample=sample,
        )
    y = sample[y_col].to_numpy(dtype=float)
    x = sample[x_cols].to_numpy(dtype=float)
    x = np.column_stack([np.ones(len(sample)), x])
    return RegressionData(terms=terms, y=y, x=x, sample=sample)


def _fit_beta(y: np.ndarray, x: np.ndarray) -> np.ndarray:
    beta, *_ = np.linalg.lstsq(x, y, rcond=None)
    return beta


def _derived_seed(seed: int, terms: list[str], n_obs: int) -> int:
    payload = "|".join([str(seed), str(n_obs), *terms]).encode("utf-8")
    digest = hashlib.blake2b(payload, digest_size=8).digest()
    return int.from_bytes(digest, "little") % (2**32)


def circular_moving_block_indices(
    n_obs: int,
    *,
    block_length: int,
    repetitions: int,
    seed: int = DEFAULT_BOOTSTRAP_SEED,
) -> np.ndarray:
    """Generate circular moving-block bootstrap index arrays."""
    if n_obs <= 0:
        return np.empty((0, 0), dtype=int)
    if repetitions <= 0:
        return np.empty((0, n_obs), dtype=int)
    block_length = min(max(int(block_length), 1), int(n_obs))
    n_blocks = int(np.ceil(n_obs / block_length))
    rng = np.random.default_rng(seed)
    starts = rng.integers(0, n_obs, size=(int(repetitions), n_blocks))
    offsets = np.arange(block_length)
    indices = (starts[:, :, None] + offsets[None, None, :]) % n_obs
    return indices.reshape(int(repetitions), n_blocks * block_length)[:, :n_obs]


def moving_block_bootstrap_regression(
    y: np.ndarray,
    x: np.ndarray,
    beta_original: np.ndarray,
```

```python
    *,
    terms: list[str],
    block_length: int,
    repetitions: int = DEFAULT_BOOTSTRAP_REPETITIONS,
    seed: int = DEFAULT_BOOTSTRAP_SEED,
) -> pd.DataFrame:
    """Pairs moving-block bootstrap inference for an OLS regression."""
    n_obs, n_params = x.shape
    block_length = min(max(int(block_length), 1), n_obs) if n_obs else 0
    if n_obs <= n_params or repetitions <= 1 or block_length <= 0:
        return pd.DataFrame(
            {
                "term": terms,
                "bootstrap_std_error": np.nan,
                "bootstrap_t_stat": np.nan,
                "bootstrap_p_value": np.nan,
                "bootstrap_repetitions": int(repetitions),
                "bootstrap_block_length": block_length,
                "bootstrap_seed": int(seed),
                "bootstrap_valid_repetitions": 0,
            }
        )

    indices = circular_moving_block_indices(
        n_obs,
        block_length=block_length,
        repetitions=repetitions,
        seed=_derived_seed(seed, terms, n_obs),
    )
    betas = np.full((int(repetitions), n_params), np.nan, dtype=float)
    for rep_idx, sample_idx in enumerate(indices):
        x_boot = x[sample_idx]
        y_boot = y[sample_idx]
        if np.linalg.matrix_rank(x_boot) < n_params:
            continue
        betas[rep_idx] = _fit_beta(y_boot, x_boot)

    valid = np.isfinite(betas).all(axis=1)
    valid_betas = betas[valid]
    valid_repetitions = len(valid_betas)
    if valid_repetitions <= 1:
        bootstrap_std_error = np.full(n_params, np.nan)
        bootstrap_t_stat = np.full(n_params, np.nan)
        bootstrap_p_value = np.full(n_params, np.nan)
    else:
        bootstrap_std_error = np.nanstd(valid_betas, axis=0, ddof=1)
        bootstrap_t_stat = np.divide(
            beta_original,
            bootstrap_std_error,
            out=np.full(n_params, np.nan, dtype=float),
            where=bootstrap_std_error > 0,
        )
        less_equal_zero = np.mean(valid_betas <= 0.0, axis=0)
        greater_equal_zero = np.mean(valid_betas >= 0.0, axis=0)
        bootstrap_p_value = np.minimum(1.0, 2.0 * np.minimum(less_equal_zero, greater_equal_zero))

    return pd.DataFrame(
        {
            "term": terms,
            "bootstrap_std_error": bootstrap_std_error,
            "bootstrap_t_stat": bootstrap_t_stat,
            "bootstrap_p_value": bootstrap_p_value,
```

```
            "bootstrap_repetitions": int(repetitions),
            "bootstrap_block_length": block_length,
            "bootstrap_seed": int(seed),
            "bootstrap_valid_repetitions": int(valid_repetitions),
        }
    )


def run_ols_with_hac_and_bootstrap(
    frame: pd.DataFrame,
    y_col: str,
    x_cols: Iterable[str],
    *,
    hac_lags: int | None = None,
    min_hac_lags: int = 0,
    return_horizon_months: int = 1,
    bootstrap_repetitions: int = DEFAULT_BOOTSTRAP_REPETITIONS,
    bootstrap_seed: int = DEFAULT_BOOTSTRAP_SEED,
    bootstrap_block_length: int | None = None,
) -> pd.DataFrame:
    """Run OLS with HAC inference and moving-block bootstrap robustness inference."""
    regression_data = prepare_regression_data(frame, y_col, x_cols)
    terms = regression_data.terms
    y = regression_data.y
    x = regression_data.x
    n_obs = len(y)
    if n_obs <= len(terms):
        block_length = default_bootstrap_block_length(
            n_obs,
            return_horizon_months=return_horizon_months,
        )
        return pd.DataFrame(
            [
                {
                    "term": "insufficient_observations",
                    "coef": np.nan,
                    "std_error": np.nan,
                    "t_stat": np.nan,
                    "n_obs": n_obs,
                    "r2": np.nan,
                    "standard_error_type": "HAC_Newey_West",
                    "hac_lags": np.nan,
                    "bootstrap_std_error": np.nan,
                    "bootstrap_t_stat": np.nan,
                    "bootstrap_p_value": np.nan,
                    "bootstrap_repetitions": int(bootstrap_repetitions),
                    "bootstrap_block_length": block_length,
                    "bootstrap_seed": int(bootstrap_seed),
                    "bootstrap_valid_repetitions": 0,
                }
            ]
        )

    beta = _fit_beta(y, x)
    resid = y - x @ beta
    if hac_lags is None:
        hac_lags = max(automatic_newey_west_lags(n_obs), int(min_hac_lags))
    cov = newey_west_covariance(x, resid, hac_lags)
    se = np.sqrt(np.clip(np.diag(cov), 0.0, np.inf))
    t_stat = np.divide(beta, se, out=np.full_like(beta, np.nan), where=se > 0)
    ss_total = float(((y - y.mean()) @ (y - y.mean())))
    ss_resid = float(resid @ resid)
```

```python
    r2 = np.nan if ss_total == 0 else 1.0 - ss_resid / ss_total
    result = pd.DataFrame(
        {
            "term": terms,
            "coef": beta,
            "std_error": se,
            "t_stat": t_stat,
            "n_obs": n_obs,
            "r2": r2,
            "standard_error_type": "HAC_Newey_West",
            "hac_lags": int(hac_lags),
        }
    )
    block_length = (
        default_bootstrap_block_length(n_obs, return_horizon_months=return_horizon_months)
        if bootstrap_block_length is None
        else min(max(int(bootstrap_block_length), 1), n_obs)
    )
    bootstrap = moving_block_bootstrap_regression(
        y,
        x,
        beta,
        terms=terms,
        block_length=block_length,
        repetitions=bootstrap_repetitions,
        seed=bootstrap_seed,
    )
    return result.merge(bootstrap, on="term", how="left")


def save_table(frame: pd.DataFrame, name: str) -> pd.DataFrame:
    if SAVE_OUTPUTS:
        frame.to_csv(ANALYSIS_OUTPUT_DIR / f"{name}.csv", index=False)
    return frame


def save_figure(name: str) -> None:
    if SAVE_OUTPUTS:
        plt.savefig(ANALYSIS_OUTPUT_DIR / f"{name}.png", dpi=160, bbox_inches="tight")


def hac_only(frame: pd.DataFrame) -> pd.DataFrame:
    bootstrap_columns = [
        "bootstrap_std_error",
        "bootstrap_t_stat",
        "bootstrap_p_value",
        "bootstrap_repetitions",
        "bootstrap_block_length",
        "bootstrap_seed",
        "bootstrap_valid_repetitions",
    ]
    return frame.drop(columns=bootstrap_columns, errors="ignore") if isinstance(frame, pd.DataFrame) else frame


def concat_non_empty(frames: list[pd.DataFrame]) -> pd.DataFrame:
    non_empty = [frame for frame in frames if frame is not None and not frame.empty]
    return pd.concat(non_empty, ignore_index=True) if non_empty else pd.DataFrame()


def run_predictive_regression(
    frame: pd.DataFrame,
    *,
```

```
    y_col: str,
    x_cols: list[str],
    return_horizon_months: int,
) -> pd.DataFrame:
    return run_ols_with_hac_and_bootstrap(
        frame,
        y_col,
        x_cols,
        min_hac_lags=max(int(return_horizon_months) - 1, 0),
        return_horizon_months=int(return_horizon_months),
        bootstrap_repetitions=BOOTSTRAP_REPETITIONS,
        bootstrap_seed=BOOTSTRAP_SEED,
    )


def run_regression_suite(
    panel: pd.DataFrame,
    *,
    y_col: str,
    group_columns: list[str],
    result_set: str,
    sentiment_specs: list[dict[str, object]] | None = None,
) -> pd.DataFrame:
    if panel is None or panel.empty:
        return pd.DataFrame()
    specs = sentiment_specs or SENTIMENT_SPECS
    rows = []
    for spec in specs:
        sentiment_regressor = str(spec["sentiment_regressor"])
        if sentiment_regressor not in panel.columns:
            continue
        x_cols = [sentiment_regressor, *[column for column in CONTROL_COLUMNS if column in panel.columns]]
        for group_key, group in panel.groupby(group_columns, dropna=False, sort=True):
            if not isinstance(group_key, tuple):
                group_key = (group_key,)
            metadata = dict(zip(group_columns, group_key))
            horizon = int(metadata.get("return_horizon_months", 1))
            result = run_predictive_regression(group, y_col=y_col, x_cols=x_cols, return_horizon_months=horizon)
            for column, value in reversed(list(metadata.items())):
                result.insert(0, column, value)
            result.insert(0, "result_set", result_set)
            result.insert(1, "sentiment_index_family", spec["sentiment_index_family"])
            result.insert(2, "sentiment_index_label", spec["sentiment_index_label"])
            result.insert(3, "sentiment_regressor", sentiment_regressor)
            result.insert(4, "macro_orthogonalized_flag", bool(spec["macro_orthogonalized_flag"]))
            result.insert(5, "main_specification_flag", bool(spec["main_specification_flag"]))
            result.insert(6, "dependent_variable", y_col)
            result.insert(7, "model", f"{y_col}_on_{sentiment_regressor}_and_factors")
            rows.append(result)
    out = concat_non_empty(rows)
    return add_interpretation_columns(out)


def add_interpretation_columns(results: pd.DataFrame) -> pd.DataFrame:
    if results is None or results.empty:
        return pd.DataFrame(results)
    out = results.copy()
    sentiment_term = out["term"].eq(out["sentiment_regressor"])
    out["sentiment_term_flag"] = sentiment_term
    out["expected_coef_sign"] = np.nan
    if "return_leg" in out.columns:
        out.loc[sentiment_term & out["return_leg"].isin(["directional_spread", "long_leg"]), "expected_coef_sign"
```

```python
    ] = -1.0
    else:
        out.loc[sentiment_term, "expected_coef_sign"] = -1.0
    out["expected_signed_t_stat"] = out["expected_coef_sign"] * pd.to_numeric(out["t_stat"], errors="coerce")
    out["expected_signed_bootstrap_t_stat"] = out["expected_coef_sign"] * pd.to_numeric(
        out.get("bootstrap_t_stat"), errors="coerce"
    )
    out["theory_consistent_sign"] = out["expected_signed_t_stat"].gt(0)
    out["theory_consistent_hac_5pct"] = out["expected_signed_t_stat"].ge(1.96)
    out["theory_consistent_bootstrap_5pct"] = out["expected_signed_bootstrap_t_stat"].ge(1.96)
    out["mechanical_overlap_warning"] = out.get("sort_variable", pd.Series(pd.NA, index=out.index)).isin(["NON_D_PAYER", "BE_ME"])
    out["mechanical_overlap_note"] = np.where(
        out["mechanical_overlap_warning"],
        "Interpret cautiously: dividend-premium sentiment indices contain information related to dividend status and valuation.",
        "",
    )
    return out


def summarize_sentiment_terms(results: pd.DataFrame, group_columns: list[str]) -> pd.DataFrame:
    if results is None or results.empty:
        return pd.DataFrame()
    terms = results.loc[results["term"].eq(results["sentiment_regressor"])].copy()
    if terms.empty:
        return terms
    return (
        terms.groupby(group_columns, dropna=False)
        .agg(
            model_count=("term", "size"),
            mean_coef=("coef", "mean"),
            mean_hac_t=("t_stat", "mean"),
            mean_expected_signed_t=("expected_signed_t_stat", "mean"),
            mean_bootstrap_t=("bootstrap_t_stat", "mean"),
            mean_expected_signed_bootstrap_t=("expected_signed_bootstrap_t_stat", "mean"),
            hac_theory_consistent_5pct_share=("theory_consistent_hac_5pct", "mean"),
            bootstrap_theory_consistent_5pct_share=("theory_consistent_bootstrap_5pct", "mean"),
            mean_r2=("r2", "mean"),
            min_n_obs=("n_obs", "min"),
            median_n_obs=("n_obs", "median"),
        )
        .reset_index()
    )


def sentiment_comparison_summary(results: pd.DataFrame, group_columns: list[str]) -> pd.DataFrame:
    summary = summarize_sentiment_terms(
        results,
        [*group_columns, "sentiment_index_label", "sentiment_regressor", "macro_orthogonalized_flag", "main_specification_flag"],
    )
    if summary.empty:
        return summary
    return summary.sort_values([*group_columns, "main_specification_flag"], ascending=[*[True] * len(group_columns), False]).reset_index(drop=True)


def sentiment_term_rows(results: pd.DataFrame) -> pd.DataFrame:
    if results is None or results.empty:
        return pd.DataFrame()
    return results.loc[results["term"].eq(results["sentiment_regressor"])].copy()
```

```
PREPARED_TABLES = [
    ("Sentiment numeric summary", "final_sentiment_numeric_summary.csv"),
    ("PCA lead-lag selection", "final_sentiment_pca_lead_lag_selection.csv"),
    ("Final PCA coefficients", "final_sentiment_pca_stage3_weighted_coefficients.csv"),
    ("Sibley horizon comparison from full final run", "final_sibley_sentiment_comparison_by_horizon.csv"),
]

for title, filename in PREPARED_TABLES:
    display(Markdown(f"**{title}**  \\\\n`{(PREPARED_DATA_DIR / filename).relative_to(REPO_ROOT)}`"))
    display(pd.read_csv(PREPARED_DATA_DIR / filename).head(25))

PREPARED_FIGURES = [
    ("Raw versus macro-orthogonalized sentiment", "final_sentiment_raw_vs_orth_div_premium_over_time.png"),
    ("Baseline versus Sibley-expanded sentiment", "final_sentiment_orth_vs_sibley_black_yellow_over_time.png"),
]
for title, filename in PREPARED_FIGURES:
    path = PREPARED_DATA_DIR / filename
    display(Markdown(f"**{title}**  \\\\n`{path.relative_to(REPO_ROOT)}`"))
    if path.exists():
        display(Image(filename=str(path)))


directional_results = run_regression_suite(
    regression_panel_directional_spreads,
    y_col="portfolio_return_window",
    group_columns=["return_horizon_months", "return_leg", "sort_variable", "sentiment_prone_direction", "sentiment_prone_interpretation"],
    result_set="directional_spread",
)

by_leg_results = run_regression_suite(
    regression_panel_by_leg,
    y_col="portfolio_return_window",
    group_columns=["return_horizon_months", "return_leg", "sort_variable", "portfolio_code", "sentiment_prone_direction", "sentiment_prone_interpretation"],
    result_set="directional_by_leg",
)

save_table(directional_results, "final_predictive_results_directional_spreads_bootstrap")
save_table(hac_only(directional_results), "final_predictive_results_directional_spreads")
save_table(by_leg_results, "final_predictive_results_by_leg_bootstrap")
save_table(hac_only(by_leg_results), "final_predictive_results_by_leg")

summary_directional_by_horizon = sentiment_comparison_summary(directional_results, ["return_horizon_months"])
summary_directional_by_characteristic = sentiment_comparison_summary(directional_results, ["sort_variable"])
summary_by_leg = sentiment_comparison_summary(by_leg_results, ["return_horizon_months", "return_leg"])

raw_vs_orth_directional_comparison = (
    summary_directional_by_characteristic.pivot_table(
        index="sort_variable",
        columns="sentiment_index_label",
        values="mean_expected_signed_t",
        aggfunc="last",
    )
    .reset_index()
)
if {"SENT_ORTH_DIV_PREMIUM", "SENT_DIV_PREMIUM"}.issubset(raw_vs_orth_directional_comparison.columns):
    raw_vs_orth_directional_comparison["orth_minus_raw_mean_expected_signed_t"] = (
        raw_vs_orth_directional_comparison["SENT_ORTH_DIV_PREMIUM"]
        - raw_vs_orth_directional_comparison["SENT_DIV_PREMIUM"]
```

```
    )

SIBLEY_COMPARISON_LABELS = ["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH_SIBLEY_DIV_PREMIUM"]
sibley_summary_by_horizon = summary_directional_by_horizon.loc[
    summary_directional_by_horizon["sentiment_index_label"].isin(SIBLEY_COMPARISON_LABELS)
].copy()
sibley_summary_by_characteristic = summary_directional_by_characteristic.loc[
    summary_directional_by_characteristic["sentiment_index_label"].isin(SIBLEY_COMPARISON_LABELS)
].copy()

top_directional = (
    directional_results.loc[directional_results["term"].eq(directional_results["sentiment_regressor"])]
    .sort_values("expected_signed_t_stat", ascending=False)
    .head(40)
)

save_table(summary_directional_by_horizon, "final_summary_directional_by_horizon")
save_table(summary_directional_by_characteristic, "final_summary_directional_by_characteristic")
save_table(summary_by_leg, "final_summary_by_leg")
save_table(raw_vs_orth_directional_comparison, "final_raw_vs_orth_directional_comparison_by_characteristic")
save_table(sibley_summary_by_horizon, "final_sibley_sentiment_comparison_by_horizon")
save_table(sibley_summary_by_characteristic, "final_sibley_sentiment_comparison_by_characteristic")
save_table(top_directional, "final_top_directional_sentiment_tests")

display(summary_directional_by_horizon)
display(summary_directional_by_characteristic.sort_values(["sort_variable", "main_specification_flag"], ascending=[True, False]))
display(top_directional[["sentiment_index_label", "return_horizon_months", "sort_variable", "coef", "t_stat", "bootstrap_t_stat", "expected_signed_t_stat", "expected_signed_bootstrap_t_stat", "n_obs", "r2", "mechanical_overlap_warning"]])


if not summary_directional_by_characteristic.empty:
    pivot = summary_directional_by_characteristic.pivot_table(
        index="sort_variable",
        columns="sentiment_index_label",
        values="mean_expected_signed_t",
        aggfunc="last",
    ).sort_values("SENT_ORTH_DIV_PREMIUM", ascending=True)
    fig, ax = plt.subplots(figsize=(9, 5.5))
    plot_columns = [column for column in ["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH_SIBLEY_DIV_PREMIUM", "SENT_DIV_PREMIUM"] if column in pivot.columns]
    plot_frame = pivot[plot_columns].rename(
        columns={
            "SENT_ORTH_DIV_PREMIUM": "$SENT^{\\perp}_{t}$",
            "SENT_ORTH_SIBLEY_DIV_PREMIUM": "$SENT^{S}_{t}$",
            "SENT_DIV_PREMIUM": "$SENT_{t}$",
        }
    )
    plot_frame.plot.barh(ax=ax)
    ax.axvline(0, color="black", linewidth=0.8)
    ax.axvline(1.96, color="#D62728", linewidth=0.8, linestyle="--", label="1.96")
    ax.set_xlabel("Mean expected-signed HAC t-statistic")
    ax.set_ylabel("Characteristic")
    ax.grid(axis="x", alpha=0.25)
    plt.tight_layout()
    save_figure("final_raw_vs_orth_mean_expected_signed_t_by_characteristic")
    plt.show()

main_terms = directional_results.loc[
    directional_results["term"].eq(directional_results["sentiment_regressor"])
    & directional_results["sentiment_regressor"].eq(MAIN_SENTIMENT_REGRESSOR)
```

```
].copy()
if not main_terms.empty:
    heat = main_terms.pivot_table(
        index="sort_variable",
        columns="return_horizon_months",
        values="expected_signed_t_stat",
        aggfunc="mean",
    )
    fig, ax = plt.subplots(figsize=(7, 5))
    image = ax.imshow(heat.values, aspect="auto", cmap="RdBu_r", vmin=-4, vmax=4)
    ax.set_xticks(range(len(heat.columns)), labels=heat.columns)
    ax.set_yticks(range(len(heat.index)), labels=heat.index)
    ax.set_xlabel("Forward return horizon, months")
    for i in range(heat.shape[0]):
        for j in range(heat.shape[1]):
            value = heat.iloc[i, j]
            if pd.notna(value):
                ax.text(j, i, f"{value:.1f}", ha="center", va="center", fontsize=8)
    fig.colorbar(image, ax=ax, label="Expected-signed t-statistic")
    plt.tight_layout()
    save_figure("final_directional_expected_signed_t_heatmap")
    plt.show()


BASELINE_SENTIMENT_SPECS = [spec for spec in SENTIMENT_SPECS if spec["sentiment_regressor"] ==
    MAIN_SENTIMENT_REGRESSOR]
INNOVATION_REGRESSOR = "SENT_ORTH_DIV_PREMIUM_innovation_lag"
INNOVATION_SENTIMENT_SPECS = [
    {
        "sentiment_index_family": "dividend_premium_macro_adjusted_fixed_pca_innovation",
        "sentiment_index_label": "SENT_ORTH_DIV_PREMIUM_INNOVATION",
        "sentiment_level_column": "SENT_ORTH_DIV_PREMIUM_innovation",
        "sentiment_regressor": INNOVATION_REGRESSOR,
        "macro_orthogonalized_flag": True,
        "main_specification_flag": False,
    }
]


def summarize_nonoverlap_results(results: pd.DataFrame) -> tuple[pd.DataFrame, pd.DataFrame]:
    terms = sentiment_term_rows(results)
    if terms.empty:
        return pd.DataFrame(), pd.DataFrame()
    group_columns = [
        "return_horizon_months",
        "sort_variable",
        "sentiment_prone_direction",
        "sentiment_prone_interpretation",
    ]
    by_characteristic = (
        terms.groupby(group_columns, dropna=False)
        .agg(
            offset_count=("nonoverlap_offset", "nunique"),
            estimated_model_count=("coef", "count"),
            median_coef=("coef", "median"),
            median_hac_t=("t_stat", "median"),
            median_expected_signed_t=("expected_signed_t_stat", "median"),
            median_bootstrap_t=("bootstrap_t_stat", "median"),
            median_expected_signed_bootstrap_t=("expected_signed_bootstrap_t_stat", "median"),
            theory_consistent_sign_share=("theory_consistent_sign", "mean"),
            hac_theory_consistent_5pct_share=("theory_consistent_hac_5pct", "mean"),
            bootstrap_theory_consistent_5pct_share=("theory_consistent_bootstrap_5pct", "mean"),
```

```
            min_n_obs=("n_obs", "min"),
            median_n_obs=("n_obs", "median"),
            median_r2=("r2", "median"),
        )
        .reset_index()
        .sort_values(["return_horizon_months", "sort_variable"])
    )
    by_horizon = (
        terms.groupby(["return_horizon_months"], dropna=False)
        .agg(
            characteristic_count=("sort_variable", "nunique"),
            offset_model_count=("coef", "count"),
            median_coef=("coef", "median"),
            median_hac_t=("t_stat", "median"),
            median_expected_signed_t=("expected_signed_t_stat", "median"),
            median_bootstrap_t=("bootstrap_t_stat", "median"),
            median_expected_signed_bootstrap_t=("expected_signed_bootstrap_t_stat", "median"),
            theory_consistent_sign_share=("theory_consistent_sign", "mean"),
            hac_theory_consistent_5pct_share=("theory_consistent_hac_5pct", "mean"),
            bootstrap_theory_consistent_5pct_share=("theory_consistent_bootstrap_5pct", "mean"),
            min_n_obs=("n_obs", "min"),
            median_n_obs=("n_obs", "median"),
            median_r2=("r2", "median"),
        )
        .reset_index()
        .sort_values("return_horizon_months")
    )
    return by_characteristic, by_horizon


def build_nonoverlap_panel(panel: pd.DataFrame, horizons: tuple[int, ...]) -> pd.DataFrame:
    if panel is None or panel.empty:
        return pd.DataFrame()
    base = panel.copy()
    base["month_end_date"] = pd.to_datetime(base["month_end_date"], errors="coerce")
    month_index = pd.Series(
        range(base["month_end_date"].dropna().nunique()),
        index=sorted(base["month_end_date"].dropna().unique()),
    )
    base["nonoverlap_month_index"] = base["month_end_date"].map(month_index).astype("Int64")
    frames = []
    for horizon in horizons:
        horizon = int(horizon)
        horizon_base = base.loc[base["return_horizon_months"].eq(horizon)].copy()
        for offset in range(horizon):
            sample = horizon_base.loc[horizon_base["nonoverlap_month_index"].mod(horizon).eq(offset)].copy()
            if sample.empty:
                continue
            duplicate_keys = ["return_horizon_months", "sort_variable", "month_end_date"]
            if sample.duplicated(duplicate_keys).any():
                duplicates = sample.loc[sample.duplicated(duplicate_keys, keep=False), duplicate_keys].head()
                raise ValueError(f"Non-overlap sample has duplicated regression rows: {duplicates.to_dict('records')}")
            sample["nonoverlap_offset"] = offset
            sample["nonoverlap_offset_label"] = f"offset_{offset:02d}_of_{horizon:02d}"
            frames.append(sample)
    return concat_non_empty(frames)


nonoverlap_panel = build_nonoverlap_panel(regression_panel_directional_spreads, PREDICTIVE_HORIZONS)
nonoverlap_results = run_regression_suite(
    nonoverlap_panel,
```

```
    y_col="portfolio_return_window",
    group_columns=[
        "return_horizon_months",
        "nonoverlap_offset",
        "nonoverlap_offset_label",
        "return_leg",
        "sort_variable",
        "sentiment_prone_direction",
        "sentiment_prone_interpretation",
    ],
    result_set="nonoverlap_directional_spread",
    sentiment_specs=BASELINE_SENTIMENT_SPECS,
)
nonoverlap_summary_by_characteristic, nonoverlap_summary_by_horizon = summarize_nonoverlap_results(nonoverlap_results)

save_table(nonoverlap_results, "final_nonoverlap_horizon_results_bootstrap")
save_table(hac_only(nonoverlap_results), "final_nonoverlap_horizon_results")
save_table(nonoverlap_summary_by_characteristic, "final_nonoverlap_horizon_summary_by_characteristic")
save_table(nonoverlap_summary_by_horizon, "final_nonoverlap_horizon_summary_by_horizon")

display(nonoverlap_summary_by_horizon)


sentiment_innovation_monthly = sentiment_panel[
    ["country_code", "month_end_date", "SENT_ORTH_DIV_PREMIUM", "SENT_ORTH_DIV_PREMIUM_lag"]
].copy()
ar1_sample = sentiment_innovation_monthly.dropna(subset=["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH_DIV_PREMIUM_lag"]).copy()
y = ar1_sample["SENT_ORTH_DIV_PREMIUM"].to_numpy(dtype=float)
x_lag = ar1_sample["SENT_ORTH_DIV_PREMIUM_lag"].to_numpy(dtype=float)
x = np.column_stack([np.ones(len(ar1_sample)), x_lag])
beta, *_ = np.linalg.lstsq(x, y, rcond=None)
ar1_sample["SENT_ORTH_DIV_PREMIUM_fitted"] = x @ beta
ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"] = y - ar1_sample["SENT_ORTH_DIV_PREMIUM_fitted"]
ss_total = float(((y - y.mean()) @ (y - y.mean())))
ss_resid = float((ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].to_numpy(dtype=float) @ ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].to_numpy(dtype=float)))
ar1_r2 = np.nan if ss_total == 0 else 1.0 - ss_resid / ss_total

sentiment_innovation_monthly = sentiment_innovation_monthly.merge(
    ar1_sample[
        [
            "month_end_date",
            "SENT_ORTH_DIV_PREMIUM_fitted",
            "SENT_ORTH_DIV_PREMIUM_innovation",
        ]
    ],
    on="month_end_date",
    how="left",
)
sentiment_innovation_monthly = sentiment_innovation_monthly.sort_values("month_end_date").reset_index(drop=True)
sentiment_innovation_monthly["SENT_ORTH_DIV_PREMIUM_innovation_lag"] = sentiment_innovation_monthly[
    "SENT_ORTH_DIV_PREMIUM_innovation"
].shift(1)

innovation_diagnostics = pd.DataFrame(
    [
        {
            "sentiment_index_label": "SENT_ORTH_DIV_PREMIUM",
            "model": "SENT_t_on_intercept_and_SENT_lag",
            "intercept": beta[0],
```

```python
            "rho": beta[1],
            "r2": ar1_r2,
            "n_obs": len(ar1_sample),
            "innovation_mean": ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].mean(),
            "innovation_std": ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].std(),
            "innovation_min": ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].min(),
            "innovation_max": ar1_sample["SENT_ORTH_DIV_PREMIUM_innovation"].max(),
            "lagged_innovation_count": int(sentiment_innovation_monthly["SENT_ORTH_DIV_PREMIUM_innovation_lag"].
    notna().sum()),
        }
    ]
)

innovation_panel = regression_panel_directional_spreads.merge(
    sentiment_innovation_monthly[["month_end_date", "SENT_ORTH_DIV_PREMIUM_innovation_lag"]],
    on="month_end_date",
    how="left",
)
innovation_results = run_regression_suite(
    innovation_panel,
    y_col="portfolio_return_window",
    group_columns=[
        "return_horizon_months",
        "return_leg",
        "sort_variable",
        "sentiment_prone_direction",
        "sentiment_prone_interpretation",
    ],
    result_set="sentiment_innovation_directional_spread",
    sentiment_specs=INNOVATION_SENTIMENT_SPECS,
)
innovation_summary_by_characteristic = sentiment_comparison_summary(innovation_results, ["sort_variable"])
innovation_summary_by_horizon = sentiment_comparison_summary(innovation_results, ["return_horizon_months"])

save_table(sentiment_innovation_monthly, "final_sentiment_innovation_monthly")
save_table(innovation_diagnostics, "final_sentiment_innovation_ar1_diagnostics")
save_table(innovation_results, "final_sentiment_innovation_results_bootstrap")
save_table(hac_only(innovation_results), "final_sentiment_innovation_results")
save_table(innovation_summary_by_characteristic, "final_sentiment_innovation_summary_by_characteristic")
save_table(innovation_summary_by_horizon, "final_sentiment_innovation_summary_by_horizon")

display(innovation_diagnostics)
display(innovation_summary_by_horizon)


def persistence_rows_from_terms(terms: pd.DataFrame, method: str) -> pd.DataFrame:
    if terms is None or terms.empty:
        return pd.DataFrame()
    out = terms.copy()
    return pd.DataFrame(
        {
            "robustness_method": method,
            "return_horizon_months": out["return_horizon_months"],
            "sort_variable": out["sort_variable"],
            "sentiment_prone_direction": out["sentiment_prone_direction"],
            "sentiment_prone_interpretation": out["sentiment_prone_interpretation"],
            "coef": out["coef"],
            "expected_signed_hac_t_stat": out["expected_signed_t_stat"],
            "expected_signed_bootstrap_t_stat": out["expected_signed_bootstrap_t_stat"],
            "theory_consistent_sign_flag": out["theory_consistent_sign"],
            "theory_consistent_sign_share": out["theory_consistent_sign"].astype(float),
            "theory_consistent_hac_5pct_share": out["theory_consistent_hac_5pct"].astype(float),
```

```
            "theory_consistent_bootstrap_5pct_share": out["theory_consistent_bootstrap_5pct"].astype(float),
            "n_obs": out["n_obs"],
            "model_count": 1,
        }
    )


baseline_terms = sentiment_term_rows(directional_results)
baseline_terms = baseline_terms.loc[baseline_terms["sentiment_regressor"].eq(MAIN_SENTIMENT_REGRESSOR)].copy()
innovation_terms = sentiment_term_rows(innovation_results)

nonoverlap_persistence = pd.DataFrame()
if not nonoverlap_summary_by_characteristic.empty:
    nonoverlap_persistence = pd.DataFrame(
        {
            "robustness_method": "nonoverlapping_horizon_median_offset",
            "return_horizon_months": nonoverlap_summary_by_characteristic["return_horizon_months"],
            "sort_variable": nonoverlap_summary_by_characteristic["sort_variable"],
            "sentiment_prone_direction": nonoverlap_summary_by_characteristic["sentiment_prone_direction"],
            "sentiment_prone_interpretation": nonoverlap_summary_by_characteristic["sentiment_prone_interpretation"],
            "coef": nonoverlap_summary_by_characteristic["median_coef"],
            "expected_signed_hac_t_stat": nonoverlap_summary_by_characteristic["median_expected_signed_t"],
            "expected_signed_bootstrap_t_stat": nonoverlap_summary_by_characteristic["median_expected_signed_bootstrap_t"],
            "theory_consistent_sign_flag": nonoverlap_summary_by_characteristic["theory_consistent_sign_share"].ge(0.5),
            "theory_consistent_sign_share": nonoverlap_summary_by_characteristic["theory_consistent_sign_share"],
            "theory_consistent_hac_5pct_share": nonoverlap_summary_by_characteristic["hac_theory_consistent_5pct_share"],
            "theory_consistent_bootstrap_5pct_share": nonoverlap_summary_by_characteristic["bootstrap_theory_consistent_5pct_share"],
            "n_obs": nonoverlap_summary_by_characteristic["median_n_obs"],
            "model_count": nonoverlap_summary_by_characteristic["estimated_model_count"],
        }
    )

persistence_robustness_summary = concat_non_empty(
    [
        persistence_rows_from_terms(baseline_terms, "overlapping_baseline"),
        nonoverlap_persistence,
        persistence_rows_from_terms(innovation_terms, "sentiment_ar1_innovation"),
    ]
).sort_values(["return_horizon_months", "sort_variable", "robustness_method"])

save_table(persistence_robustness_summary, "final_persistence_robustness_summary")
display(persistence_robustness_summary.head(30))


def compare_saved_output(filename: str, key_columns: list[str], numeric_columns: list[str]) -> pd.DataFrame:
    original_path = PREPARED_DATA_DIR / filename
    recomputed_path = ANALYSIS_OUTPUT_DIR / filename
    if not original_path.exists() or not recomputed_path.exists():
        return pd.DataFrame(
            [{"file": filename, "status": "missing", "original_exists": original_path.exists(), "recomputed_exists": recomputed_path.exists()}]
        )
    original = pd.read_csv(original_path)
    recomputed = pd.read_csv(recomputed_path)
    merged = original.merge(recomputed, on=key_columns, how="outer", suffixes=("_original", "_recomputed"), indicator=True)
    rows = [
```

```
        {
            "file": filename,
            "status": "compared",
            "original_rows": len(original),
            "recomputed_rows": len(recomputed),
            "merged_rows": len(merged),
            "unmatched_rows": int(merged["_merge"].ne("both").sum()),
        }
    ]
    for column in numeric_columns:
        left = f"{column}_original"
        right = f"{column}_recomputed"
        if left in merged.columns and right in merged.columns:
            diff = pd.to_numeric(merged[left], errors="coerce") - pd.to_numeric(merged[right], errors="coerce")
            rows[0][f"{column}_max_abs_diff"] = diff.abs().max()
    return pd.DataFrame(rows)


comparison_checks = concat_non_empty(
    [
        compare_saved_output(
            "final_summary_directional_by_horizon.csv",
            ["return_horizon_months", "sentiment_index_label", "sentiment_regressor"],
            ["mean_coef", "mean_expected_signed_t", "mean_expected_signed_bootstrap_t"],
        ),
        compare_saved_output(
            "final_summary_directional_by_characteristic.csv",
            ["sort_variable", "sentiment_index_label", "sentiment_regressor"],
            ["mean_coef", "mean_expected_signed_t", "mean_expected_signed_bootstrap_t"],
        ),
        compare_saved_output(
            "final_persistence_robustness_summary.csv",
            ["robustness_method", "return_horizon_months", "sort_variable"],
            ["coef", "expected_signed_hac_t_stat", "expected_signed_bootstrap_t_stat"],
        ),
    ]
)
save_table(comparison_checks, "analysis_only_recomputed_vs_final_output_checks")
display(comparison_checks)


analysis_output_inventory = pd.DataFrame(
    [{"file": path.name, "bytes": path.stat().st_size} for path in sorted(ANALYSIS_OUTPUT_DIR.glob("*"))]
)
save_table(analysis_output_inventory, "analysis_only_output_inventory")
display(analysis_output_inventory)
```

Listing 1: Analysis-only notebook script for Swedish sentiment regressions and robustness checks

```
from __future__ import annotations

import numpy as np
import pandas as pd


def _infer_date_column(frame: pd.DataFrame) -> str:
    for candidate in ["month_end_date", "ym", "date"]:
        if candidate in frame.columns:
            return candidate
    raise KeyError("No supported date column found.")


```

```python
def _bucketize(series: pd.Series, n_portfolios: int) -> pd.Series:
    valid = series.dropna()
    if valid.nunique() < 2:
        return pd.Series(pd.NA, index=series.index, dtype="Int64")
    ranks = valid.rank(method="first")
    buckets = pd.qcut(ranks, n_portfolios, labels=False, duplicates="drop") + 1
    out = pd.Series(pd.NA, index=series.index, dtype="Int64")
    out.loc[valid.index] = buckets.astype("Int64")
    return out


def add_lagged_columns(
    frame: pd.DataFrame,
    columns: list[str],
    *,
    entity_column: str = "security_id",
    date_column: str | None = None,
    suffix: str = "_lag",
) -> pd.DataFrame:
    """Attach one-period lagged columns within each entity."""
    if frame is None or frame.empty:
        return pd.DataFrame(frame).copy()

    out = frame.copy()
    date_col = date_column or _infer_date_column(out)
    required = {entity_column, date_col, *columns}
    missing = sorted(required.difference(out.columns))
    if missing:
        raise KeyError(f"Missing required columns for lagging: {missing}")

    out["_original_order"] = range(len(out))
    ordered = out.sort_values([entity_column, date_col, "_original_order"]).copy()
    for column in columns:
        ordered[f"{column}{suffix}"] = ordered.groupby(entity_column, dropna=False)[column].shift(1)
    ordered = ordered.sort_values("_original_order").drop(columns=["_original_order"])
    return ordered.reset_index(drop=True)


def add_lagged_sort_signals(
    frame: pd.DataFrame,
    sort_columns: list[str],
    *,
    entity_column: str = "security_id",
    date_column: str | None = None,
    suffix: str = "_lag",
) -> pd.DataFrame:
    """Attach one-period lagged sort signals within each security."""
    return add_lagged_columns(
        frame,
        sort_columns,
        entity_column=entity_column,
        date_column=date_column,
        suffix=suffix,
    )


def build_lagged_sort_signal_coverage(
    frame: pd.DataFrame,
    sort_specs: list[tuple[str, str]],
    *,
    value_column: str = "ret",
    date_column: str | None = None,
```

```
) -> pd.DataFrame:
    """Summarize monthly eligibility for lagged-signal portfolio sorts."""
    columns = [
        "sort_variable",
        "month_end_date",
        "rows",
        "lagged_signal_non_null",
        "return_non_null",
        "eligible_rows",
        "formed_portfolios",
    ]
    if frame is None or frame.empty:
        return pd.DataFrame(columns=columns)

    out = frame.copy()
    date_col = date_column or _infer_date_column(out)
    rows: list[dict[str, object]] = []
    for sort_variable, lagged_column in sort_specs:
        if lagged_column not in out.columns or value_column not in out.columns:
            continue
        for month_end_date, group in out.groupby(date_col, sort=True):
            signal = pd.to_numeric(group[lagged_column], errors="coerce")
            returns = pd.to_numeric(group[value_column], errors="coerce")
            eligible = signal.notna() & returns.notna()
            rows.append(
                {
                    "sort_variable": sort_variable,
                    "month_end_date": month_end_date,
                    "rows": int(len(group)),
                    "lagged_signal_non_null": int(signal.notna().sum()),
                    "return_non_null": int(returns.notna().sum()),
                    "eligible_rows": int(eligible.sum()),
                    "formed_portfolios": int(_bucketize(signal.where(eligible), 10).nunique(dropna=True)),
                }
            )
    return pd.DataFrame(rows, columns=columns)


def build_forward_compounded_return_windows(
    frame: pd.DataFrame,
    *,
    group_columns: list[str],
    date_column: str,
    return_column: str,
    horizons: tuple[int, ...] = (1, 3, 6, 12),
    output_column: str = "portfolio_return_window",
    input_return_unit: str = "decimal",
    output_return_unit: str | None = None,
) -> pd.DataFrame:
    """Compute overlapping forward compounded returns within each group."""
    valid_units = {"decimal", "percent"}
    if input_return_unit not in valid_units:
        raise ValueError("input_return_unit must be either 'decimal' or 'percent'.")
    output_return_unit = input_return_unit if output_return_unit is None else output_return_unit
    if output_return_unit not in valid_units:
        raise ValueError("output_return_unit must be either 'decimal' or 'percent'.")

    columns = [*group_columns, date_column, "return_horizon_months", output_column]
    if frame is None or frame.empty:
        return pd.DataFrame(columns=columns)
    if any(horizon <= 0 for horizon in horizons):
        raise ValueError("All return horizons must be positive.")
```

```python

    required = {date_column, return_column, *group_columns}
    missing = sorted(required.difference(frame.columns))
    if missing:
        raise KeyError(f"Missing required columns for forward return windows: {missing}")

    out = frame.copy()
    out[date_column] = pd.to_datetime(out[date_column], errors="coerce")
    out[return_column] = pd.to_numeric(out[return_column], errors="coerce")
    out = out.dropna(subset=[date_column]).sort_values([*group_columns, date_column] if group_columns else [date_column])

    grouped = out.groupby(group_columns, dropna=False, sort=True) if group_columns else [((), out)]
    rows: list[dict[str, object]] = []
    for group_key, group in grouped:
        if not isinstance(group_key, tuple):
            group_key = (group_key,)
        group = group.sort_values(date_column).reset_index(drop=True)
        month_ordinals = group[date_column].dt.year.mul(12).add(group[date_column].dt.month).to_numpy()
        returns = group[return_column].to_numpy(dtype=float)
        group_values = dict(zip(group_columns, group_key))
        for row_pos, month_end_date in enumerate(group[date_column]):
            for horizon in horizons:
                value = np.nan
                end_pos = row_pos + horizon
                if end_pos <= len(group):
                    window_ordinals = month_ordinals[row_pos:end_pos]
                    expected_ordinals = np.arange(month_ordinals[row_pos], month_ordinals[row_pos] + horizon)
                    window_returns = returns[row_pos:end_pos]
                    if np.array_equal(window_ordinals, expected_ordinals) and np.isfinite(window_returns).all():
                        decimal_returns = window_returns / 100.0 if input_return_unit == "percent" else window_returns
                        compounded_decimal = float(np.prod(1.0 + decimal_returns) - 1.0)
                        value = compounded_decimal * 100.0 if output_return_unit == "percent" else compounded_decimal
                rows.append(
                    {
                        **group_values,
                        date_column: month_end_date,
                        "return_horizon_months": int(horizon),
                        output_column: value,
                    }
                )
    return pd.DataFrame(rows, columns=columns)


def build_equal_weight_portfolios(
    frame: pd.DataFrame,
    sort_column: str,
    value_column: str,
    date_column: str | None = None,
    n_portfolios: int = 10,
) -> pd.DataFrame:
    if frame is None or frame.empty:
        return pd.DataFrame(columns=["month_end_date", "portfolio_code", "ew_return"])

    out = frame.copy()
    date_col = date_column or _infer_date_column(out)
    out["portfolio_num"] = out.groupby(date_col)[sort_column].transform(lambda s: _bucketize(s, n_portfolios))
    out = out.dropna(subset=["portfolio_num", value_column])
    grouped = out.groupby([date_col, "portfolio_num"], as_index=False)[value_column].mean()
    grouped["portfolio_code"] = grouped["portfolio_num"].astype(int).map(lambda value: f"P{value:02d}")
```

```
    grouped = grouped.rename(columns={date_col: "month_end_date", value_column: "ew_return"})
    return grouped[["month_end_date", "portfolio_code", "ew_return"]]


def build_binary_flag_portfolios(
    frame: pd.DataFrame,
    sort_column: str,
    value_column: str,
    date_column: str | None = None,
    *,
    safe_code: str = "SAFE",
    prone_code: str = "PRONE",
) -> pd.DataFrame:
    if frame is None or frame.empty:
        return pd.DataFrame(columns=["month_end_date", "portfolio_code", "ew_return"])

    out = frame.copy()
    date_col = date_column or _infer_date_column(out)
    out["_binary_sort_value"] = pd.to_numeric(out[sort_column], errors="coerce")
    out = out.loc[out["_binary_sort_value"].isin([0.0, 1.0])].copy()
    out["portfolio_code"] = np.where(out["_binary_sort_value"].eq(1.0), prone_code, safe_code)
    out = out.dropna(subset=["portfolio_code", value_column])
    grouped = out.groupby([date_col, "portfolio_code"], as_index=False)[value_column].mean()
    grouped = grouped.rename(columns={date_col: "month_end_date", value_column: "ew_return"})
    return grouped[["month_end_date", "portfolio_code", "ew_return"]]


def build_long_short_spread(
    portfolios: pd.DataFrame,
    low_code: str = "P01",
    high_code: str = "P10",
) -> pd.DataFrame:
    if portfolios is None or portfolios.empty:
        return pd.DataFrame(columns=["month_end_date", "spread_code", "spread_return"])
    low = portfolios.loc[portfolios["portfolio_code"] == low_code, ["month_end_date", "ew_return"]].rename(
        columns={"ew_return": "low_return"}
    )
    high = portfolios.loc[portfolios["portfolio_code"] == high_code, ["month_end_date", "ew_return"]].rename(
        columns={"ew_return": "high_return"}
    )
    spread = high.merge(low, on="month_end_date", how="inner")
    spread["spread_code"] = f"{high_code}_{low_code}"
    spread["spread_return"] = spread["high_return"] - spread["low_return"]
    return spread[["month_end_date", "spread_code", "spread_return"]]


def build_spread_between_portfolios(
    portfolios: pd.DataFrame,
    *,
    long_code: str,
    short_code: str,
    spread_code: str = "LONG_SHORT",
) -> pd.DataFrame:
    if portfolios is None or portfolios.empty:
        return pd.DataFrame(columns=["month_end_date", "spread_code", "spread_return", "long_code", "short_code"
])
    long_leg = portfolios.loc[portfolios["portfolio_code"] == long_code, ["month_end_date", "ew_return"]].rename(
        columns={"ew_return": "long_return"}
    )
    short_leg = portfolios.loc[portfolios["portfolio_code"] == short_code, ["month_end_date", "ew_return"]].
rename(
        columns={"ew_return": "short_return"}
```

```
    )
    spread = long_leg.merge(short_leg, on="month_end_date", how="inner")
    spread["spread_code"] = spread_code
    spread["spread_return"] = spread["long_return"] - spread["short_return"]
    spread["long_code"] = long_code
    spread["short_code"] = short_code
    return spread[["month_end_date", "spread_code", "spread_return", "long_code", "short_code"]]
```

Listing 2: Portfolio sorting and directional-spread construction

```
from __future__ import annotations

import numpy as np
import pandas as pd

MARKET_CAP_FULL_TO_MILLIONS = 1_000_000.0

CHARACTERISTIC_COLUMNS = [
    "security_id",
    "company_id",
    "country_code",
    "month_end_date",
    "fiscal_period_end",
    "report_available_date",
    "effective_month_end",
    "reporting_periodicity",
    "ret",
    "ME",
    "risk",
    "IVOL_FF3",
    "BE",
    "book_equity_base_source",
    "book_equity_nonpositive_flag",
    "book_equity_zero_ceq_replaced_flag",
    "BE_ME_raw",
    "BE_ME",
    "PPE_A_raw",
    "PPE_A",
    "E_plus_BE_raw",
    "E_plus_BE",
    "D_PAYER",
    "D_PAYER_daily",
    "D_PAYER_fundamental",
    "D_PAYER_source",
    "dividend_payer_year",
    "regular_dividend_payer_fundamental",
    "regular_dividend_payer_formation_flag",
    "consecutive_nondividend_years",
    "NON_D_PAYER",
    "E_POSITIVE",
    "UNPROFITABLE",
    "GS_raw",
    "GS",
    "ILLIQ_raw",
    "ILLIQ",
    "XTURN_raw",
    "XTURN",
    "liquidity_dividend_status",
    "be_me_valid_flag",
    "be_me_invalid_reason",
    "ratio_invalid_reason",
    "age",
```

```python
    "ME10",
    "risk10",
    "IVOL_FF310",
    "BE_ME10",
    "GS10",
    "age10",
    "E_plus_BE10",
    "PPE_A10",
    "ILLIQ10",
    "XTURN10",
]


def winsorize_by_group(
    frame: pd.DataFrame,
    column: str,
    group_column: str,
    *,
    lower: float = 0.01,
    upper: float = 0.99,
) -> pd.Series:
    values = pd.to_numeric(frame[column], errors="coerce")

    def _clip_group(group: pd.Series) -> pd.Series:
        non_null = group.dropna()
        if non_null.empty:
            return group
        lower_bound = non_null.quantile(lower)
        upper_bound = non_null.quantile(upper)
        return group.clip(lower=lower_bound, upper=upper_bound)

    return values.groupby(frame[group_column]).transform(_clip_group)


def assign_deciles_with_reference_subset(
    frame: pd.DataFrame,
    column: str,
    group_column: str,
    reference_mask: pd.Series,
) -> pd.Series:
    values = pd.to_numeric(frame[column], errors="coerce")
    groups = frame[group_column]
    output = pd.Series(pd.NA, index=frame.index, dtype="Int64")

    for group_value in groups.dropna().unique():
        group_mask = groups.eq(group_value)
        reference_values = values.loc[group_mask & reference_mask.fillna(False) & values.notna()]
        if reference_values.empty:
            continue
        quantiles = reference_values.quantile(np.linspace(0.0, 1.0, 11)).to_numpy(dtype="float64")
        bins = np.unique(quantiles)
        target_values = values.loc[group_mask]

        if bins.size <= 1:
            output.loc[group_mask & target_values.notna()] = 1
            continue

        labels = list(range(1, bins.size))
        cut = pd.cut(
            target_values,
            bins=bins,
            labels=labels,
```

```
            include_lowest=True,
            duplicates="drop",
        )
        output.loc[group_mask & target_values.notna()] = cut.astype("Int64")

    return output


def _empty_characteristics() -> pd.DataFrame:
    return pd.DataFrame(columns=CHARACTERISTIC_COLUMNS)


def _to_numeric(series: pd.Series | None, index: pd.Index) -> pd.Series:
    if series is None:
        return pd.Series(np.nan, index=index, dtype="float64")
    return pd.to_numeric(series, errors="coerce")


def _safe_ratio(numerator: pd.Series | None, denominator: pd.Series | None, index: pd.Index, scale: float = 1.0)
     -> pd.Series:
    numer = _to_numeric(numerator, index=index)
    denom = _to_numeric(denominator, index=index)
    return pd.Series(
        np.where(denom.notna() & denom.ne(0), (numer / denom) * scale, np.nan),
        index=index,
        dtype="float64",
    )


def _positive_denominator_ratio(
    numerator: pd.Series | None,
    denominator: pd.Series | None,
    index: pd.Index,
    *,
    scale: float = 1.0,
    require_positive_numerator: bool = False,
) -> pd.Series:
    numer = _to_numeric(numerator, index=index)
    denom = _to_numeric(denominator, index=index)
    valid = denom.notna() & denom.gt(0) & numer.notna()
    if require_positive_numerator:
        valid &= numer.gt(0)
    return pd.Series(np.where(valid, (numer / denom) * scale, np.nan), index=index, dtype="float64")


def _join_reasons(reason_columns: pd.DataFrame) -> pd.Series:
    if reason_columns.empty:
        return pd.Series("ok", index=reason_columns.index, dtype="string")
    reasons: list[str] = []
    for _, row in reason_columns.iterrows():
        values = [str(value) for value in row.dropna().tolist() if str(value)]
        reasons.append(";".join(values) if values else "ok")
    return pd.Series(reasons, index=reason_columns.index, dtype="string")


def _market_cap_millions(frame: pd.DataFrame, index: pd.Index) -> pd.Series:
    market_cap_millions = _to_numeric(frame.get("month_end_market_cap_millions"), index)
    market_cap_raw = _to_numeric(frame.get("month_end_market_cap"), index)
    return market_cap_millions.where(market_cap_millions.notna(), market_cap_raw / MARKET_CAP_FULL_TO_MILLIONS)


def _availability_row(
```

```python
    frame: pd.DataFrame,
    *,
    column_name: str,
    source_columns: list[str],
    translation_rule: str,
) -> dict[str, object]:
    available_sources = [column for column in source_columns if column in frame.columns]
    non_null_rows = int(frame[available_sources].notna().any(axis=1).sum()) if available_sources else 0
    return {
        "column_name": column_name,
        "source_columns": ",".join(source_columns),
        "available": bool(available_sources),
        "non_null_source_rows": non_null_rows,
        "translation_rule": translation_rule,
    }


def build_sweden_availability_audit(fundamental_snapshots: pd.DataFrame) -> pd.DataFrame:
    frame = pd.DataFrame(fundamental_snapshots).copy()
    rules = [
        (
            "BE",
            ["seqq", "at", "ltq", "pstkq", "book_equity", "BE"],
            "Latest filing snapshot book equity: seqq, with at-minus-ltq fallback and preferred-stock adjustment.",
        ),
        (
            "BE_ME",
            ["seqq", "at", "ltq", "pstkq", "book_equity", "BE"],
            "Positive book equity in millions divided by positive month-end market cap in millions.",
        ),
        (
            "PPE_A",
            ["ppegt", "ppentq", "at"],
            "Net property, plant, and equipment over assets; raw ppentq is mapped to ppegt during staging.",
        ),
        (
            "E_plus_BE",
            ["earning_ttm", "earning", "ib", "book_equity", "BE"],
            "Positive trailing earnings over positive book equity.",
        ),
        (
            "GS",
            ["revenue_ttm", "lag_revenue_ttm", "GS"],
            "Trailing annual revenue growth from revtq TTM versus the prior-year revtq TTM window.",
        ),
        (
            "D_PAYER",
            [
                "regular_dividend_payer_fundamental",
                "dividend_payer_year",
                "dvtq",
                "dvty",
                "dvy",
            ],
            "Backward-looking regular dividend-payer flag from fundamentals; daily cash-distribution history is retained only as fallback/diagnostic.",
        ),
        (
            "E_POSITIVE",
            ["earning_ttm", "earning", "ib"],
            "Indicator for positive trailing earnings from ib, with non-positive earnings classified as
```

```python
 unprofitable.",
        ),
    ]
    return pd.DataFrame(
        [_availability_row(frame, column_name=name, source_columns=sources, translation_rule=rule) for name, sources, rule in rules]
    )


def build_sweden_characteristics_layers(
    market_monthly: pd.DataFrame,
    fundamental_snapshots: pd.DataFrame,
) -> tuple[pd.DataFrame, pd.DataFrame, pd.DataFrame]:
    if market_monthly is None or market_monthly.empty:
        empty = _empty_characteristics()
        return empty, empty.copy(), build_sweden_availability_audit(pd.DataFrame())

    market = market_monthly.copy()
    market["month_end_date"] = pd.to_datetime(market["month_end_date"], errors="coerce")
    market = market.sort_values(["security_id", "month_end_date"])

    if fundamental_snapshots is None or fundamental_snapshots.empty:
        merged = market.copy()
    else:
        snapshots = fundamental_snapshots.copy()
        snapshots["month_end_date"] = pd.to_datetime(snapshots["month_end_date"], errors="coerce")
        merged = market.merge(
            snapshots,
            on=["company_id", "month_end_date"],
            how="left",
            suffixes=("", "_fund"),
        )

    idx = merged.index
    merged["country_code"] = merged.get("country_code", "SWE")
    merged["ret"] = _to_numeric(merged.get("monthly_total_return"), idx) * 100.0
    merged["ME"] = _market_cap_millions(merged, idx)
    merged["risk"] = (
        merged.groupby("security_id")["ret"].rolling(window=12, min_periods=9).std().reset_index(level=0, drop=True)
    )

    first_month = merged.groupby("security_id")["month_end_date"].transform("min")
    months_since_listing = (
        (merged["month_end_date"].dt.year - first_month.dt.year) * 12
        + (merged["month_end_date"].dt.month - first_month.dt.month)
    )
    merged["age"] = months_since_listing / 12.0

    merged["BE"] = _to_numeric(merged.get("BE", merged.get("book_equity")), idx)
    me = _to_numeric(merged.get("ME"), idx)
    at = _to_numeric(merged.get("at"), idx)
    revenue_ttm = _to_numeric(merged.get("revenue_ttm"), idx)
    lag_revenue_ttm = _to_numeric(merged.get("lag_revenue_ttm"), idx)
    be_positive = merged["BE"].gt(0)
    me_positive = me.gt(0)
    at_positive = at.gt(0)
    valid_sales_growth_denominator = lag_revenue_ttm.gt(0) & revenue_ttm.ge(0)

    merged["BE_ME_raw"] = _safe_ratio(merged["BE"], me, idx)
    merged["BE_ME"] = _positive_denominator_ratio(merged["BE"], me, idx, require_positive_numerator=True)
    merged["be_me_valid_flag"] = merged["BE_ME"].notna()
```

```python
    be_me_reasons = pd.DataFrame(
        {
            "missing_be": np.where(merged["BE"].isna(), "missing_be", pd.NA),
            "nonpositive_be": np.where(merged["BE"].notna() & ~be_positive, "nonpositive_be", pd.NA),
            "missing_me": np.where(me.isna(), "missing_me", pd.NA),
            "nonpositive_me": np.where(me.notna() & ~me_positive, "nonpositive_me", pd.NA),
        },
        index=idx,
    )
    merged["be_me_invalid_reason"] = _join_reasons(be_me_reasons).where(~merged["be_me_valid_flag"], "ok")

    merged["PPE_A_raw"] = _safe_ratio(merged.get("ppegt"), at, idx, scale=100.0)
    merged["PPE_A"] = merged["PPE_A_raw"].where(at_positive)

    earning_ttm = _to_numeric(merged.get("earning_ttm"), idx)
    earning_positive = earning_ttm.clip(lower=0.0)
    merged["E_plus_BE_raw"] = _safe_ratio(earning_positive, merged["BE"], idx, scale=100.0)
    merged["E_plus_BE"] = merged["E_plus_BE_raw"].where(be_positive)

    regular_dividend_payer = _to_numeric(merged.get("regular_dividend_payer_fundamental"), idx)
    merged["D_PAYER_fundamental"] = regular_dividend_payer
    merged["D_PAYER"] = merged["D_PAYER_fundamental"]
    merged["D_PAYER_source"] = np.where(merged["D_PAYER"].notna(), "fundamentals", pd.NA)

    merged["GS_raw"] = _to_numeric(merged.get("GS"), idx)
    merged["GS"] = merged["GS_raw"].where(valid_sales_growth_denominator)
    merged["E_POSITIVE"] = np.where(earning_ttm.notna(), earning_ttm.gt(0).astype(float), np.nan)
    merged["UNPROFITABLE"] = np.where(earning_ttm.notna(), earning_ttm.le(0).astype(float), np.nan)
    merged["NON_D_PAYER"] = np.where(merged["D_PAYER"].notna(), 1.0 - merged["D_PAYER"], np.nan)
    ratio_reasons = pd.DataFrame(
        {
            "be_me_invalid": np.where(merged["BE_ME_raw"].notna() & merged["BE_ME"].isna(), "be_me_invalid", pd.
     NA),
            "book_equity_denominator_invalid": np.where(
                merged["E_plus_BE_raw"].notna() & merged["E_plus_BE"].isna(),
                "book_equity_denominator_invalid",
                pd.NA,
            ),
            "asset_denominator_invalid": np.where(
                merged["PPE_A_raw"].notna() & merged["PPE_A"].isna(),
                "asset_denominator_invalid",
                pd.NA,
            ),
            "sales_growth_denominator_invalid": np.where(
                merged["GS_raw"].notna() & merged["GS"].isna(),
                "sales_growth_denominator_invalid",
                pd.NA,
            ),
        },
        index=idx,
    )
    merged["ratio_invalid_reason"] = _join_reasons(ratio_reasons)
    merged["fyear"] = merged["month_end_date"].dt.year

    for column in ["ME", "risk", "E_plus_BE", "PPE_A", "BE_ME", "GS"]:
        merged[column] = winsorize_by_group(merged, column, "month_end_date")
    merged["age"] = winsorize_by_group(merged, "age", "fyear")

    reference_mask = merged["security_id"].notna()
    for column in ["ME", "risk", "BE_ME", "GS"]:
        merged[f"{column}10"] = assign_deciles_with_reference_subset(merged, column, "month_end_date",
     reference_mask)
```

```python
    merged["age10"] = assign_deciles_with_reference_subset(merged, "age", "fyear", reference_mask)
    for column in ["E_plus_BE", "PPE_A"]:
        merged[f"{column}10"] = assign_deciles_with_reference_subset(merged, column, "month_end_date", reference_mask)

    merged.loc[earning_ttm.le(0) | earning_ttm.isna(), "E_plus_BE10"] = 0
    merged.loc[_to_numeric(merged.get("ppegt"), idx).fillna(0.0).eq(0), "PPE_A10"] = 0

    merged["me"] = merged["ME"]
    merged["be_me"] = merged["BE_ME"]
    merged["gs"] = merged["GS"]
    merged["me10"] = merged["ME10"]
    merged["be_me10"] = merged["BE_ME10"]
    merged["gs10"] = merged["GS10"]

    ordered = [column for column in CHARACTERISTIC_COLUMNS if column in merged.columns]
    passthrough = [column for column in merged.columns if column not in ordered]
    monthly = (
        merged.sort_values(["security_id", "month_end_date", "fiscal_period_end"])
        .drop_duplicates(["security_id", "month_end_date"], keep="last")[ordered + passthrough]
        .reset_index(drop=True)
    )

    december = monthly.loc[monthly["month_end_date"].dt.month == 12].copy()
    if december.empty:
        december = monthly.sort_values("month_end_date").groupby(
            ["security_id", monthly["month_end_date"].dt.year], as_index=False
        ).tail(1)
    december["year_num"] = december["month_end_date"].dt.year

    return monthly, december.reset_index(drop=True), build_sweden_availability_audit(fundamental_snapshots)


def attach_liquidity_dividend_to_characteristics(
    characteristics_monthly: pd.DataFrame,
    liquidity_dividend_monthly: pd.DataFrame,
) -> tuple[pd.DataFrame, pd.DataFrame]:
    monthly = pd.DataFrame(characteristics_monthly).copy()
    if monthly.empty:
        return monthly, monthly.copy()

    if "D_PAYER" not in monthly.columns:
        monthly["D_PAYER"] = np.nan
    if "D_PAYER_fundamental" not in monthly.columns:
        monthly["D_PAYER_fundamental"] = pd.to_numeric(monthly["D_PAYER"], errors="coerce")
    if "D_PAYER_source" not in monthly.columns:
        monthly["D_PAYER_source"] = np.where(monthly["D_PAYER"].notna(), "fundamentals", pd.NA)

    liquidity = pd.DataFrame(liquidity_dividend_monthly).copy()
    if not liquidity.empty:
        liquidity["month_end_date"] = pd.to_datetime(liquidity["month_end_date"], errors="coerce")
        merge_columns = [
            column
            for column in [
                "security_id",
                "month_end_date",
                "ILLIQ_raw",
                "ILLIQ",
                "XTURN_raw",
                "XTURN",
                "D_PAYER_daily",
                "daily_dividend_observed_flag",
```

```
                "liquidity_dividend_status",
                "n_daily_obs",
                "n_illiq_obs",
                "monthly_trading_volume",
                "month_end_shares_outstanding_liquidity",
                "monthly_traded_value_millions",
            ]
            if column in liquidity.columns
        ]
        monthly = monthly.merge(liquidity[merge_columns], on=["security_id", "month_end_date"], how="left")
    else:
        for column in [
            "ILLIQ_raw",
            "ILLIQ",
            "XTURN_raw",
            "XTURN",
            "D_PAYER_daily",
            "daily_dividend_observed_flag",
            "liquidity_dividend_status",
            "n_daily_obs",
            "n_illiq_obs",
        ]:
            if column not in monthly.columns:
                monthly[column] = np.nan

    daily_d_payer = pd.to_numeric(monthly.get("D_PAYER_daily"), errors="coerce")
    fundamental_d_payer = pd.to_numeric(monthly.get("D_PAYER_fundamental"), errors="coerce")
    monthly["D_PAYER"] = fundamental_d_payer.where(fundamental_d_payer.notna(), daily_d_payer)
    monthly["D_PAYER_source"] = np.select(
        [fundamental_d_payer.notna(), fundamental_d_payer.isna() & daily_d_payer.notna()],
        ["fundamentals", "daily"],
        default=pd.NA,
    )
    monthly["NON_D_PAYER"] = np.where(monthly["D_PAYER"].notna(), 1.0 - monthly["D_PAYER"], np.nan)
    for column in ["ILLIQ", "XTURN"]:
        if column in monthly.columns:
            monthly[column] = winsorize_by_group(monthly, column, "month_end_date")

    reference_mask = monthly["security_id"].notna()
    for column in ["ILLIQ", "XTURN"]:
        if column in monthly.columns:
            monthly[f"{column}10"] = assign_deciles_with_reference_subset(
                monthly,
                column,
                "month_end_date",
                reference_mask,
            )

    ordered = [column for column in CHARACTERISTIC_COLUMNS if column in monthly.columns]
    passthrough = [column for column in monthly.columns if column not in ordered]
    monthly = monthly[ordered + passthrough].sort_values(["security_id", "month_end_date"]).reset_index(drop=True
     )

    annual = monthly.loc[monthly["month_end_date"].dt.month == 12].copy()
    if annual.empty:
        annual = monthly.sort_values("month_end_date").groupby(
            ["security_id", monthly["month_end_date"].dt.year],
            as_index=False,
        ).tail(1)
    annual["year_num"] = annual["month_end_date"].dt.year
    return monthly, annual.reset_index(drop=True)

```

```python

def attach_ivol_to_characteristics(
    characteristics_monthly: pd.DataFrame,
    ivol_monthly: pd.DataFrame,
) -> tuple[pd.DataFrame, pd.DataFrame]:
    monthly = pd.DataFrame(characteristics_monthly).copy()
    if monthly.empty:
        return monthly, monthly.copy()

    ivol = pd.DataFrame(ivol_monthly).copy()
    if ivol.empty:
        if "year_num" in monthly.columns:
            annual = monthly.loc[monthly["month_end_date"].dt.month == 12].copy()
            return monthly, annual.reset_index(drop=True)
        annual = monthly.loc[monthly["month_end_date"].dt.month == 12].copy()
        return monthly, annual.reset_index(drop=True)

    ivol["month_end_date"] = pd.to_datetime(ivol["month_end_date"], errors="coerce")
    monthly = monthly.merge(
        ivol[["security_id", "month_end_date", "ivol_ff3"]],
        on=["security_id", "month_end_date"],
        how="left",
    )
    monthly = monthly.rename(columns={"ivol_ff3": "IVOL_FF3"})
    monthly["IVOL_FF3"] = winsorize_by_group(monthly, "IVOL_FF3", "month_end_date")

    reference_mask = monthly["security_id"].notna()
    monthly["IVOL_FF310"] = assign_deciles_with_reference_subset(monthly, "IVOL_FF3", "month_end_date",
     reference_mask)

    ordered = [column for column in CHARACTERISTIC_COLUMNS if column in monthly.columns]
    passthrough = [column for column in monthly.columns if column not in ordered]
    monthly = monthly[ordered + passthrough].sort_values(["security_id", "month_end_date"]).reset_index(drop=True
     )

    annual = monthly.loc[monthly["month_end_date"].dt.month == 12].copy()
    if annual.empty:
        annual = monthly.sort_values("month_end_date").groupby(
            ["security_id", monthly["month_end_date"].dt.year], as_index=False
        ).tail(1)
    annual["year_num"] = annual["month_end_date"].dt.year
    return monthly, annual.reset_index(drop=True)


def build_characteristics_monthly(
    market_monthly: pd.DataFrame,
    fundamental_snapshots: pd.DataFrame,
) -> pd.DataFrame:
    monthly, _, _ = build_sweden_characteristics_layers(market_monthly, fundamental_snapshots)
    return monthly
```

Listing 3: Firm-characteristic construction and decile assignment

```python
from __future__ import annotations

from pathlib import Path

import numpy as np
import pandas as pd
import statsmodels.api as sm


```

```
OUTPUT_DIR = Path("outputs/final_sentiment_sweden")
PANEL_PATH = OUTPUT_DIR / "final_predictive_regression_panel_directional_spreads.csv"
FULL_OUT = OUTPUT_DIR / "final_factor_control_robustness.csv"
SUMMARY_OUT = OUTPUT_DIR / "final_factor_control_robustness_summary_by_horizon.csv"
SELECTED_OUT = OUTPUT_DIR / "final_factor_control_robustness_selected_12m.csv"

SENTIMENT_REGRESSOR = "SENT_ORTH_DIV_PREMIUM_lag"
DEPENDENT_VARIABLE = "portfolio_return_window"
FACTOR_TERMS = ["RMRF", "SMB", "HML", "UMD"]

SELECTED_SORTS = [
    "risk",
    "IVOL_MKT",
    "IVOL_FF3",
    "PPE_A",
    "NON_D_PAYER",
    "ME",
    "UNPROFITABLE",
]

SORT_LABELS = {
    "risk": "Total risk",
    "IVOL_MKT": "IVOL (market)",
    "IVOL_FF3": "IVOL (FF3)",
    "PPE_A": "Low tangibility",
    "NON_D_PAYER": "Non-dividend payer",
    "ME": "Small firms",
    "UNPROFITABLE": "Unprofitable",
    "age": "Young firms",
    "ILLIQ": "Illiquidity",
    "E_plus_BE": "Low profitability",
    "XTURN": "Low turnover",
}


def hac_lags(horizon: int) -> int:
    return max(4, int(horizon) - 1)


def fit_model(frame: pd.DataFrame, terms: list[str], horizon: int) -> dict[str, float]:
    x = sm.add_constant(frame[terms], has_constant="add")
    y = frame[DEPENDENT_VARIABLE]
    model = sm.OLS(y, x).fit(cov_type="HAC", cov_kwds={"maxlags": hac_lags(horizon)})
    return {
        "coef": float(model.params[SENTIMENT_REGRESSOR]),
        "std_error": float(model.bse[SENTIMENT_REGRESSOR]),
        "t_stat": float(model.tvalues[SENTIMENT_REGRESSOR]),
        "r2": float(model.rsquared),
    }


def main() -> None:
    panel = pd.read_csv(PANEL_PATH)
    rows: list[dict[str, object]] = []

    group_cols = [
        "sort_variable",
        "sentiment_prone_direction",
        "sentiment_prone_interpretation",
        "return_horizon_months",
    ]

```

```
    required_cols = [DEPENDENT_VARIABLE, SENTIMENT_REGRESSOR, *FACTOR_TERMS]
    for keys, group in panel.groupby(group_cols, dropna=False, sort=True):
        sort_variable, prone_direction, prone_interpretation, horizon = keys
        sample = group[required_cols].replace([np.inf, -np.inf], np.nan).dropna()
        if len(sample) < 30:
            continue

        sentiment_only = fit_model(sample, [SENTIMENT_REGRESSOR], int(horizon))
        factor_adjusted = fit_model(sample, [SENTIMENT_REGRESSOR, *FACTOR_TERMS], int(horizon))
        coef_change = factor_adjusted["coef"] - sentiment_only["coef"]

        rows.append(
            {
                "sort_variable": sort_variable,
                "characteristic": SORT_LABELS.get(str(sort_variable), str(sort_variable)),
                "sentiment_prone_direction": prone_direction,
                "sentiment_prone_interpretation": prone_interpretation,
                "return_horizon_months": int(horizon),
                "n_obs": int(len(sample)),
                "sentiment_only_coef": sentiment_only["coef"],
                "factor_adjusted_coef": factor_adjusted["coef"],
                "coef_change": coef_change,
                "sentiment_only_t": sentiment_only["t_stat"],
                "factor_adjusted_t": factor_adjusted["t_stat"],
                "sentiment_only_r2": sentiment_only["r2"],
                "factor_adjusted_r2": factor_adjusted["r2"],
                "delta_r2": factor_adjusted["r2"] - sentiment_only["r2"],
                "sentiment_sign_change": bool(
                    np.sign(sentiment_only["coef"]) != np.sign(factor_adjusted["coef"])
                ),
                "hac_lags": hac_lags(int(horizon)),
            }
        )

    full = pd.DataFrame(rows).sort_values(["return_horizon_months", "sort_variable"])
    full.to_csv(FULL_OUT, index=False)

    summary = (
        full.groupby("return_horizon_months", as_index=False)
        .agg(
            n_characteristics=("sort_variable", "count"),
            mean_sentiment_only_coef=("sentiment_only_coef", "mean"),
            mean_factor_adjusted_coef=("factor_adjusted_coef", "mean"),
            mean_coef_change=("coef_change", "mean"),
            mean_sentiment_only_r2=("sentiment_only_r2", "mean"),
            mean_factor_adjusted_r2=("factor_adjusted_r2", "mean"),
            mean_delta_r2=("delta_r2", "mean"),
            sign_change_share=("sentiment_sign_change", "mean"),
        )
        .sort_values("return_horizon_months")
    )
    summary.to_csv(SUMMARY_OUT, index=False)

    selected = full[
        full["return_horizon_months"].eq(12) & full["sort_variable"].isin(SELECTED_SORTS)
    ].copy()
    selected["selection_order"] = selected["sort_variable"].map(
        {key: i for i, key in enumerate(SELECTED_SORTS)}
    )
    selected = selected.sort_values("selection_order").drop(columns="selection_order")
    selected.to_csv(SELECTED_OUT, index=False)

```

```python
    print(f"Wrote {FULL_OUT} with {len(full)} rows")
    print(f"Wrote {SUMMARY_OUT} with {len(summary)} rows")
    print(f"Wrote {SELECTED_OUT} with {len(selected)} rows")


if __name__ == "__main__":
    main()
```

Listing 4: Factor-control robustness table construction

```python
#!/usr/bin/env python3
"""Build descriptive statistics for raw sentiment proxy inputs."""

from __future__ import annotations

import csv
from pathlib import Path

import pandas as pd


ROOT = Path(__file__).resolve().parents[1]
OUTPUT_DIR = ROOT / "outputs" / "final_sentiment_sweden"
SOURCE = OUTPUT_DIR / "final_sentiment_monthly.csv"
CSV_OUT = OUTPUT_DIR / "final_raw_sentiment_proxy_descriptive_statistics.csv"
TEX_OUT = OUTPUT_DIR / "final_raw_sentiment_proxy_descriptive_statistics_table.tex"
INVENTORY = OUTPUT_DIR / "final_output_inventory.csv"


VARIABLES = [
    ("ESI", "Economic sentiment index", "\\var{ESI_t}"),
    ("CCI", "Consumer confidence index", "\\var{CCI_t}"),
    ("TURN", "Turnover", "\\var{TURN_t}"),
    ("NIPO", "IPO number", "\\var{NIPO_t}"),
    ("RIPO", "IPO initial return", "\\var{RIPO_t}"),
    ("ED_RATIO", "Equity-to-debt ratio", "\\var{ED\\_RATIO_t}"),
    ("DIV_PREMIUM", "Dividend premium", "\\var{DIVP_t}"),
]


def fmt(value: float) -> str:
    if pd.isna(value):
        return ""
    return f"{float(value):.4f}"


def build_table() -> pd.DataFrame:
    df = pd.read_csv(SOURCE)
    rows = []
    for column, label, tex_label in VARIABLES:
        series = pd.to_numeric(df[column], errors="coerce").dropna()
        rows.append(
            {
                "variable": column,
                "label": label,
                "tex_label": tex_label,
                "mean": series.mean(),
                "std": series.std(ddof=1),
                "min": series.min(),
                "max": series.max(),
                "median": series.median(),
                "obs": int(series.count()),
```

```
            }
        )
    return pd.DataFrame(rows)


def make_latex(rows: pd.DataFrame) -> str:
    lines = [
        "\\begin{table}[H]",
        "\\centering",
        "\\caption{Descriptive statistics for raw sentiment proxy inputs}",
        "\\label{tab:appendix_raw_sentiment_proxy_descriptives}",
        "\\small",
        "\\begin{tabular}{lrrrrrr}",
        "\\toprule",
        "Variable & Mean & Std. dev. & Min. & Max. & Median & Obs. \\\\",
        "\\midrule",
    ]
    for row in rows.itertuples(index=False):
        label = f"{row.label} ({row.tex_label})"
        lines.append(
            " & ".join(
                [
                    label,
                    fmt(row.mean),
                    fmt(row.std),
                    fmt(row.min),
                    fmt(row.max),
                    fmt(row.median),
                    str(row.obs),
                ]
            )
            + " \\\\"
        )
    lines.extend(
        [
            "\\bottomrule",
            "\\end{tabular}",
            "\\tablenote{Statistics are computed from the unstandardized monthly proxy series before PCA and before macro residualization. \\var{RIPO_t} is observed only in months with observed IPO initial-return information. \\var{NIPO_t} equals zero in months without IPO activity. \\var{TURN_t} and \\var{DIV\\_PREMIUM_t} are logarithmic proxy constructions.}",
            "\\end{table}",
            "",
        ]
    )
    return "\n".join(lines)


def update_inventory() -> None:
    files = sorted(p for p in OUTPUT_DIR.iterdir() if p.is_file())
    with INVENTORY.open("w", newline="", encoding="utf-8") as handle:
        writer = csv.writer(handle)
        writer.writerow(["file", "bytes"])
        for path in files:
            writer.writerow([path.name, path.stat().st_size])


def main() -> None:
    rows = build_table()
    rows.to_csv(CSV_OUT, index=False)
    TEX_OUT.write_text(make_latex(rows), encoding="utf-8")
    update_inventory()
```

```python
    print(f"Wrote {CSV_OUT.relative_to(ROOT)}")
    print(f"Wrote {TEX_OUT.relative_to(ROOT)}")
    print(f"Rows: {len(rows)}")


if __name__ == "__main__":
    main()
```

Listing 5: Raw sentiment proxy descriptive statistics

```python
from __future__ import annotations

import os
import sys
from pathlib import Path

os.environ.setdefault("ARROW_USER_SIMD_LEVEL", "NONE")

import numpy as np
import pandas as pd


REPO_ROOT = Path(__file__).resolve().parents[1]
SRC_ROOT = REPO_ROOT / "nordic_sentiment" / "src"
if str(SRC_ROOT) not in sys.path:
    sys.path.insert(0, str(SRC_ROOT))

from nordic_sentiment import load_project_configs  # noqa: E402
from nordic_sentiment.io.compustat_sweden import (  # noqa: E402
    load_sweden_macro_control_cpi,
    load_sweden_macro_control_em,
    load_sweden_macro_control_ip,
    load_sweden_macro_control_ppi,
)
from nordic_sentiment.regressions import run_ols_with_hac_and_bootstrap  # noqa: E402
from nordic_sentiment.sentiment.sweden import (  # noqa: E402
    STV_SWEDEN_SENTIMENT_PROXIES,
    build_sweden_macro_controls_monthly,
    build_sweden_sentiment_index,
    build_sweden_sentiment_index_with_paper_lead_lag,
    build_sweden_sentiment_proxy_mart,
    build_sweden_stv_sentiment_index,
    default_sweden_paper_spec_audit,
)


FINAL_OUTPUT_DIR = REPO_ROOT / "outputs" / "final_sentiment_sweden"
OUTPUT_DIR = REPO_ROOT / "outputs" / "alternative_sentiment_index_robustness"
BOOTSTRAP_REPETITIONS = 1_000
BOOTSTRAP_SEED = 20260510
CONTROL_COLUMNS = ["RMRF", "SMB", "HML", "UMD"]

INDEX_SPECS = [
    {
        "sentiment_index_label": "SENT_ORTH_DIV_PREMIUM",
        "sentiment_regressor": "SENT_ORTH_DIV_PREMIUM_lag",
        "sentiment_index_family": "dividend_premium_macro_adjusted_fixed_pca",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": True,
    },
    {
        "sentiment_index_label": "SENT_ORTH",
```

```python
        "sentiment_regressor": "SENT_ORTH_lag",
        "sentiment_index_family": "six_proxy_macro_adjusted_fixed_pca",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": False,
    },
    {
        "sentiment_index_label": "SENT_ORTH_PAPER_LEAD_LAG",
        "sentiment_regressor": "SENT_ORTH_PAPER_LEAD_LAG_lag",
        "sentiment_index_family": "paper_lead_lag_macro_adjusted_fixed_pca",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": False,
    },
    {
        "sentiment_index_label": "STV_SWE",
        "sentiment_regressor": "STV_SWE_lag",
        "sentiment_index_family": "rolling_36_month_macro_adjusted_pc1",
        "macro_orthogonalized_flag": True,
        "main_specification_flag": False,
    },
]

LABELS = {
    "SENT_ORTH_DIV_PREMIUM": r"\var{SENT^{\perp}}",
    "SENT_ORTH": r"\var{SENT^{\perp,noDIV}}",
    "SENT_ORTH_PAPER_LEAD_LAG": r"\var{SENT^{\perp,PLL}}",
    "STV_SWE": r"\var{STV^{SWE}}",
}


def require_file(path: Path) -> Path:
    if not path.exists():
        raise FileNotFoundError(f"Required input is missing: {path}")
    return path


def save_table(frame: pd.DataFrame, name: str) -> pd.DataFrame:
    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
    frame.to_csv(OUTPUT_DIR / f"{name}.csv", index=False)
    return frame


def final_sentiment_proxy_mart() -> pd.DataFrame:
    final = pd.read_csv(require_file(FINAL_OUTPUT_DIR / "final_sentiment_monthly.csv"))
    final["month_end_date"] = pd.to_datetime(final["month_end_date"], errors="coerce")
    proxy_columns = [column for column in STV_SWEDEN_SENTIMENT_PROXIES if column in final.columns]
    rows = []
    for proxy in proxy_columns:
        part = final[["country_code", "month_end_date", proxy]].rename(columns={proxy: "raw_value"}).copy()
        part["proxy_code"] = proxy
        part["paper_proxy_name"] = proxy
        part["source_name"] = "final_sentiment_sweden_proxy_snapshot"
        part["build_status"] = np.where(proxy == "DIV_PREMIUM", "adapted", "exact")
        part["exact_replication_flag"] = proxy != "DIV_PREMIUM"
        rows.append(part)
    return build_sweden_sentiment_proxy_mart(pd.concat(rows, ignore_index=True))


def load_standard_macro_controls() -> pd.DataFrame:
    configs = load_project_configs()
    return build_sweden_macro_controls_monthly(
        cpi=load_sweden_macro_control_cpi(configs),
        ppi=load_sweden_macro_control_ppi(configs),
```

```python
        ip=load_sweden_macro_control_ip(configs),
        em=load_sweden_macro_control_em(configs),
    )


def build_alternative_indices() -> tuple[pd.DataFrame, dict[str, pd.DataFrame]]:
    proxy_mart = final_sentiment_proxy_mart()
    macro_controls = load_standard_macro_controls()
    paper_spec = default_sweden_paper_spec_audit()
    paper_spec.loc[paper_spec["required_for_final_index"].fillna(False), "build_status"] = "exact"

    no_div_monthly, _, no_div_diagnostics = build_sweden_sentiment_index(
        paper_spec,
        proxy_mart,
        macro_controls=macro_controls,
    )
    paper_monthly, _, paper_diagnostics = build_sweden_sentiment_index_with_paper_lead_lag(
        paper_spec,
        proxy_mart,
        macro_controls=macro_controls,
    )
    stv_monthly, _, stv_diagnostics = build_sweden_stv_sentiment_index(
        paper_spec,
        proxy_mart,
        macro_controls=macro_controls,
        window_months=36,
    )
    baseline = pd.read_csv(require_file(FINAL_OUTPUT_DIR / "final_sentiment_monthly.csv"))
    baseline["month_end_date"] = pd.to_datetime(baseline["month_end_date"], errors="coerce")

    monthly = baseline[["country_code", "month_end_date", "SENT_ORTH_DIV_PREMIUM"]].copy()
    monthly = monthly.merge(
        no_div_monthly[["country_code", "month_end_date", "SENT_ORTH"]],
        on=["country_code", "month_end_date"],
        how="left",
    )
    monthly = monthly.merge(
        paper_monthly[["country_code", "month_end_date", "SENT_ORTH_PAPER_LEAD_LAG"]],
        on=["country_code", "month_end_date"],
        how="left",
    )
    monthly = monthly.merge(
        stv_monthly[["country_code", "month_end_date", "STV_SWE"]],
        on=["country_code", "month_end_date"],
        how="left",
    )
    for column in ["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH", "SENT_ORTH_PAPER_LEAD_LAG", "STV_SWE"]:
        monthly[f"{column}_lag"] = monthly.groupby("country_code")[column].shift(1)

    diagnostics = {
        "no_div_stage3_eigenvalues": no_div_diagnostics.get("stage3_eigenvalues", pd.DataFrame()),
        "paper_lead_lag_selected_variables": paper_diagnostics.get("paper_lead_lag_selected_variables", pd.
    DataFrame()),
        "paper_lead_lag_stage3_eigenvalues": paper_diagnostics.get("stage3_eigenvalues", pd.DataFrame()),
        "stv_swe_rolling_eigenvalues": stv_diagnostics.get("stv_swe_rolling_eigenvalues", pd.DataFrame()),
        "stv_swe_rolling_loadings": stv_diagnostics.get("stv_swe_rolling_loadings", pd.DataFrame()),
    }
    return monthly, diagnostics


def add_interpretation_columns(results: pd.DataFrame) -> pd.DataFrame:
    if results.empty:
```

```
        return results
    out = results.copy()
    sentiment_term = out["term"].eq(out["sentiment_regressor"])
    out["sentiment_term_flag"] = sentiment_term
    out["expected_coef_sign"] = np.nan
    out.loc[sentiment_term, "expected_coef_sign"] = -1.0
    out["expected_signed_t_stat"] = out["expected_coef_sign"] * pd.to_numeric(out["t_stat"], errors="coerce")
    out["expected_signed_bootstrap_t_stat"] = out["expected_coef_sign"] * pd.to_numeric(
        out.get("bootstrap_t_stat"), errors="coerce"
    )
    out["theory_consistent_sign"] = out["expected_signed_t_stat"].gt(0)
    out["theory_consistent_hac_5pct"] = out["expected_signed_t_stat"].ge(1.96)
    out["theory_consistent_bootstrap_5pct"] = out["expected_signed_bootstrap_t_stat"].ge(1.96)
    return out


def run_regressions(panel: pd.DataFrame) -> pd.DataFrame:
    rows = []
    group_columns = [
        "return_horizon_months",
        "return_leg",
        "sort_variable",
        "sentiment_prone_direction",
        "sentiment_prone_interpretation",
    ]
    for spec in INDEX_SPECS:
        sentiment_regressor = str(spec["sentiment_regressor"])
        if sentiment_regressor not in panel.columns:
            continue
        x_cols = [sentiment_regressor, *[column for column in CONTROL_COLUMNS if column in panel.columns]]
        for group_key, group in panel.groupby(group_columns, dropna=False, sort=True):
            metadata = dict(zip(group_columns, group_key, strict=True))
            horizon = int(metadata["return_horizon_months"])
            result = run_ols_with_hac_and_bootstrap(
                group,
                "portfolio_return_window",
                x_cols,
                min_hac_lags=max(horizon - 1, 0),
                return_horizon_months=horizon,
                bootstrap_repetitions=BOOTSTRAP_REPETITIONS,
                bootstrap_seed=BOOTSTRAP_SEED,
            )
            for column, value in reversed(list(metadata.items())):
                result.insert(0, column, value)
            result.insert(0, "result_set", "alternative_sentiment_index_directional_spread")
            result.insert(1, "sentiment_index_family", spec["sentiment_index_family"])
            result.insert(2, "sentiment_index_label", spec["sentiment_index_label"])
            result.insert(3, "sentiment_regressor", sentiment_regressor)
            result.insert(4, "macro_orthogonalized_flag", bool(spec["macro_orthogonalized_flag"]))
            result.insert(5, "main_specification_flag", bool(spec["main_specification_flag"]))
            result.insert(6, "dependent_variable", "portfolio_return_window")
            result.insert(7, "model", f"portfolio_return_window_on_{sentiment_regressor}_and_factors")
            rows.append(result)
    combined = pd.concat(rows, ignore_index=True) if rows else pd.DataFrame()
    return add_interpretation_columns(combined)


def summarize_sentiment_terms(results: pd.DataFrame, group_columns: list[str]) -> pd.DataFrame:
    terms = results.loc[results["term"].eq(results["sentiment_regressor"])].copy()
    if terms.empty:
        return terms
    return (
```

```
        terms.groupby(group_columns, dropna=False)
        .agg(
            model_count=("term", "size"),
            mean_coef=("coef", "mean"),
            mean_hac_t=("t_stat", "mean"),
            mean_expected_signed_t=("expected_signed_t_stat", "mean"),
            mean_bootstrap_t=("bootstrap_t_stat", "mean"),
            mean_expected_signed_bootstrap_t=("expected_signed_bootstrap_t_stat", "mean"),
            hac_theory_consistent_5pct_share=("theory_consistent_hac_5pct", "mean"),
            bootstrap_theory_consistent_5pct_share=("theory_consistent_bootstrap_5pct", "mean"),
            mean_r2=("r2", "mean"),
            min_n_obs=("n_obs", "min"),
            median_n_obs=("n_obs", "median"),
        )
        .reset_index()
    )


def latex_escape(text: str) -> str:
    return (
        str(text)
        .replace("\\", "\\textbackslash{}")
        .replace("&", "\\&")
        .replace("%", "\\%")
        .replace("_", "\\_")
    )


def fmt(value: float, digits: int = 2) -> str:
    if pd.isna(value):
        return "--"
    return f"{float(value):.{digits}f}"


def fmt_pct(value: float) -> str:
    if pd.isna(value):
        return "--"
    return f"{100.0 * float(value):.0f}\\%"


def build_latex_summary_table(summary_by_horizon: pd.DataFrame, correlation: pd.DataFrame, stv_eigen: pd.DataFrame) -> str:
    columns = ["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH", "SENT_ORTH_PAPER_LEAD_LAG", "STV_SWE"]
    rows: list[tuple[str, dict[str, str]]] = []

    corr_lookup = correlation.set_index("variable") if not correlation.empty and "variable" in correlation.columns else pd.DataFrame()
    rows.append(
        (
            r"Correlation with \var{SENT^{\perp}}",
            {
                column: fmt(corr_lookup.loc[column, "SENT_ORTH_DIV_PREMIUM"], 3)
                if column in corr_lookup.index and "SENT_ORTH_DIV_PREMIUM" in corr_lookup.columns
                else "--"
                for column in columns
            },
        )
    )
    for horizon in [1, 3, 6, 12]:
        h = summary_by_horizon.loc[summary_by_horizon["return_horizon_months"].eq(horizon)]
        lookup = h.set_index("sentiment_index_label")
        rows.append(
```

```
            (
                f"Mean expected-signed HAC $t$, {horizon}-month horizon",
                {
                    column: fmt(lookup.loc[column, "mean_expected_signed_t"], 2) if column in lookup.index else "--"
                    for column in columns
                },
            )
        )
    h12 = summary_by_horizon.loc[summary_by_horizon["return_horizon_months"].eq(12)].set_index("sentiment_index_label")
    rows.append(
        (
            r"Theory-consistent 5\% HAC share, 12-month horizon",
            {
                column: fmt_pct(h12.loc[column, "hac_theory_consistent_5pct_share"]) if column in h12.index else "--"
                for column in columns
            },
        )
    )
    stv_pc1 = stv_eigen["explained_variance_ratio"].mean() if not stv_eigen.empty else np.nan
    rows.append(
        (
            "Average rolling PC1 variance share",
            {
                "SENT_ORTH_DIV_PREMIUM": "--",
                "SENT_ORTH": "--",
                "SENT_ORTH_PAPER_LEAD_LAG": "--",
                "STV_SWE": fmt_pct(stv_pc1),
            },
        )
    )

    lines = [
        r"\begin{table}[H]",
        r"\centering",
        r"\caption{Alternative sentiment-index robustness comparison}",
        r"\label{tab:appendix_alternative_sentiment_indices}",
        r"\small",
        r"\begin{tabularx}{\textwidth}{>{\raggedright\arraybackslash}Xrrrr}",
        r"\toprule",
        "Measure & " + " & ".join(LABELS[column] for column in columns) + r" \\",
        r"\midrule",
    ]
    for measure, values in rows:
        lines.append(measure + " & " + " & ".join(values[column] for column in columns) + r" \\")
    lines.extend(
        [
            r"\bottomrule",
            r"\end{tabularx}",
            r"\tablenote{All regressions use the current 11-characteristic directional-spread panel and the same factor controls as the main specification. Expected-signed \(t\)-statistics multiply the sentiment coefficient \(t\)-statistic by the predicted negative sign of the sentiment-prone spread. \var{STV^{SWE}} is a 36-month rolling, macro-residualized first-component PCA index adapted from the enhanced sentiment-index methodology.}",
            r"\end{table}",
        ]
    )
    return "\n".join(lines) + "\n"
```

```python
def main() -> None:
    OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
    monthly, diagnostics = build_alternative_indices()
    save_table(monthly, "alternative_sentiment_monthly")
    index_columns = ["SENT_ORTH_DIV_PREMIUM", "SENT_ORTH", "SENT_ORTH_PAPER_LEAD_LAG", "STV_SWE"]
    correlation = monthly[index_columns].corr().reset_index().rename(columns={"index": "variable"})
    save_table(correlation, "alternative_sentiment_correlation_matrix")
    for name, frame in diagnostics.items():
        save_table(frame, f"alternative_{name}")

    panel = pd.read_csv(require_file(FINAL_OUTPUT_DIR / "final_predictive_regression_panel_directional_spreads.csv"))
    panel["month_end_date"] = pd.to_datetime(panel["month_end_date"], errors="coerce")
    merge_columns = ["country_code", "month_end_date", *index_columns, *[f"{column}_lag" for column in index_columns]]
    panel = panel.drop(columns=[column for column in merge_columns if column not in {"country_code", "month_end_date"}], errors="ignore")
    panel = panel.merge(monthly[merge_columns], on=["country_code", "month_end_date"], how="left")

    sorts = set(panel["sort_variable"].dropna().unique())
    if "GS" not in sorts or "IVOL_MKT" in sorts:
        raise RuntimeError("Alternative comparison must use the current 11-characteristic panel with GS and without IVOL_MKT.")

    results = run_regressions(panel)
    save_table(results.drop(columns=[c for c in results.columns if c.startswith("_")], errors="ignore"), "alternative_predictive_results_directional_spreads_bootstrap")
    bootstrap_columns = [
        "bootstrap_std_error",
        "bootstrap_t_stat",
        "bootstrap_p_value",
        "bootstrap_repetitions",
        "bootstrap_block_length",
        "bootstrap_seed",
        "bootstrap_valid_repetitions",
    ]
    save_table(results.drop(columns=bootstrap_columns, errors="ignore"), "alternative_predictive_results_directional_spreads")

    summary_by_horizon = summarize_sentiment_terms(
        results,
        ["return_horizon_months", "sentiment_index_label", "sentiment_regressor", "main_specification_flag"],
    )
    summary_by_characteristic = summarize_sentiment_terms(
        results,
        ["sort_variable", "sentiment_index_label", "sentiment_regressor", "main_specification_flag"],
    )
    save_table(summary_by_horizon, "alternative_summary_by_horizon")
    save_table(summary_by_characteristic, "alternative_summary_by_characteristic")

    table = build_latex_summary_table(
        summary_by_horizon,
        correlation,
        diagnostics.get("stv_swe_rolling_eigenvalues", pd.DataFrame()),
    )
    (OUTPUT_DIR / "alternative_sentiment_appendix_table.tex").write_text(table)

    print(f"Wrote alternative sentiment-index appendix outputs to {OUTPUT_DIR}")


if __name__ == "__main__":
    main()
```

Listing 6: Alternative sentiment-index appendix table generation

References

Baker, M. and Stein, J. C. (2004). Market liquidity as a sentiment indicator. *Journal of Financial Markets*, 7(3):271–299.

Baker, M. and Wurgler, J. (2000). The equity share in new issues and aggregate stock returns. *The Journal of Finance*, 55(5):2219–2257.

Baker, M. and Wurgler, J. (2004). A catering theory of dividends. *The Journal of Finance*, 59(3):1125–1165.

Baker, M. and Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance*, 61(4):1645–1680.

Baker, M. and Wurgler, J. (2007). Investor sentiment in the stock market. *Journal of Economic Perspectives*, 21(2):129–151.

Baker, M., Wurgler, J., and Yuan, Y. (2012). Global, local, and contagious investor sentiment. *Journal of Financial Economics*, 104(2):272–287.

Barber, B. M. and Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21(2):785–818.

Benveniste, L. M., Ljungqvist, A., Wilhelm, W. J., and Yu, X. (2003). Evidence of information spillovers in the production of investment banking services. *The Journal of Finance*, 58(2):577–608.

Blume, L. and Easley, D. (2006). If you're so smart, why aren't you rich? belief selection in complete and incomplete markets. *Econometrica*, 74(4):929–966.

Board of Governors of the Federal Reserve System (2026). Financial accounts of the united states, z.1, fourth quarter 2025. Statistical release. Released 19 March 2026.

Brown, G. W. and Cliff, M. T. (2004). Investor sentiment and the near-term stock market. *Journal of Empirical Finance*, 11(1):1–27.

Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82.

De Long, J. B., Shleifer, A., Summers, L. H., and Waldmann, R. J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4):703–738.

Ding, W., Mazouz, K., and Wang, Q. (2019). Investor sentiment and the cross-section of stock returns: New theory and evidence. *Review of Quantitative Finance and Accounting*, 53:493–525.

Euroclear Sweden (2026). Aktieägandet i sverige 2025: Rekord i aktieägande – millennials tar över och männen drar ifrån. Press release and annual share-ownership report. Published 11 February 2026.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417.

Fama, E. F. (1991). Efficient capital markets: Ii. *The Journal of Finance*, 46(5):1575–1617.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2001). Disappearing dividends: Changing firm characteristics or lower propensity to pay? *Journal of Financial Economics*, 60(1):3–43.

Finter, P., Niessen-Ruenzi, A., and Ruenzi, S. (2012). The impact of investor sentiment on the german stock market. *Zeitschrift fur Betriebswirtschaft*, 82:133–163.

Guo, P., Tian, L., and Shi, J. (2026). Economic policy uncertainty, investor sentiment and industry stock market volatility in china: A quantile regression approach. *International Journal of Finance & Economics*, 31:1682–1710.

Ibbotson, R. G. and Jaffe, J. F. (1975). Hot issue markets. *The Journal of Finance*, 30(4):1027–1042.

Investment Company Institute (2026). 2026 investment company fact book. Annual statistical fact book.

Kumar, A. and Lee, C. M. C. (2006). Retail investor sentiment and return comovements. *The Journal of Finance*, 61(5):2451–2486.

Li, H. (2026). ECIN replication package for "Re-Examining Investor Sentiment and Stock Returns: A Replication and Extension of Baker & Wurgler (2006)". Inter-university Consortium for Political and Social Research (ICPSR), Ann Arbor, MI. Distributed February 25, 2026.

Lowry, M. and Schwert, G. W. (2002). IPO market cycles: Bubbles or sequential learning? *The Journal of Finance*, 57(3):1171–1200.

Miller, E. M. (1977). Risk, uncertainty, and divergence of opinion. *The Journal of Finance*, 32(4):1151–1168.

Nasdaq (2026). Omx stockholm 30 index overview. Nasdaq Global Indexes. Accessed 13 May 2026.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

OECD (2026). The swedish equity market, assessment and policy recommendations 2026. OECD Capital Market Series, OECD Publishing, Paris. Published 28 April 2026.

Qiu, L. and Welch, I. (2006). Investor sentiment measures. NBER Working Paper 10794, National Bureau of Economic Research. Revised July 2006.

Ritter, J. R. (1991). The long-run performance of initial public offerings. *The Journal of Finance*, 46(1):3–27.

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3):425–442.

Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *The Journal of Finance*, 52(1):35–55.

Shoaib, G. (2019). Investor sentiment in the nordic stock markets. Master's thesis, University of Jyvaskyla, School of Business and Economics.

Sibley, S. E., Xing, Y., and Zhang, X. (2012). Is "sentiment" sentimental? Working paper.

S&P Dow Jones Indices (2026). U.s. equities market attributes: December 2025. Market commentary. Data as of 31 December 2025.

Stambaugh, R. F., Yu, J., and Yuan, Y. (2012). The short of it: Investor sentiment and anomalies. *Journal of Financial Economics*, 104(2):288–302.

Stambaugh, R. F., Yu, J., and Yuan, Y. (2014). The long of it: Odds that investor sentiment spuriously predicts anomaly returns. *Journal of Financial Economics*, 114(3):613–619.

Stambaugh, R. F., Yu, J., and Yuan, Y. (2015). Arbitrage asymmetry and the idiosyncratic volatility puzzle. *The Journal of Finance*, 70(5):1903–1948.

Statistics Sweden (2026). Shareholders statistics. Statistical database and statistical news. Updated 26 February 2026.

Ung, S. N., Gebka, B., and Anderson, R. D. J. (2024). An enhanced investor sentiment index. *The European Journal of Finance*, 30(8):827–864.

Wheat, C. and Eckerd, G. (2025). A decade in the market: How retail investing behavior has shifted since 2015. JPMorganChase Institute research report. Published 27 August 2025.

Yi, Z. and Mao, N. (2009). Research on measuring investor sentiment in chinese stock market: The construction of cicsi. *Journal of Financial Research*. Referenced for the CICSI cumulative-variance PCA construction rule.

Zhang, X. and Zhang, W. (2023). The construction of an investor sentiment index for sweden and its impact on the stock market. Master's thesis, Lund University School of Economics and Management.